Winding Embeddings and Prime-Power Pisot Floors

Integral Intersections, Composite Windows, and Local Obstructions

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24 October 2026

Unified and extended research manuscript

Abstract

We consolidate three research manuscripts on winding embeddings of Euler–Bessel value fields and on composite values in prime-power Pisot sequences. The analytic part develops polynomial dependence on winding, exact integral-intersection arguments for two full Bessel blocks, three harmonic coordinates, and affine projections, with imported independence and nonvanishing inputs stated explicitly. The arithmetic part gives simultaneous composite windows, shifted coefficient sieves, Pell-family certificates, and field obstructions for rational powers of Mills' constant. The extensions include a composite-progression theorem for polynomial observables in every degree and every prime-power clock, a finite exact winding criterion in bounded degree, and a closed formula for the adjacent-derivative determinant of a normalized Bessel equation. We also separate the two floor branches in finite-prime survival. All reproduced proofs and finite certificates are contained in one source file. Claims depending on unavailable earlier manuscripts remain explicitly qualified. Neither Euler's constant nor Mills' constant is proved transcendental, and the full cubic Pisot problem is not resolved.

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1 Scope, dependencies, and results of this edition

This paper combines the supplied manuscripts [6, 27, 28] into one English article, with one bibliography and globally unique labels. The two subjects share a concern with the exact reach of algebraic and local certificates, but their proofs are independent; no transfer between them is asserted merely because they occur in one paper. Source mathematical results and proofs are retained, apart from duplicate statements replaced by internal references and corrected descriptions of scope and provenance.

Logical status. The continued phase construction is derived from Fischler–Rivoal's preprint and standard connection formulas in Section 2. Its extension to a continuous winding parameter does not require joint algebraic independence. The exact analytic intersection classifications, however, use the joint-independence assertions explicitly listed as (I5) and their complex-scale or multi-Ein analogues imported from [1]. That manuscript is not among the attachments. These classifications are therefore proved here *under those stated inputs*, rather than being presented as independently established unconditional arithmetic theorems. The fractional-order algebraicity-locus statement likewise invokes the unprovided all-derivative nonvanishing theorem of [7]. The determinant identities added below are proved directly and do not depend on either manuscript.

The Pisot progression arguments, the coefficient sieve, the reproduced Pell certificates, and the new prime-clock theorem have proofs in this paper using standard published tools. The discriminant-1524 corollary retains its explicitly identified earlier trace theorem as an assumption. Finite prime patterns that invoke Schinzel's Hypothesis H remain conditional. Historical status claims about earlier numbered research notes are records of the supplied manuscript, not independent verification of absent files.

Further results proved in this edition. The composite-window mechanism extends from cubic traces to polynomial observables of root-power coefficients in arbitrary degree, for every prime clock. In particular, a Pisot number of any degree with a noninteger three-adic trace limit has arbitrarily wide fixed windows of simultaneous composite floors on an eventual arithmetic progression of indices. The analytic additions give a bounded number of exact winding tests under the stated independence inputs and an unconditional closed derivative determinant. Additional finite-sieve constructions clarify why the two floor branches have different local field restrictions. These are extensions of the supplied arguments; no historical priority or new resolution of a previously named global conjecture is asserted.

Notation. The analytic coefficient field is $k = \overline{\mathbb{Q}}$ and $\overline{\mathbf{E}}$ denotes integral closure of the ring of E-values. The arithmetic finite field is written $\mathbb{F}_p$; the calligraphic Euler–Bessel field $\mathfrak{F}$ is a different object. All positive rational powers of Mills' constant refer to its real branch. The words "composite progression" always concern indices, with finite initial exceptions removed after growth is established.

Part I
Winding embeddings and special-function value fields

2 A direct foundation for the winding construction

This section separates the phase construction from the joint algebraic independence hypothesis. The former follows from the published-source construction of Fischler–Rivoal [2, Theorem 2.3 and Proposition 3.1] and the standard connection identities below. It does not require the unprovided Euler–Bessel manuscript. Joint independence, where used in later intersection arguments, remains a separate input.

Write $k = \overline{\mathbb{Q}}$ and let $\mathcal{R} = \mathbb{C}[k_{\mathrm{add}}]$ be the ring of finite exponential sums. Fix a direction $-\pi/2 < \theta < \pi/2$, using the right-hand convention at an anti-Stokes direction. The cited phase construction assigns canonical components f_ρ to an E-function f, with $f(z) = \sum_\rho f_\rho(z)e^{\rho z}$. Each component admits analytic continuation along paths in $\mathbb{C}^\times$: the products $f_\rho(z)e^{\rho z}$ solve an E-operator whose only possible finite singularity is zero. Denote continuation around a positively oriented loop about zero, repeated $j \in \mathbb{Z}$ times, by a superscript $[j]$.

Proposition 2.1 (Continued phase morphisms)**.** *For every integer j the rule*

$$\Phi^{[j]}(f(1))(X) = \sum_\rho f_\rho^{[j]}(1)e^{\rho X}$$

is a well-defined injective k-algebra homomorphism $\mathbf{E} \longrightarrow \mathcal{R}$ satisfying $\Phi^{[j]}(\xi)(1) = \xi$ and $\Phi^{[j]}(e^a) = e^{aX}$ for algebraic a.

Proof. If $f(1) = g(1)$, Beukers' division theorem, as used in the proof of [2, Proposition 3.1], gives an E-function h with $f - g = (z-1)h$. Uniqueness of the canonical expansions gives $f_\rho - g_\rho = (z-1)h_\rho$ componentwise. Continuing this identity along the same loop and evaluating at 1 gives $f_\rho^{[j]}(1) = g_\rho^{[j]}(1)$. Thus the rule is well defined. The component identities for sums and products continue along the loop; for products the phase-ρ component is the finite convolution $\sum_{\alpha+\beta=\rho} f_\alpha g_\beta$. Consequently the rule respects addition and multiplication. Constants are fixed. Continuing the identity $f(z) = \sum_\rho f_\rho(z)e^{\rho z}$ and using that f is entire gives $\sum_\rho f_\rho^{[j]}(1)e^\rho = f(1)$. Evaluation at $X = 1$ is therefore a left inverse, proving injectivity. The exponential identity follows from its single constant component. □

Lemma 2.2 (Normality and subgroup intersections)**.** *The ring $\mathcal{R}$ is integrally closed in its fraction field. For every torsion-free abelian group G and subgroup Λ,*

$$\mathbb{C}[G] \cap \mathrm{Frac}(\mathbb{C}[\Lambda]) = \mathbb{C}[\Lambda].$$

Consequently, if $\xi \in \mathrm{Frac}(\mathbf{E})$ is integral over $\mathbf{E}$, then $\Phi^{[j]}(\xi) \in \mathcal{R}$ for every j.

Proof. For the normality assertion, put the finite supports of a fraction and of the coefficients of its monic integral equation in one finitely generated subgroup of k_{add}. That subgroup is free abelian. Its group algebra is a Laurent polynomial ring over $\mathbb{C}$ and is integrally closed. The fraction therefore belongs to this ring, hence to $\mathcal{R}$. For the intersection assertion, write $u = f/g \in \mathbb{C}[G]$ with $f, g \in \mathbb{C}[\Lambda]$ and $g \neq 0$. Decompose the finite sum u according to its G/Λ cosets. Multiplication by g preserves every coset. The identity $gu = f$ and the fact that the group algebra of a torsion-free abelian group is a domain force every nonidentity coset component of u to vanish. Thus $u \in \mathbb{C}[\Lambda]$. Finally, applying a continued phase homomorphism to a monic equation for ξ makes its image integral over $\mathcal{R}$; normality applies. □

2.1 Positive-argument component formulas

For $t > 0$ choose $\theta = -\pi/4$ and put $p = \pi i$. The connection identity $K_0(ze^{\pi i}) = K_0(z) - \pi i I_0(z)$ shows that the two exponential components of $I_0(tz)$ are

$$F_+(z) = I_0(tz) - K_0(tz)/p, \qquad F_-(z) = K_0(tz)/p,$$

with phases t and $-t$, respectively; thus the actual component functions are $e^{-tz}F_+(z)$ and $e^{tz}F_-(z)$. They have the canonical Gevrey asymptotic expansions in a common large sector about $-\pi/4$, by the standard K asymptotics and connection formula [9, §§10.34, 10.40]. The power series identity

$$Y(tz) = (\gamma + \log(tz/2))I_0(tz) + K_0(tz)$$

then gives the phase components $g(z)F_+(z)$ and $(g(z)+p)F_-(z)$, where $g(z) = \gamma + \log(tz/2)$.

Under j positive loops,

$$K_0 \longmapsto K_0 - 2\pi i j I_0, \qquad K_1 \longmapsto K_1 + 2\pi i j I_1, \qquad g \longmapsto g + 2\pi i j.$$

The first two identities are [9, 10.34.5]; differentiation of the two I_0 components gives the corresponding I_1 components. Writing $I = I_0(t)$, $J = I_1(t)$, $K = K_0(t)$, and $L = K_1(t)$ therefore yields

$$\begin{aligned}
\Phi^{[j]}(I_0(t)) &= A_j e^{tX} + B_j e^{-tX},\\
\Phi^{[j]}(I_1(t)) &= A'_j e^{tX} + B'_j e^{-tX},\\
\Phi^{[j]}(Y(t)) &= (g + j\tau)A_j e^{tX} + (g + j\tau + p)B_j e^{-tX},
\end{aligned}$$

where $g = \gamma + \log(t/2)$, $\tau = 2\pi i$, and

$$\begin{aligned}
A_j &= e^{-t}((1+2j)I - K/p), & B_j &= e^{t}(K/p - 2jI),\\
A'_j &= e^{-t}((1+2j)J + L/p), & B'_j &= e^{t}(-2jJ - L/p).
\end{aligned}$$

These are the positive-scale formulas used in the winding line. For $a > 0$, the identity $\mathrm{Ein}(az) = \gamma + \log(az) + E_1(az)$, together with the phases $0, -a$, gives directly

$$\Phi^{[j]}(\mathrm{Ein}(a)) = \gamma + \log a + j\tau + (\mathrm{Ein}(a) - \gamma - \log a - j\tau)e^{a(1-X)}.$$

The identity and its branch convention are recorded in [9, 6.2.4]. These derivations concern positive arguments; complex-scale normalizations must be justified with a common summation direction and the corresponding branches separately.

2.2 Continuous winding without algebraic independence

Proposition 2.3 (Interpolation of genuine embeddings)**.** *Let $u_1, \dots, u_N$ be generators of a subfield F of* Frac($\mathbf{E}$)*, and suppose their continued phase images are rational functions $\Psi_i(z, s)$ in a finite phase torus z and a parameter s, at infinitely many distinct parameter values s_j. Suppose also that, for every $s_0 \in \mathbb{C}$, all $\Psi_i(z, s_0)$ are defined as rational functions, are regular at $z = \mathbf{1}$, and satisfy $\Psi_i(\mathbf{1}, s_0) = u_i$. Then substitution defines a field embedding $\Phi_{s_0} : F \to \mathbb{C}(z)$ for every s_0, regardless of algebraic relations among the u_i.*

Proof. If $P(u_1, \dots, u_N) = 0$ for $P \in k[T_1, \dots, T_N]$, then the genuine embeddings give $P(\Psi(z, s_j)) = 0$ at infinitely many s_j. Clearing rational denominators shows that $P(\Psi(z, s)) = 0$ identically: a polynomial in s over $\mathbb{C}(z)$ cannot have infinitely many distinct roots. For a fixed s_0, the assumed regular substitution therefore preserves every algebraic relation. Conversely, when $Q(u_1, \dots, u_N) \neq 0$, the rational function $Q(\Psi(z, s_0))$ is nonzero because its value at $z = \mathbf{1}$ is that same nonzero number. Substitution thus extends to the fraction field and is injective. □

This proposition applies to the explicit positive Euler–Bessel images. It removes joint independence from the construction of Φ_s and from the polynomial-winding proof: the latter only uses genuine embeddings, normality, subgroup intersections, and the absence of parameter-only denominators, all supplied above. Joint independence is still used when the later integral-intersection arguments identify the coordinate fields with rational function fields and compare coefficients. It cannot be inferred merely from the existence of the phase embeddings.

3 Introduction

The manuscript [1] studies the integral closure $\overline{\mathbf{E}}$ of the ring $\mathbf{E}$ of E-values through the continued phase embeddings $\Phi^{[j]} = \Phi^{[j]}_{-\pi/4}$, $j \in \mathbb{Z}$, obtained from the phase embedding of Fischler and Rivoal [2] by continuing the canonical components of an E-function j times around the origin. An element of $\mathrm{Frac}(\mathbf{E})$ whose image has a finite pole is not integral over $\mathbf{E}$. The manuscript closes with four questions [1, §"Scope"]:

(Q1) the arithmetic nature of γ, δ, and of any specified member of $K_0(2), K_0(4), K_0(8)$;

(Q2) algebraic independence of $\gamma, \delta, \kappa = K_0(2)/I_0(2)$ and $C = I_1(2)/I_0(2)$;

(Q3) integral-intersection classifications with two or more harmonic Bessel coordinates at dependent scales, for example $\overline{\mathbb{Q}}(\beta(2), C(2), \beta(4), C(4))$;

(Q4) an effective criterion selecting the density-one set of integer arguments of [1, Theorem jden:density].

The following sections develop the winding construction and its consequences. The missing joint-independence input is retained explicitly, as specified in Section 1.

Throughout, $k = \overline{\mathbb{Q}}$, $p = i\pi$, $\tau = 2\pi i$, and $\mathcal{R} = \mathbb{C}[k_{\mathrm{add}}]$ is the ring of finite sums $\sum c_\rho e^{\rho X}$ with $\rho \in k$. For positive algebraic t we use $I_0(t), I_1(t), K_0(t)$, the harmonic series $Y(t) = \sum_{n\geq 0} H_n (t/2)^{2n}/(n!)^2$, and

$$\beta(t) = \frac{Y(t)}{I_0(t)} = \gamma + \log\frac{t}{2} + \frac{K_0(t)}{I_0(t)}, \qquad C(t) = \frac{I_1(t)}{I_0(t)}, \qquad h(t) = \frac{1}{tI_0(t)^2}.$$

Theorem A (the winding line; Proposition 4.3, Theorem 4.5)**.** *On the Euler–Bessel fields $\mathfrak{F}$ of Definition 4.2 the embeddings $\Phi^{[j]}$ are the members $s \in \gamma + \tau\mathbb{Z}$ of a family of field embeddings Φ_s, $s \in \mathbb{C}$, whose defining formulas do not contain γ. If $\xi \in \mathfrak{F}$ is integral over $\mathbf{E}$, its image, written in exponential coordinates, is a Laurent polynomial whose coefficients are polynomials in s.*

Theorem B (Theorem 5.1)**.** *For positive algebraic a and distinct positive algebraic t_1, t_2,*

$$\overline{\mathbf{E}} \cap k(e^a, \mathrm{Ein}(a), \beta(t_1), C(t_1), \beta(t_2), C(t_2)) = k[e^a, e^{-a}, \mathrm{Ein}(a)],$$

and $\overline{\mathbf{E}} \cap k(\beta(t_1), C(t_1), \beta(t_2), C(t_2)) = k$.

Proposition C (Proposition 6.1)**.** *On the fields $\mathfrak{F}$, algebraic relations among E-values, joint independence and the conclusion of Theorem A remain true when γ is replaced, in the definitions of δ and $K_0(t)$, by an arbitrary complex number. Arguments using only these inputs cannot settle* (Q1) *or* (Q2)*.*

The pure two-block intersection in Theorem B is already reported in the current version of [1]. The retained exponential–Ein pair is the additional feature of the mixed statement. For rationally independent scales it recovers the case $m = 2$ of [1, Theorem mbmulti:main]. Remark 6.4 explains why three or more harmonic coordinates at mutually dependent scales need

a further idea. Proposition C extends the elementary model of [1, §"Scope"], which concerns algebraic input only, to the integrality information carried by the phase embeddings on $\mathfrak{F}$. Within that information, γ enters only through the position $s = \gamma$ of the genuine embedding on the line (Remark 6.2). Other inputs of [1] do see γ, for example the positivity of $K_0(t)$ and the critical approximation limits; Proposition C says nothing about them.

4 The winding line

4.1 Inputs

We use the following results of [1].

(I1) [1, Lemma mbc:continued] For $j \in \mathbb{Z}$, $\Phi^{[j]} \colon \mathbf{E} \to \mathcal{R}$ is an injective k-algebra homomorphism with $\Phi^{[j]}(\xi)(1) = \xi$ and $\Phi^{[j]}(e^a) = e^{aX}$; it extends to $\mathrm{Frac}(\mathbf{E})$.

(I2) [1, Lemma mbc:pole] If $\xi \in \mathrm{Frac}(\mathbf{E}) \cap \overline{\mathbf{E}}$, then $\Phi^{[j]}(\xi) \in \mathcal{R}$ for every j.

(I3) [1, (mbc:Iimage)–(mbc:Yimage), (dcr:eta-image)] the explicit winding images, recalled in Lemma 4.1.

(I4) [1, Lemma mbmulti:cosets] For a torsion-free abelian group G and a subgroup Λ, $\mathbb{C}[G] \cap \mathrm{Frac}(\mathbb{C}[\Lambda]) = \mathbb{C}[\Lambda]$.

(I5) [1, Theorem cf:joint] For positive algebraic a and distinct positive algebraic $t_1, \dots, t_m$, the numbers $e^a, \mathrm{Ein}(a), I_0(t_i), C(t_i), \beta(t_i)$ are algebraically independent over k.

4.2 Closed form of the images

Lemma 4.1. *Let $t > 0$ be algebraic, $j \in \mathbb{Z}$ and $s = \gamma + j\tau$. Put*

$$v = \beta(t) - \log\frac{t}{2} - s = \frac{K_0(t)}{I_0(t)} - j\tau, \qquad q = e^{2t(1-X)}, \qquad \mu = \frac{q-1}{p + v(q-1)}.$$

Then

$$\Phi^{[j]}(\beta(t)) = \beta(t) - v(v-p)\mu, \qquad \Phi^{[j]}(C(t)) = C(t) - h(t)\mu. \tag{1}$$

For algebraic $a > 0$ and $v_a = \mathrm{Ein}(a) - \log a - s$,

$$\Phi^{[j]}(\mathrm{Ein}(a)) = \mathrm{Ein}(a) + v_a(e^{a(1-X)} - 1). \tag{2}$$

Proof. Write I, J, K, L for $I_0(t), I_1(t), K_0(t), K_1(t)$, $\kappa_t = K/I$, $g = \gamma + \log(t/2)$ and $\varepsilon = 1/p$. By (I3), $\Phi^{[j]}(I_0(t)) = A_j e^{tX} + B_j e^{-tX}$, $\Phi^{[j]}(I_1(t)) = A'_j e^{tX} + B'_j e^{-tX}$ and $\Phi^{[j]}(Y(t)) = (g + j\tau)A_j e^{tX} + (g + j\tau + p)B_j e^{-tX}$, with

$$A_j = e^{-t}((1+2j)I - \varepsilon K),\ B_j = e^{t}(\varepsilon K - 2jI),\ A'_j = e^{-t}((1+2j)J + \varepsilon L),\ B'_j = e^{t}(-2jJ - \varepsilon L).$$

Since $e^{-2tX} = e^{-2t}q$ and $2j - \varepsilon\kappa_t = (j\tau - \kappa_t)/p = -v/p$,

$$A_j + B_j e^{-2t}q = e^{-t}I\Big(1 + \frac{v}{p}(q-1)\Big), \qquad pB_j e^{-2t}q = e^{-t}(K - j\tau I)q = e^{-t}Ivq.$$

Hence $\Phi^{[j]}(\beta(t)) = g + j\tau + pqv/(p + v(q-1))$. As $g + j\tau = \beta(t) - v$, this equals $\beta(t) + v(pq - p - v(q-1))/(p + v(q-1)) = \beta(t) - v(v-p)\mu$. Similarly $A'_j + B'_j e^{-2t}q = e^{-t}(J + (2jJ + \varepsilon L)(1-q))$, and the Wronskian $IL + JK = 1/t$ gives $\Phi^{[j]}(C(t)) - C(t) = \varepsilon(1-q)/(tI^2(1 + (v/p)(q-1))) = -h(t)\mu$. Formula (2) is [1, (dcr:eta-image)]: there $\eta_j = g_a + j\tau + d_j e^{a(1-X)}$ with $g_a = \gamma + \log a$ and $d_j = \mathrm{Ein}(a) - g_a - j\tau = v_a$. □

4.3 The family

Definition 4.2. Fix positive algebraic a and distinct positive algebraic $t_1, \dots, t_m$, and write $\eta_a = \mathrm{Ein}(a)$, $\beta_i = \beta(t_i)$, $C_i = C(t_i)$, $h_i = h(t_i)$. By (I5) the field

$$\mathfrak{F} = k(e^a, \eta_a, \beta_1, C_1, \dots, \beta_m, C_m)$$

is purely transcendental on these generators; identify it with $k(E, U, x_1, y_1, \dots, x_m, y_m)$. For $s \in \mathbb{C}$ put

$$v_i(s) = \beta_i - \log\frac{t_i}{2} - s, \quad v_a(s) = \eta_a - \log a - s, \quad q_i = e^{2t_i(1-X)}, \quad \mu_i(s) = \frac{q_i - 1}{p + v_i(s)(q_i - 1)},$$

and let Φ_s send

$$E \mapsto e^{aX}, \quad U \mapsto \eta_a + v_a(s)\big(e^{a(1-X)} - 1\big), \quad x_i \mapsto \beta_i - v_i(s)\big(v_i(s) - p\big)\mu_i(s), \quad y_i \mapsto C_i - h_i\mu_i(s).$$

Proposition 4.3. *Let $\mathcal{M}$ be the field of meromorphic functions of $X \in \mathbb{C}$.*

1. *For every $s \in \mathbb{C}$, Φ_s extends to a field embedding $\mathfrak{F} \to \mathcal{M}$ with $\Phi_s(\xi)(1) = \xi$.*
2. *$\Phi_{\gamma + j\tau} = \Phi^{[j]}|_{\mathfrak{F}}$ for every $j \in \mathbb{Z}$.*
3. *The data defining Φ_s are e^a, η_a, $\log a$, β_i, C_i, h_i, $\log(t_i/2)$, π and s. Euler's constant does not occur.*

Proof. At $X = 1$ one has $q_i = 1$, $\mu_i = 0$ and $e^{a(1-X)} = 1$, so each generator is mapped to a function whose value at $X = 1$ is the generator itself, for every s. A nonzero polynomial Q is therefore mapped to a function with value $Q(e^a, \eta_a, \beta, C) \neq 0$ at $X = 1$, by (I5). Thus the substitution homomorphism is injective and extends to $\mathfrak{F}$, with $\Phi_s(\xi)(1) = \xi$. Part (2) holds because both sides are field homomorphisms agreeing on the generators (Lemma 4.1). Part (3) is read off from the formulas. □

Remark 4.4. On $\mathfrak{F}$ the images are rational functions of s, and they depend on the winding number only through s because the only monodromy comes from $\log z$, carried by K_0 and Ein. For a general E-value the components in [2] may have non-integral rational exponents at 0; then $j \mapsto \Phi^{[j]}(\xi)$ is a quasi-polynomial, polynomial on each residue class of j modulo a common denominator.

4.4 Polynomial winding

Let $\Lambda \subset \mathbb{R}$ be the group generated by $a, 2t_1, \dots, 2t_m$, choose a basis $\lambda_1, \dots, \lambda_r$ and put $z_k = e^{\lambda_k(1-X)}$. The λ_k are $\mathbb{Q}$-linearly independent, so $z_1, \dots, z_r$ are algebraically independent functions; q_i and $e^{a(1-X)}$ are Laurent monomials in z, and $e^{aX} = e^a e^{-a(1-X)}$. Write $\Psi(z, s)$ for the tuple of images of the generators, a tuple of rational functions of (z, s). For $\xi = P(e^a, \eta_a, \beta, C)/Q(e^a, \eta_a, \beta, C)$ with coprime $P, Q \in k[E, U, x, y]$, put

$$F_\xi = P(\Psi)/Q(\Psi) \in \mathbb{C}(z_1, \dots, z_r, s),$$

so that $\Phi_s(\xi)(X) = F_\xi(z(X), s)$; since the $z_k(X)$ are algebraically independent, F_ξ is determined by ξ.

Theorem 4.5 (polynomial winding)**.** *If $\xi \in \mathfrak{F} \cap \overline{\mathbf{E}}$, then $F_\xi \in \mathbb{C}[z_1^{\pm 1}, \dots, z_r^{\pm 1}, s]$. In particular, for every one-parameter subgroup $w \mapsto (w^{u_1}, \dots, w^{u_r})$, the restriction of F_ξ lies in $\mathbb{C}[w, w^{-1}, s]$.*

Proof. Work in the factorial ring $\mathbb{C}[z^{\pm 1}, s]$, whose units are the elements cz^b with $c \in \mathbb{C}^\times$. Write $F_\xi = N/D$ with $N, D \in \mathbb{C}[z^{\pm 1}, s]$ coprime, and let $s_j = \gamma + j\tau$.

Integer points. By Proposition 4.3(2) and (I2), $\Phi_{s_j}(\xi) = \Phi^{[j]}(\xi) \in \mathcal{R}$. It also lies in $\mathrm{Frac}(\mathbb{C}[\Lambda])$, since the images of the generators do, so by (I4) it lies in $\mathbb{C}[\Lambda] = \mathbb{C}[z^{\pm 1}]$. For all but finitely many j the specialization $D(\cdot, s_j)$ is not identically zero, and then $N(\cdot, s_j)/D(\cdot, s_j) \in \mathbb{C}[z^{\pm 1}]$.

Denominators. Let D_0 be an irreducible factor of D and suppose D_0 is not associate to an element of $\mathbb{C}[s]$. Then D_0 has at least two monomials in z with coefficients not identically zero, so for all but finitely many s_0 the Laurent polynomial $D_0(\cdot, s_0)$ is not a monomial and has zeros in the torus $(\mathbb{C}^\times)^r$; its zero set there has dimension $r-1$. The common zero set of D_0 and N in $(\mathbb{C}^\times)^r \times \mathbb{C}$ has dimension at most $r-1$, so its fibre over all but finitely many s_0 has dimension at most $r-2$. Hence for all but finitely many j there is a point of the torus where $D_0(\cdot, s_j)$ vanishes and $N(\cdot, s_j)$ does not, and $N(\cdot, s_j)/D(\cdot, s_j)$ has a pole there. This contradicts the first step. So $D = c(s)$ up to a unit, with $c \in \mathbb{C}[s] \setminus \{0\}$.

Every s_0. Fix $s_0 \in \mathbb{C}$. The tuple $\Psi(\cdot, s_0)$ consists of well-defined rational functions of z, and $Q(\Psi(\cdot, s_0)) \not\equiv 0$ because its value at $z = (1, \dots, 1)$ is $Q(e^a, \eta_a, \beta, C) \neq 0$. Hence $G_0 = P(\Psi(\cdot, s_0))/Q(\Psi(\cdot, s_0))$ is a rational function of z, and $F_\xi(z, s) \to G_0(z)$ as $s \to s_0$ for z outside a proper Zariski-closed set. If $c(s_0) = 0$, coprimality gives $N(\cdot, s_0) \not\equiv 0$, and $|F_\xi(z, s)| \to \infty$ as $s \to s_0$ for z outside a proper closed set, which is impossible. Therefore c has no zeros and $F_\xi \in \mathbb{C}[z^{\pm 1}, s]$. Restriction to a one-parameter subgroup preserves this form. □

Thus integrality is a polynomial identity in the winding variable. The pole criterion can be applied at every complex s, and the asymptotic charts of [1], taken there along integer windings $j \to \infty$, become limits of polynomials in (w, s).

5 Two harmonic coordinates at arbitrary scales

Theorem 5.1. *Let a be positive algebraic and $t_1 \neq t_2$ positive algebraic. Then*

$$\overline{\mathbf{E}} \cap k(e^a, \mathrm{Ein}(a), \beta(t_1), C(t_1), \beta(t_2), C(t_2)) = k[e^a, e^{-a}, \mathrm{Ein}(a)].$$

Corollary 5.2. *For distinct positive algebraic t_1, t_2, $\overline{\mathbf{E}} \cap k(\beta(t_1), C(t_1), \beta(t_2), C(t_2)) = k$. In particular $\overline{\mathbf{E}} \cap k(\beta(2), C(2), \beta(4), C(4)) = k$.*

Proof. Choose any a and put $G = k(\beta(t_1), C(t_1), \beta(t_2), C(t_2))$. By the theorem, $\overline{\mathbf{E}} \cap G$ lies in $k[e^{\pm a}, \mathrm{Ein}(a)] \cap G$. This is k, because e^a and $\mathrm{Ein}(a)$ are algebraically independent over G by (I5). □

The inclusion $\supseteq$ in Theorem 5.1 holds since $e^{\pm a}$ and $\mathrm{Ein}(a)$ are E-values. The rest of this section proves $\subseteq$.

5.1 Normalization on a one-parameter subgroup

In the notation of §2 with $m = 2$, write $q_i = z^{n_i}$ and $e^{a(1-X)} = z^{m_0}$. For an integer L let $\mathbf{u} \in \mathbb{Z}^r$ have k-th entry the nearest integer to $L\lambda_k$; then $\langle n_i, \mathbf{u}\rangle = 2Lt_i + O(1)$ and $\langle m_0, \mathbf{u}\rangle = La + O(1)$. For large L the restriction to the one-parameter subgroup $w \mapsto w^{\mathbf{u}}$ gives

$$q_i \mapsto w^{N_i}, \qquad e^{a(1-X)} \mapsto w^M, \qquad N_1 \neq N_2,\ N_1, N_2, M \in \mathbb{Z}_{>0}.$$

Let $v = v_1(s)$; this is an affine bijection of $\mathbb{C}$. Put $\Delta = v_2 - v_1 = K_0(t_2)/I_0(t_2) - K_0(t_1)/I_0(t_1)$ and $\Delta_a = v_a - v_1$; both are constants. On the subgroup the images of $(E, U, x_1, y_1, x_2, y_2)$ form

$$\Psi(w, v) = (e^a w^{-M},\ \eta_a + (v + \Delta_a)(w^M - 1),\ B_1, \Gamma_1, B_2, \Gamma_2),$$

$$B_i = \beta_i - v_i(v_i - p)\mu_i, \quad \Gamma_i = C_i - h_i\mu_i, \quad \mu_i = \frac{w^{N_i} - 1}{D_i}, \quad D_i = p + v_i(w^{N_i} - 1),$$

with $v_1 = v$ and $v_2 = v + \Delta$. Let $\xi = R(e^a, \eta_a, \beta_1, C_1, \beta_2, C_2)$ lie in $\overline{\mathbf{E}}$, where $R = P/Q$ with coprime $P, Q \in k[E, U, x_1, y_1, x_2, y_2]$. By Theorem 4.5,

$$F := R \circ \Psi \in \mathbb{C}[w, w^{-1}, v].$$

We use this in the following form.

(∗) *There is no point* $(w_0, v_0) \in \mathbb{C}^\times \times \mathbb{C}$ *with all* $D_i(w_0, v_0) \neq 0$, $Q(\Psi(w_0, v_0)) = 0$ *and* $P(\Psi(w_0, v_0)) \neq 0$.

Indeed F would have a pole there.

5.2 Independence

Lemma 5.3. *1. The numbers* $e^a, \eta_a, \beta_1, C_1, \beta_2, C_2, h_1, h_2$ *are algebraically independent over* k.

2. For $i \in \{1, 2\}$ *and* $u_i^* = \eta_a - Mp/N_i$, *the numbers* $e^a, C_1, C_2, h_1, h_2, u_i^*$ *are algebraically independent over* k.

Proof. (1) is (I5), since h_i is a nonzero algebraic multiple of $I_0(t_i)^{-2}$. For (2), let S be a subset of $\{e^a, C_1, C_2, h_1, h_2\}$. The field $k(S)$ is stable under complex conjugation. If u_i^* were algebraic over $k(S)$, so would be its conjugate $\eta_a + Mp/N_i$, hence η_a, contradicting (1). If adjoining u_i^* to the independent set $\{e^a, C_1, C_2, h_1, h_2\}$ produced a relation, some element x of the set would be algebraic over $k(S)(u_i^*)$, with S the other elements, but not over $k(S)$. Exchange would make u_i^* algebraic over $k(S \cup \{x\})$, which was just excluded. □

5.3 Asymptotic regimes

Lemma 5.4. *Let* $|v| \to \infty$. *The following hold uniformly for* c *in compact subsets of the indicated sets.*

(A) $w = 1 + c\,v^{-2}$, $c \in \mathbb{C}$: $\Psi \to (e^a, \eta_a, \beta_1 - N_1 c/p,\ C_1,\ \beta_2 - N_2 c/p,\ C_2)$, *and* $D_i \to p$.

(B) $w = 1 + c\,v^{-3/2}$ *for a fixed branch of* $v^{1/2}$, $c \in \mathbb{C}$: $(E, U, \Gamma_1, \Gamma_2) \to (e^a, \eta_a, C_1, C_2)$, $D_i \to p$, *and*

$$B_i = \beta_i - N_i \frac{c}{p}\, v^{1/2} + N_i^2 \Big(\frac{c}{p}\Big)^2 + O(|v|^{-1/2}).$$

(C$_i$) w *the branch near* 1 *of* $(1 - p/v_i + c/v_i^2)^{1/N_i}$, $c \in \mathbb{C}^\times$, *and* $l \neq i$: $(E, U, \Gamma_i, \Gamma_l) \to (e^a, u_i^*, C_i + ph_i/c, C_l)$, $D_i = c/v_i$, $D_l \to p(N_i - N_l)/N_i$, *and*

$$B_i = \frac{p}{c}\, v^2 + O(|v|), \qquad B_l = \frac{N_l}{N_i - N_l}\, v + O(1).$$

(P$_i$) *On* $\Pi_i = \{D_i = 0\}$, *parametrized by* $w = 1 + \omega$ *with* $\omega \to 0$ *(so* $v_i = p/(1 - (1+\omega)^{N_i}) \to \infty$*), and* $l \neq i$: $(E, U, \Gamma_l) \to (e^a, u_i^*, C_l)$, $D_l \to p(N_i - N_l)/N_i$,

$$v_i(v_i - p) = \frac{p^2}{N_i^2 \omega^2} + O(|\omega|^{-1}), \qquad B_l = -\frac{pN_l}{N_i(N_i - N_l)}\, \omega^{-1} + O(1).$$

Proof. Put $\theta_i = w^{N_i} - 1$, so $\mu_i = \theta_i/(p + v_i\theta_i)$ and $B_i - \beta_i = -v_i(v_i - p)\mu_i$.

(A) Here $\theta_i = N_i c v^{-2} + O(|v|^{-4})$, $v_i\theta_i = O(|v|^{-1})$, $\mu_i = N_i c/(pv^2) + O(|v|^{-3})$ and $v_i(v_i - p) = v^2 + O(|v|)$. Also $(v + \Delta_a)(w^M - 1) = O(|v|^{-1})$.

(B) Here $\theta_i = N_i c v^{-3/2} + O(|v|^{-3})$ and $v_i\theta_i = N_i c v^{-1/2} + O(|v|^{-3/2})$. Expanding $\mu_i = (\theta_i/p)(1 - v_i\theta_i/p + (v_i\theta_i/p)^2 - \cdots)$ gives $v_i(v_i - p)\mu_i = N_i(c/p)v^{1/2} - N_i^2(c/p)^2 + O(|v|^{-1/2})$. Moreover $\Gamma_i - C_i = O(|v|^{-3/2})$ and $U - \eta_a = O(|v|^{-1/2})$.

(C_i) By construction $\theta_i = -p/v_i + c/v_i^2$, so $D_i = c/v_i$, $\mu_i = -p/c + 1/v_i$, $\Gamma_i \to C_i + ph_i/c$ and $B_i = \beta_i + v_i(v_i - p)(p/c - 1/v_i) = (p/c)v^2 + O(|v|)$. For $l \neq i$, $\theta_l = (N_l/N_i)\theta_i + O(|v|^{-2})$, hence $v_l\theta_l = -(N_l/N_i)p + O(|v|^{-1})$, $D_l \to p(N_i - N_l)/N_i$, $\mu_l = -N_l/((N_i - N_l)v) + O(|v|^{-2})$, $\Gamma_l \to C_l$ and $B_l = N_l v/(N_i - N_l) + O(1)$. Finally $w^M - 1 = (M/N_i)\theta_i + O(|v|^{-2})$ gives $U \to \eta_a - Mp/N_i = u_i^*$.

(P_i) Here $\theta_i = N_i\omega + O(\omega^2)$ and $v_i = -p/\theta_i = -p/(N_i\omega) + O(1)$. For $l \neq i$, $\theta_l = (N_l/N_i)\theta_i + O(\omega^2)$ and $v_l = v_i + O(1)$, so $v_l\theta_l \to -(N_l/N_i)p$, $D_l \to p(N_i - N_l)/N_i$ and $\mu_l = N_iN_l\omega/(p(N_i - N_l)) + O(\omega^2)$. This gives the formula for B_l and $\Gamma_l \to C_l$. Also $(v + \Delta_a)(w^M - 1) \to -pM/N_i$. □

5.4 Proof of Theorem 5.1

Write $T = (e^a, \eta_a, \beta_1, C_1, \beta_2, C_2)$ and $N' = (0, 0, N_1, 0, N_2, 0)$. Throughout, f denotes an irreducible factor of Q in $k[E, U, x, y]$; since $Q = f\,Q_1$, every zero of $f \circ \Psi$ at which $Q_1 \circ \Psi$ is finite is a zero of $Q \circ \Psi$.

Step 1: every f is invariant under translation by N'. Put $\varphi(c) = f(T - (c/p)N')$, the limit of $f \circ \Psi$ in regime (A). Suppose φ is not constant and let c^* be a zero. By Hurwitz's theorem $f \circ \Psi(1 + cv^{-2}, v)$ has zeros $c_v \to c^*$ for large $|v|$. The set $V = Z(P) \cap Z(f)$ has dimension at most 4, and the Zariski closure of $V + \mathbb{C}N'$ is a proper subvariety of $\mathbb{C}^6$ defined over k. By Lemma 5.3(1) the point T does not lie on it, so $P(T - (c^*/p)N') \neq 0$. Hence $P \circ \Psi \neq 0$ at these zeros for large $|v|$, while $Q \circ \Psi = 0$ and $D_i \neq 0$, contradicting $(*)$. Thus $\lambda \mapsto f(T + \lambda N')$ is constant, and since T is generic over k, $f(X + \lambda N') = f(X)$ identically.

Step 2: Q does not depend on x_1, x_2. Put $\varpi = N_2x_1 - N_1x_2$ and $\sigma = (N_1x_1 + N_2x_2)/(N_1^2 + N_2^2)$. By Step 1, $Q \in k[E, U, \varpi, y_1, y_2]$. Let f have $\deg_\varpi f > 0$. In regime (B),

$$\varpi(B) \to \varpi_0 + \kappa_N(c/p)^2, \qquad \varpi_0 = N_2\beta_1 - N_1\beta_2, \quad \kappa_N = N_1N_2(N_1 - N_2) \neq 0,$$

and $\sigma(B) = -(c/p)v^{1/2} + O(1)$. The polynomial $f(e^a, \eta_a, W, C_1, C_2)$ has positive degree in W, its leading coefficient being nonzero at an independent point. Its roots W^* are algebraic over $k(e^a, \eta_a, C_1, C_2)$, while ϖ_0 is transcendental over that field. So $\varphi(c) = f(e^a, \eta_a, \varpi_0 + \kappa_N(c/p)^2, C_1, C_2)$ has a zero $c^* \neq 0$, and Hurwitz gives zeros $c_v \to c^*$ of $f \circ \Psi$. The other factors of Q converge, so $Q \circ \Psi = 0$ at these zeros, and $D_i \neq 0$ there. Write $P = \sum_n P_{(n)}(E, U, \varpi, y)\sigma^n$ and let n_0 be the largest n with $f \nmid P_{(n)}$; it exists because $f \nmid P$. At the zeros the terms with $f \mid P_{(n)}$ vanish exactly. Since f is irreducible of positive degree in ϖ and $f \nmid P_{(n_0)}$, the resultant in ϖ is nonzero; evaluated at the independent point (e^a, η_a, C_1, C_2), it shows $P_{(n_0)}(e^a, \eta_a, W^*, C_1, C_2) \neq 0$. Therefore

$$P \circ \Psi = P_{(n_0)}(e^a, \eta_a, W^*, C_1, C_2)\,(-c^*/p)^{n_0}\,v^{n_0/2}(1 + o(1)) \neq 0,$$

contradicting $(*)$. Hence $Q \in k[E, U, y_1, y_2]$.

Step 3: Q does not depend on y_1, y_2. Let f have $\deg_{y_1} f > 0$ and use regime (C_1). There $f \circ \Psi$ converges to $\varphi(c) = f(e^a, u_1^*, C_1 + ph_1/c, C_2)$. By Lemma 5.3(2), $f(e^a, u_1^*, Y, C_2)$ has positive degree in Y, and each root Y^* is algebraic over $k(e^a, u_1^*, C_2)$, hence $Y^* \neq C_1$. So φ has the zero $c^* = ph_1/(Y^* - C_1) \in \mathbb{C}^\times$, and Hurwitz gives zeros $c_v \to c^*$ of $f \circ \Psi$; there $Q \circ \Psi = 0$, $D_1 = c_v/v_1 \neq 0$ and $D_2 \neq 0$. Write $P = \sum_{a',b'} P_{(a',b')}(E, U, y_1, y_2)x_1^{a'}x_2^{b'}$ and let W_0 be the largest value of $2a' + b'$ with $f \nmid P_{(a',b')}$. Terms with $f \mid P_{(a',b')}$ vanish exactly at the zeros, and

$B_1^{a'}B_2^{b'} = (p/c)^{a'}\varrho^{b'}v^{2a'+b'}(1+O(|v|^{-1}))$ with $\varrho = N_2/(N_1-N_2)$. Hence at the zeros

$$v^{-W_0}P\circ\Psi \longrightarrow \sum_{\substack{2a'+b'=W_0\\ f\nmid P_{(a',b')}}} P_{(a',b')}(e^a,u_1^*,Y^*,C_2)\Big(\frac{Y^*-C_1}{h_1}\Big)^{a'}\varrho^{b'},$$

because $p/c^* = (Y^*-C_1)/h_1$. Each displayed coefficient $P_{(a',b')}(e^a,u_1^*,Y^*,C_2)$ is nonzero, by the resultant argument at the independent point (e^a,u_1^*,C_2). The limit is a polynomial in h_1^{-1} with coefficients in $\overline{k(e^a,u_1^*,C_1,C_2)}$, not all zero, and h_1 is transcendental over that field by Lemma 5.3(2). So the limit is nonzero, contradicting $(*)$. Regime (C_2) treats y_2 in the same way. Hence $Q\in k[E,U]$.

Step 4: P does not depend on x_1,y_1,x_2,y_2. Now $Q\circ\Psi = Q(e^aw^{-M},\eta_a+(v+\Delta_a)(w^M-1))$ lies in $\mathbb{C}[w^{\pm1},v]$, so $P\circ\Psi = F\cdot Q\circ\Psi\in\mathbb{C}[w^{\pm1},v]$ has no pole along Π_1. Write $P=\sum_{n=0}^d P_n$ with P_n homogeneous of degree n in (x_1,y_1), coefficients in $k[E,U,x_2,y_2]$, and $P_d\neq0$. Near a point of Π_1 off Π_2, all coordinates of Ψ other than (B_1,Γ_1) are regular, and $(B_1,\Gamma_1) = (\beta_1,C_1)-\mu_1\,(v_1(v_1-p),h_1)$, where μ_1 has a simple pole along Π_1 (D_1 is linear in v with coefficient $w^{N_1}-1\neq0$ there). Thus $P\circ\Psi=\sum_n g_n\mu_1^n$ with regular g_n and $g_d = (-1)^dP_d(v_1(v_1-p),h_1;B_2,\Gamma_2,E,U)$. If $d>0$, regularity forces $g_d|_{\Pi_1}\equiv0$. Write $P_d=\sum_\nu c_\nu(x_2,y_2,E,U)Z_1^\nu Z_2^{d-\nu}$ in the variables (Z_1,Z_2) standing for (x_1,y_1), let $e_\nu=\deg_{x_2}c_\nu$ and $\widehat{E}=\max\{e_\nu+2\nu : c_\nu\neq0\}$. In regime (P_1), with $\tilde b=-N_2/(N_1(N_1-N_2))$ and ℓc_ν the leading coefficient of c_ν in x_2 (a polynomial in y_2,E,U),

$$\omega^{\widehat{E}}g_d \longrightarrow (-1)^dp^{\widehat{E}}\sum_{e_\nu+2\nu=\widehat{E}}\ell c_\nu(C_2,e^a,u_1^*)\,\tilde b^{\,e_\nu}N_1^{-2\nu}h_1^{d-\nu}.$$

The powers of p combine to $p^{\widehat{E}}$ because each term contributes $p^{2\nu}$ from $v_1(v_1-p)$ and p^{e_ν} from B_2. The sum is a polynomial in h_1 whose coefficients $\ell c_\nu(C_2,e^a,u_1^*)$ are nonzero, by Lemma 5.3(2), and h_1 is transcendental over $k(C_2,e^a,u_1^*)$. So the limit is nonzero and $g_d|_{\Pi_1}\not\equiv0$, a contradiction. Hence P does not depend on (x_1,y_1); regime (P_2) removes (x_2,y_2). Thus $R\in k(E,U)$.

Step 5: $R\in k[E^{\pm1},U]$. Now $F(w,v)=R(e^aw^{-M},\eta_a+(v+\Delta_a)(w^M-1))\in\mathbb{C}[w^{\pm1},v]$. Let f be an irreducible factor of Q not associate to E. Its zero set is a curve not contained in $\{E=0\}$, nor in $\{E=e^a\}$ since $e^a\notin k$. Choose a point (E_0,U_0) on it with $E_0\notin\{0,e^a\}$ and $P(E_0,U_0)\neq0$; only finitely many points are excluded. Take w_0 with $w_0^M=e^a/E_0\neq1$ and $v_0=(U_0-\eta_a)/(w_0^M-1)-\Delta_a$. The images of E and U are regular everywhere and take the values (E_0,U_0) at (w_0,v_0), so F has a pole there. Hence Q is a monomial in E and $R\in k[E^{\pm1},U]$, which proves the theorem. □

Remark 5.5. Step 5 proves, modulo Theorem 4.5 and (I1)–(I4), the one-point case of [1, Theorem ner:main] for a positive argument: on the winding line the coordinate $U=\mathrm{Ein}(a)$ moves freely, so only Laurent polynomials survive. Theorem 5.1 also gives the units: $(\overline{\mathbf{E}}\cap\mathfrak{F})^\times = k^\times e^{a\mathbb{Z}}$ for $m=2$.

6 What the winding line cannot see

Proposition 6.1 (γ-blindness on $\mathfrak{F}$)**.** *For $\gamma'\in\mathbb{C}$ define the formal counterparts*

$$\delta' = e(\mathrm{Ein}(1)-\gamma'),\qquad K_0'(t) = Y(t)-\Big(\gamma'+\log\frac{t}{2}\Big)I_0(t)\quad(t>0).$$

1. *Algebraic relations over k among E-values, the joint independence (I5), the family $(\Phi_s)_{s\in\mathbb{C}}$ on the fields $\mathfrak{F}$, and Theorem 4.5 for every $\xi\in\mathfrak{F}\cap\overline{\mathbf{E}}$ do not involve γ.*
2. *For $\gamma'=1/2$: $\gamma'\in k$ and $\operatorname{trdeg}_k k(\gamma',\delta',K_0'(2)/I_0(2),C(2))=3$. For $\gamma'=\mathrm{Ein}(1)-d/e$ with $d\in k^\times$: $\delta'=d$. For $\gamma'=\beta(n)-\log(n/2)-b/I_0(n)$ with $b\in k$: $K_0'(n)=b$.*

Consider an argument whose only information about γ consists of the identities

$$\delta = e(\mathrm{Ein}(1) - \gamma), \qquad K_0(t) = Y(t) - \Big(\gamma + \log\frac{t}{2}\Big) I_0(t),$$

and whose other inputs are among those in (1)*. It proves its conclusion for every $\gamma' \in \mathbb{C}$. By* (2)*, no such argument can show that γ, δ or a specified $K_0(n)$ is transcendental, or that* $\mathrm{trdeg}_k\, k(\gamma, \delta, \kappa, C) = 4$.

Proof. (1) is Proposition 4.3(3) together with the fact that the other inputs concern E-values only. (2) For $\gamma' = 1/2$ the field $k(e, \mathrm{Ein}(1), I_0(2), C(2), \beta(2))$ has transcendence degree 5 by (I5) and contains $\gamma', \delta', \beta(2) - \gamma', C(2)$; the first is algebraic and the other three are independent, as $\delta' = e(\mathrm{Ein}(1) - 1/2)$, $\beta(2) - 1/2$ and $C(2)$ are nonconstant functions of disjoint sets of independent coordinates. The other two choices are direct substitutions. □

Remark 6.2 (where γ enters). On $\mathfrak{F}$ the genuine embeddings are $\Phi_{\gamma + j\tau} = \Phi^{[j]}$; the analytic construction singles out these points of the line. An argument settling (Q1) or (Q2) through phase embeddings must use a property of the point $s = \gamma$ that generic points of the line lack. An example is the rigidity

$$\xi \in \mathbf{E}, \quad \Phi^{[0]}(\xi) \in \mathbb{C} \implies \xi \in k. \tag{3}$$

It implies $\gamma \notin k$. Indeed, if $\gamma = g \in k$, then $\delta = e(\mathrm{Ein}(1) - g)$ lies in $\mathbf{E}$ and in $\mathfrak{F}$ for $a = 1$, and by Definition 4.2

$$\Phi_s(\delta) = (s - g)e^X + e(\mathrm{Ein}(1) - s),$$

which is the constant δ exactly at $s = g = \gamma$, that is, for $\Phi^{[0]}$. Since δ is then transcendental by (I5), (3) fails. Property (3) is of the same nature as the conjecture $\mathbf{D} \cap \mathbf{E} = k$ formulated by Fischler and Rivoal [3], which already excludes $\gamma \in k$: if $\gamma \in k$, then δ, a value of the Euler series, lies in $\mathbf{D} \cap \mathbf{E}$ and is transcendental.

Other inputs of [1] are γ-sensitive and lie outside Proposition 6.1: the positivity $K_0(t) > 0$, used for instance in [1, Theorem jden:rank], and the critical approximation limits of [1, §limits]. The latter identify $\gamma + 2\kappa$ and $\gamma - 2\kappa$ with limits of explicit rational sequences, with errors $O(n^{-2})$ and worse, and $B - \gamma$ with the limit of $R_N(1/N) + \log N$. They would become decisive only with arithmetic control of these sequences, which is not available.

Remark 6.3 (on (Q4)). An effective selection of the density-one set of [1, Theorem jden:density] would exhibit an explicit integer n with $K_0(n)$ transcendental, which is a statement of type (Q1). By Proposition 6.1(2) (third choice) such a statement does not follow from the inputs listed in Proposition 6.1(1) alone.

Remark 6.4 (three or more harmonic coordinates). The proof of Theorem 5.1 uses $m = 2$ in two places. In Step 2 one intermediate regime suffices because the quotient of the x-plane by N is one-dimensional; for m scales one needs the regimes $w = 1 + cv^{-1-1/K}$, $K = 2, \dots, m$, in which the coordinates along $N, \dots, N^{\circ(K-1)}$ (with $N^{\circ k} = (N_i^k)_i$) diverge at different rates, and terms of equal weight can cancel. In regimes (C_i) and (P_i) the coordinates B_l, $l \neq i$, diverge at a common rate with rational ratios $N_l/(N_i - N_l)$. For $m = 2$ only one such coordinate occurs, so the leading coefficients cannot cancel. For $m \geq 3$ the leading forms can vanish along these rational directions, and the next-order terms contain the shifts $v_l - v_i = K_0(t_l)/I_0(t_l) - K_0(t_i)/I_0(t_i) = \beta(t_l) - \beta(t_i) - \log(t_l/t_i)$. Separating them from the harmonic coordinates is a problem about E-values and logarithms of algebraic numbers. For rationally independent scales [1, Theorem mbmulti:main] already covers every m. The three-harmonic case is treated in Theorem 10.1; the obstruction discussed here remains relevant to larger full blocks.

Remark 6.5 (routes outside the phase embeddings). (i) p-adic: δ is the Borel sum of the archimedeanly divergent series $\sum(-1)^n n!$, which converges in every $\mathbb{Q}_p$. Global-relation theorems for Euler's factorial series [4] exclude one rational value common to all places of a large set of primes, but say nothing at the archimedean place, and the defining exponential and Ein series do not converge at 1 in the usual p-adic topology, so this does not supply a p-adic evaluation of the displayed identity for γ. (ii) Logarithmic forms: γ also occurs as $-\psi(1)$ and in Euler–Lehmer constants, as γ plus linear forms in logarithms; Baker's theorem gives "at most one algebraic" statements [5]. Combining them with E-values requires separating E-values from logarithms and π, which is open already for e and π. (iii) Mahler-type: Catalan's integral $\gamma = \int_0^1 (1+x)^{-1} \sum_{n\geq 1} x^{2^n-1}\, dx$ expresses γ as an integral of a Mahler function rather than a value at an algebraic point.

7 Numerical checks

The following numerical checks and table are retained from the supplied winding-line manuscript. They used `mpmath` at 40–80 digits; the original randomized runs were not independently replayed for this edition. They are illustrations and are not used as proofs.

1. The closed forms (1) and (2) agree with the images computed from the coefficients A_j, B_j, A'_j, B'_j of [1, §mbc] (themselves the continued components of $I_0(tz) - \varepsilon K_0(tz)$ and $\varepsilon K_0(tz)$), for $t \in \{2, 4, \sqrt{5}, 1/3\}$, $a \in \{1, 3/7\}$, random $j \in [-40, 40]$ and random complex X, to better than 10^{-77}. At $X = 1$ the formulas return $\beta(t), C(t)$ for complex s.

2. Symbolically, after $j = (s-\gamma)/\tau$ the coefficients A_j, B_j and the two coefficients of the Y-image are free of γ.

3. The limits of Lemma 5.4 hold numerically. For the actual constants at $(t_1, t_2) = (2, 4)$ with $(N_1, N_2) = (1, 2)$ and the model exponent $M = 3$ (the asymptotics depend only on the exponents), the discrepancies are:

$\lvert v\rvert$	(A)	(B)	(C_1), (C_2)	(P_1), (P_2)
10^6	$2.7 \cdot 10^{-6}$	$6.5 \cdot 10^{-4}$	$2.8 \cdot 10^{-5}$, $1.1 \cdot 10^{-5}$	$9.9 \cdot 10^{-6}$, $2.5 \cdot 10^{-6}$
10^9	$2.7 \cdot 10^{-9}$	$2.1 \cdot 10^{-5}$	$2.8 \cdot 10^{-8}$, $1.1 \cdot 10^{-8}$	$9.9 \cdot 10^{-9}$, $2.5 \cdot 10^{-9}$

The decay is $O(|v|^{-1})$, and $O(|v|^{-1/2})$ in regime (B), as predicted. The same decay holds for the exponent pairs $(3,5), (7,2), (1,6), (2,3), (5,11)$ up to $|v| = 10^{18}$. The leading coefficients used in Steps 3 and 4 were checked symbolically and numerically on test polynomials, and for random denominators the zeros of $f \circ \Psi$ appear at the points c^* predicted in Steps 2 and 3.

8 A finite winding criterion in bounded degree

The polynomial-winding argument can be made finite when the degrees of its input rational function are bounded. The resulting criterion concerns exact identities of rational functions; it is not a numerical algorithm for deciding equality of arbitrary special-function constants.

Lemma 8.1 (Exceptional specializations). *Let $A, B \in \mathbb{C}[w, v]$ satisfy*

$$\deg_w A, \deg_w B \leq W, \qquad \deg_v A, \deg_v B \leq S, \qquad B(1, v) = b \in \mathbb{C}^\times.$$

Put $F = A/B$. Either $F \in \mathbb{C}[w, w^{-1}, v]$, or there are at most $2WS + 2S$ distinct numbers $v_0 \in \mathbb{C}$ for which $F(w, v_0) \in \mathbb{C}[w, w^{-1}]$.

Proof. The case $A = 0$ is immediate. Otherwise cancel the greatest common polynomial divisor and write $F = N/D$ with $N, D \in \mathbb{C}[w, v]$ coprime. Removing a power of w from D does not affect whether a specialization is Laurent; denote the remaining denominator by D_0. The separate degree bounds remain at most W, S after cancellation. Because $B(1, v)$ is a nonzero constant, every polynomial factor of B has nonzero constant value at $w = 1$. In particular D_0 has no nonconstant factor depending on v alone, and $D_0(w, v_0)$ is not identically zero at any v_0.

If $\deg_w D_0 = 0$, it follows that D_0 is constant, and the first alternative holds. Suppose instead that $q = \deg_w D_0 > 0$. Write $d_0(v) = D_0(0, v)$ and let $d_q(v)$ be the leading coefficient in w. Both are nonzero polynomials. Since N and D_0 are coprime,

$$H(v) = d_0(v)d_q(v) \operatorname{Res}_w(N, D_0)$$

is a nonzero polynomial. The standard determinant formula for the resultant gives

$$\deg_v H \le 2S + (\deg_w N)(\deg_v D_0) + (\deg_w D_0)(\deg_v N) \le 2WS + 2S.$$

If $H(v_0) \neq 0$, then $D_0(w, v_0)$ has positive degree, nonzero constant coefficient, and no common root with $N(w, v_0)$. It therefore has a nonzero complex root at which $F(w, v_0)$ has a pole. Consequently this specialization is not Laurent. Every Laurent specialization must be a zero of H, proving the bound. □

Fix the positive-scale setting of the winding family, with $m \ge 1$ Bessel blocks and one exponential–Ein pair. Choose a one-parameter subgroup for which

$$e^{a(1-X)} = w^M, \qquad q_i = w^{N_i}, \qquad M, N_i \in \mathbb{Z}_{>0},$$

with the N_i distinct, as in the proof of Theorem 5.1. Put $v = v_1$, $v_i = v + \Delta_i$, $v_a = v + \Delta_a$, and write the restricted images as

$$E \longmapsto e^a w^{-M}, \quad U \longmapsto \eta_a + (v + \Delta_a)(w^M - 1), \quad x_i \longmapsto B_i, \quad y_i \longmapsto \Gamma_i,$$

where

$$\begin{aligned} D_i &= p + (v + \Delta_i)(w^{N_i} - 1), \\ B_i &= \frac{\beta_i D_i - (v + \Delta_i)(v + \Delta_i - p)(w^{N_i} - 1)}{D_i}, \\ \Gamma_i &= \frac{C_i D_i - h_i(w^{N_i} - 1)}{D_i}. \end{aligned}$$

Denote this tuple by $\Psi(w, v)$.

Proposition 8.2 (Finite recognition of polynomial winding)**.** *Let $R = P/Q$ with $P, Q \in k[E, U, x_1, y_1, \dots, x_m, y_m]$, $Q \neq 0$, and total degrees at most $d \ge 1$. Set*

$$W = d\left(2M + \sum_{i=1}^{m} N_i\right), \qquad S = (m + 1)d, \qquad B_d = 2S(W + 1).$$

If $R(\Psi(w, v_j))$ is Laurent in w at more than B_d distinct values v_j, then

$$R(\Psi(w, v)) \in \mathbb{C}[w, w^{-1}, v].$$

Otherwise, if this polynomial-winding conclusion fails, at most B_d values $v_j \in \mathbb{C}$ can give Laurent specializations.

Proof. Multiply both substituted polynomials by

$$L(w,v) = w^{Md} \prod_{i=1}^{m} D_i(w,v)^d.$$

Then $A = LP(\Psi)$ and $B = LQ(\Psi)$ belong to $\mathbb{C}[w,v]$. To check the bounds, consider a monomial of P or Q having exponents e, u, r_i, c_i on E, U, x_i, y_i respectively. The power w^{Md} clears w^{-Me}, since $e \le d$. The total w-degree is bounded by

$$Md + Mu + d\sum_i N_i \le W.$$

For the v-degree, the ith denominator contributes $d - r_i - c_i$, the numerator of $B_i^{r_i}$ contributes at most $2r_i$, and that of $\Gamma_i^{c_i}$ at most c_i. Thus the bound is

$$md + u + \sum_i r_i \le (m+1)d = S.$$

Finally, at $w = 1$ the image tuple equals the original tuple T, and every D_i equals p. Joint independence gives

$$B(1,v) = p^{md} Q(T) \neq 0,$$

independently of v. Lemma 8.1 applies. □

Theorem 8.3 (Finite exact test for the mixed two-block intersection)**.** *Assume the phase and joint-independence inputs used in Theorem 5.1. Let*

$$\xi = R(e^a, \eta_a, \beta_1, C_1, \beta_2, C_2), \qquad \deg P, \deg Q \le d, \qquad d \ge 1,$$

and choose positive subgroup exponents M, N_1, N_2 with $N_1 \neq N_2$. Set

$$B_d = 6d\big(d(2M + N_1 + N_2) + 1\big).$$

For any $B_d + 1$ distinct integers j, the following are equivalent:

1. *ξ is integral over $\mathbf{E}$;*
2. *$\xi \in k[e^a, e^{-a}, \eta_a]$;*
3. *every restricted image $R(\Psi(w, v_j))$ in the selected finite list is Laurent in w, where $v_j = \beta_1 - \log(t_1/2) - \gamma - 2\pi i j$.*

If ξ lies outside the displayed ring, at most B_d windings in the entire integer sequence can pass this Laurent test.

Proof. The implication (2)⇒(1) follows because $e^{\pm a}$ and η_a are E-values. Integrality implies polynomial winding, hence (1)⇒(3). The parameters v_j are distinct, so Proposition 8.2 gives $R \circ \Psi \in \mathbb{C}[w, w^{-1}, v]$ from (3). Steps 1–5 of the proof of Theorem 5.1 establish precisely the algebraic implication

$$R \circ \Psi \in \mathbb{C}[w, w^{-1}, v] \quad \Longrightarrow \quad R \in k[E, E^{-1}, U].$$

Those steps use joint independence and the displayed subgroup formulas; after obtaining polynomial winding they use no further integrality hypothesis. Thus (3)⇒(2). The last assertion follows from the contrapositive and the exceptional-specialization bound. □

Remark 8.4. For instance, $a = 1$, $t_1 = 1/2$, $t_2 = 1$ admit $(M, N_1, N_2) = (1, 1, 2)$. For rational functions with numerator and denominator of total degree at most one, $B_1 = 36$: any 37 distinct windings give a sufficient exact test. This conservative bound is not claimed to be optimal. The theorem neither singles out Euler's constant arithmetically nor turns finite-precision pole checks into exact proofs.

9 Inputs, scope, and comparison with the preceding notes

Put $k = \overline{\mathbb{Q}}$. Let $\mathbf{E}$ denote the ring of values of E-functions at algebraic points, and let $\overline{\mathbf{E}}$ be its integral closure in $\mathbb{C}$. Here "integral" means satisfying a monic polynomial with coefficients in $\mathbf{E}$; it does not mean being a rational integer. For $t \in k^\times$ write

$$Y(t) = \sum_{n\geq 0} \frac{H_n (t/2)^{2n}}{(n!)^2}, \qquad \beta(t) = \frac{Y(t)}{I_0(t)}, \qquad C(t) = \frac{I_1(t)}{I_0(t)}.$$

Here $H_0 = 0$ and $H_n = \sum_{j=1}^{n} 1/j$. The entire integral is $\mathrm{Ein}(z) = \int_0^z (1 - e^{-u})\, du/u$.

The intersection results below use the continued phase embeddings, their pole criterion, the exact Bessel component formulas, and joint independence from [1]. These are assumptions imported from that research manuscript, not consequences of numerical tests. The polynomial-winding principle in [6] is the additional organizing input. Baker's theorem is the extra published arithmetic input used for three harmonic coordinates. The resultant result from [7] enters only the fractional-order derivative-ratio proposition in the derivative-coordinate section.

There is a necessary correction to the novelty description in [6]. The two-block field $k(\beta(2), C(2), \beta(4), C(4))$ is already classified in the current version of [1, Theorems `tbr:main`, `ctb:main`]. The additional intersection proved in [6] retains e^a and $\mathrm{Ein}(a)$ together with two full positive-scale Bessel blocks. The three-harmonic theorem and the mixed affine-projection theorem below extend in different directions; neither implies the full mixed three-block classification.

Polynomial winding. We use Theorem 4.5.

The proof in [6] uses the genuine integer winding values to rule out every denominator depending on the torus variables. Evaluation at the identity of the torus returns the original algebraically independent generators for every s, and rules out the remaining denominators in s. This is a necessary consequence of integrality, not a proved converse. The complex-scale version is detailed in the affine-projection section below.

Mixed two-block intersection. We use Theorem 5.1.

All conclusions here retain the original arithmetic coefficient field k, except for the expressly displayed exponential–Ein coefficients. No general separation theorem for E-values and logarithms is assumed.

10 Three harmonic coordinates at arbitrary positive scales

The third harmonic coordinate can be admitted without any rational independence assumption on the scales. Here the coefficient field remains $k = \overline{\mathbb{Q}}$; no derivative quotient or additional E-value is adjoined to the field in the following theorem.

Theorem 10.1. *Let t_1, t_2, t_3 be distinct positive algebraic numbers. Then*

$$\overline{\mathbf{E}} \cap k(\beta(t_1), \beta(t_2), \beta(t_3)) = k.$$

In particular, this holds for the three dependent scales $2, 4, 8$.

We use precisely two inputs from the winding-line construction: the algebraic independence of the three numbers $\beta(t_i)$ over k, and Theorem 4.5, which makes the image of an integral element a Laurent polynomial in the phase coordinates and a polynomial in the winding coordinate. The additional arithmetic input is Baker's theorem on linear forms in logarithms [8].

10.1 Phase restriction and the three charts

Put $p = i\pi$, $\beta_i = \beta(t_i)$, $\ell_i = \log(t_i/2)$ and $v_i = \beta_i - \ell_i - s$. The harmonic image is

$$B_i = \beta_i - \frac{v_i(v_i - p)(q_i - 1)}{p + v_i(q_i - 1)} = s + \ell_i + \frac{pv_iq_i}{p + v_i(q_i - 1)}. \tag{4}$$

As in the one-parameter subgroup restriction in the proof of Theorem 5.1, choose distinct positive integers N_i such that $q_i = w^{N_i}$. This is possible even when the original scales are not commensurable: if $2t_i = \sum_j n_{ij}\lambda_j$ in an integral basis of their additive group, choose integers u_j nearest to $L\lambda_j$. Then $N_i = \sum_j n_{ij}u_j = 2Lt_i + O(1)$ are positive and distinct for large L. Multiplying all the u_j by 2, if necessary, also ensures $N_i \geq 2$.

Set $v = v_1$ and $\Delta_i = v_i - v_1$, so $v_i = v + \Delta_i$ with constant Δ_i. Write

$$\mathcal{D}_i = p + (v + \Delta_i)(w^{N_i} - 1).$$

For an integral element $\xi = R(\beta_1, \beta_2, \beta_3)$, the restriction of the polynomial winding theorem gives

$$R(B_1, B_2, B_3) \in \mathbb{C}[w, w^{-1}, v]. \tag{5}$$

In particular, if $R = P/Q$ is reduced over k, there is no point $(w, v) \in \mathbb{C}^\times \times \mathbb{C}$ at which every $\mathcal{D}_i$ is nonzero and $Q(B) = 0$ but $P(B) \neq 0$.

Choose rational coefficients a_{ri}, $1 \leq r, i \leq 3$, such that

$$\sum_{i=1}^{3} a_{ri}N_i^j = \delta_{rj} \qquad (1 \leq r, j \leq 3),$$

and put $D_r(x) = \sum_i a_{ri}x_i$, $d_r = D_r(\beta)$ and $\widetilde{D}_r = D_r(B)$. This invertible change of coordinates is defined over k, and d_1, d_2, d_3 are algebraically independent over k.

Lemma 10.2. *For $K \in \{1, 2, 3\}$ put $v = H^K$ and $w = 1 + pzH^{-K-1}$, with $H \to +\infty$. Uniformly for z in a compact subset of $\mathbb{C}$,*

$$B_i = \beta_i + \sum_{j=1}^{K} (-1)^j N_i^j z^j H^{K-j} + O(H^{-1}), \qquad \mathcal{D}_i \longrightarrow p.$$

Consequently

$$\widetilde{D}_r = \begin{cases} d_r + (-1)^r z^r H^{K-r} + O(H^{-1}), & r \leq K, \\ d_r + O(H^{-1}), & r > K. \end{cases}$$

Proof. Let $\theta_i = w^{N_i} - 1$. Then $\theta_i = N_i pzH^{-K-1} + O(H^{-2K-2})$ and $(v + \Delta_i)\theta_i/p = N_i zH^{-1} + O(H^{-K-1})$. In the first expression in (4), expand $(1 + (v + \Delta_i)\theta_i/p)^{-1}$ through degree $K - 1$ in H^{-1}. The factor $(v + \Delta_i)(v + \Delta_i - p)\theta_i/p$ equals $N_i zH^{K-1} + O(H^{-1})$. Terms of order K or higher in the geometric expansion contribute $O(H^{-1})$; the other errors have the same bound. This proves the displayed expansion and then the assertions about the dual Vandermonde coordinates. □

10.2 Eliminating every denominator

Lemma 10.3. *If $R \in k(x_1, x_2, x_3)$ satisfies (5), then $R \in k[x_1, x_2, x_3]$.*

Proof. Write $R = P/Q$ in reduced form, now using the coordinates D_1, D_2, D_3. We may suppose $P \neq 0$. Every irreducible factor f of Q is coprime to P.

First chart. Suppose that f depends on D_1. The $K = 1$ chart has limit $(d_1 - z, d_2, d_3)$, and $f(d_1 - z, d_2, d_3)$ is a nonconstant polynomial in z. Let z_* be a root. The algebraic set $Z(P) \cap Z(f) \subset \mathbb{A}^3$ has dimension at most 1. Its translates in the D_1 direction have Zariski closure of dimension at most 2, defined over k. Since $d = (d_1, d_2, d_3)$ is generic over k, that closure does not contain d. Therefore $P(d_1 - z_*, d_2, d_3) \neq 0$. Uniform convergence in Lemma 10.2 and Hurwitz's theorem give zeros $z_H \to z_*$ of $f(\widetilde{D})$; at them $P(\widetilde{D}) \neq 0$ and all $\mathcal{D}_i \neq 0$ for large H. This contradicts (5). Thus $Q \in k[D_2, D_3]$.

Second chart. Suppose that an irreducible factor $f \in k[D_2, D_3]$ of Q has positive degree in D_2. Its specialization $f(W, d_3)$ has a root W_* algebraic over $k(d_3)$. As d_2 is transcendental over $k(d_3)$, choose a nonzero z_* with $z_*^2 = W_* - d_2$. The $K = 2$ chart has

$$\widetilde{D}_1 = -zH + O(1), \qquad \widetilde{D}_2 \longrightarrow d_2 + z^2, \qquad \widetilde{D}_3 \longrightarrow d_3.$$

Again Hurwitz gives zeros $z_H \to z_*$ of $f(\widetilde{D}_2, \widetilde{D}_3)$. Write $P = \sum_n P_n(D_2, D_3) D_1^n$ and take the largest n_0 such that $f \nmid P_{n_0}$. At these zeros all terms with $f \mid P_n$ vanish exactly. The resultant of f and P_{n_0} in D_2 is a nonzero polynomial in D_3: irreducibility and Gauss's lemma ensure that the two polynomials are coprime over $k(D_3)$. Its value at d_3 is nonzero, so $P_{n_0}(W_*, d_3) \neq 0$. Hence

$$H^{-n_0} P(\widetilde{D}) \longrightarrow (-z_*)^{n_0} P_{n_0}(W_*, d_3) \neq 0$$

along the zeros, a contradiction. It follows that $Q \in k[D_3]$.

Third chart. Because k is algebraically closed, an irreducible nonconstant factor of Q has the form $f = D_3 - \alpha$, with $\alpha \in k$. Choose z_* with $z_*^3 = d_3 - \alpha$. This number is nonzero and transcendental over k. The $K = 3$ chart gives

$$\widetilde{D}_1 = -zH^2 + O(1), \qquad \widetilde{D}_2 = z^2 H + O(1), \qquad \widetilde{D}_3 \longrightarrow d_3 - z^3.$$

Hurwitz gives zeros $z_H \to z_*$ with $\widetilde{D}_3 = \alpha$ exactly. The nonzero reduction $P(D_1, D_2, \alpha) = \sum_{a,b} c_{ab} D_1^a D_2^b$ has coefficients $c_{ab} \in k$. Let $W = \max\{2a + b : c_{ab} \neq 0\}$. Then along those zeros

$$H^{-W} P(\widetilde{D}) \longrightarrow G(z_*), \qquad G(Z) = \sum_{2a+b=W} c_{ab} (-1)^a Z^{a+2b}.$$

For fixed $2a + b = W$, the exponent $a + 2b = 2W - 3a$ determines a and b. Thus the displayed monomials have distinct exponents and G is a nonzero polynomial over k. Transcendence of z_* implies $G(z_*) \neq 0$, again a contradiction. Therefore Q is constant. □

10.3 The primitive-root pole and Baker's theorem

Lemma 10.4. *If $r > 0$ is algebraic and $r \neq 1$, then $\log r + i\pi A$ is transcendental for every $A \in k$.*

Proof. The real number $\log r \neq 0$ and the purely imaginary logarithm $\log(-1) = i\pi$ are linearly independent over $\mathbb{Q}$. Baker's theorem makes $1, \log r, \log(-1)$ linearly independent over k; equivalently, a nonzero algebraic linear form in these two logarithms cannot be algebraic [8]. The proposed form has coefficient 1 on $\log r$, so it is transcendental. □

Lemma 10.5. *If $P \in k[x_1, x_2, x_3]$ and $P(B_1, B_2, B_3) \in \mathbb{C}[w, w^{-1}, v]$, then P is constant.*

Proof. Suppose otherwise. Among the variables on which P depends choose x_i with maximal exponent N_i, and write

$$P = \sum_{d=0}^{d_0} c_d(x_j, x_k) x_i^d, \qquad d_0 > 0, \quad c_{d_0} \neq 0,$$

omitting from c_d any variable on which P does not depend. Consider the irreducible pole divisor

$$\Pi_i : \quad \mathcal{D}_i = 0, \qquad v_i = -\frac{p}{w^{N_i}-1}, \qquad s = \beta_i - \ell_i + \frac{p}{w^{N_i}-1}.$$

Its coordinate ring is the localization of $\mathbb{C}[w, w^{-1}]$ at $w^{N_i} - 1$. At its generic point B_i has a simple pole, with nonzero coefficient, while the other coordinates occurring in P are regular. We prove that $c_{d_0}(B_j, B_k)$ does not vanish identically on Π_i.

Choose a primitive N_i-th root of unity ζ. Since $N_i \geq 2$, $\zeta \neq 1$. Each other exponent occurring in P is strictly smaller than N_i, so $\zeta^{N_l} \neq 1$ for these indices l. On Π_i, as $w \to \zeta$, the variable s tends to infinity and (4) gives

$$B_l = s + \ell_l + p\frac{\zeta^{N_l}}{\zeta^{N_l}-1} + o(1). \tag{6}$$

In particular, the other denominators are not identically zero on Π_i. If both other variables occur, their difference tends to

$$L_{jk} = \log\frac{t_j}{t_k} + p\left(\frac{\zeta^{N_j}}{\zeta^{N_j}-1} - \frac{\zeta^{N_k}}{\zeta^{N_k}-1}\right).$$

The parenthesized number is algebraic. Since $t_j/t_k \neq 1$, Lemma 10.4 implies that L_{jk} is transcendental. Write the nonzero polynomial c_{d_0} uniquely as

$$c_{d_0}(x_j, x_k) = \sum_{b=0}^{b_*} H_b(x_j - x_k)x_k^b, \qquad H_{b_*} \neq 0.$$

By (6),

$$B_k^{-b_*} c_{d_0}(B_j, B_k) \longrightarrow H_{b_*}(L_{jk}) \neq 0.$$

Thus this leading coefficient is not identically zero on Π_i. If only one other variable occurs, (6) makes it tend to infinity, which gives the same conclusion for a nonzero univariate polynomial. If neither occurs, the conclusion is immediate.

At the generic point of Π_i, the term $c_{d_0}(B_j, B_k)B_i^{d_0}$ therefore has a pole of order d_0, whereas every lower term has smaller pole order. This contradicts $P(B) \in \mathbb{C}[w, w^{-1}, v]$, since Π_i lies in $\mathbb{C}^\times \times \mathbb{C}$. Consequently P is constant. □

Proof of Theorem 10.1. An element of the displayed intersection has a unique expression $R(\beta_1, \beta_2, \beta_3)$ over k by joint independence. The polynomial winding theorem yields (5). Lemma 10.3 makes R a polynomial and Lemma 10.5 makes it constant. The reverse inclusion holds because algebraic numbers are E-values. □

Remark 10.6. The theorem settles the three-coordinate harmonic subfield of the dependent-scale integral-intersection question. It does not classify the field obtained by adjoining all three derivative quotients $C(t_i)$, nor does it extend the coefficient field to arbitrary E-values. For four harmonic coordinates the present proof has two distinct obstacles: the fourth chart permits equal leading monomials such as D_1D_3 and D_2^2, and the primitive-root boundary presents two logarithmic differences whose algebraic independence does not follow from Baker's theorem. No claim about the arithmetic nature of γ, δ, or an individual $K_0(t)$ is made.

11 An exponential–Ein field with an arbitrary affine Bessel projection

The following result permits arbitrarily many dependent complex scales, but retains only one affine projection of their Bessel coordinates. It does not assert the corresponding full multivariate intersection.

Theorem 11.1. *Let $k = \overline{\mathbb{Q}}$, let $a > 0$ be algebraic, and set $E = e^a$ and $\eta = \mathrm{Ein}(a)$. Let $t_1, \dots, t_m \in k^\times$ be distinct modulo sign. For $\ell_0, u_i, v_i \in k$ put*

$$L = \ell_0 + \sum_{i=1}^{m} (u_i \beta(t_i) + v_i C(t_i)).$$

Assume that at least one pair (u_i, v_i) is nonzero. Then

$$\boxed{\overline{\mathbf{E}} \cap k(E, \eta, L) = k[E, E^{-1}, \eta].}$$

There is no rational-independence condition on $a, t_1, \dots, t_m$.

We first record why polynomial winding remains available for complex scales. Choose a common nonexceptional summation direction. The exact complex component normalization gives constants $p_i \in \{\pi i, -\pi i\}$. With $q_i = e^{2t_i(1-X)}$ the continued images have the form

$$B_i = \beta_i - \frac{b_i(b_i - p_i)(q_i - 1)}{p_i + b_i(q_i - 1)}, \qquad \Gamma_i = C_i - \frac{h_i(q_i - 1)}{p_i + b_i(q_i - 1)}, \tag{7}$$

$$E \longmapsto E e^{-a(1-X)}, \qquad \eta \longmapsto \eta + b_a(e^{a(1-X)} - 1), \tag{8}$$

where $h_i = 1/(t_i I_0(t_i)^2) \neq 0$. On the genuine winding sequence, all b_i and b_a are a common affine parameter plus a fixed constant: $b_i = v + \Delta_i$, $b_a = v + \Delta_a$, with $v = -2\pi i j + O(1)$. In particular, the signs p_i do not change the slope of the winding parameter. These are the normalized complex component formulas; the constants Δ_i are not assumed to be algebraically independent. A change of the common summation direction can add a fixed branch constant to Δ_a; it does not change the common winding slope.

Let $H = \mathbb{Z}a + \sum_i \mathbb{Z}(2t_i)$, and choose a basis of this finite-rank free abelian group. Write its normalized characters as $z_1, \dots, z_d$. Substitution in (7)– (8) defines a rational function $F_\xi(z, v)$ for every $\xi \in k(E, \eta, \beta_i, C_i : 1 \leq i \leq m)$. Evaluation at $z = (1, \dots, 1)$ returns the original generators, independently of v. Joint independence therefore makes this a field embedding for every fixed complex v. If ξ is integral over $\mathbf{E}$, the polynomial-winding argument gives

$$F_\xi \in \mathbb{C}[z_1^{\pm 1}, \dots, z_d^{\pm 1}, v]. \tag{9}$$

For completeness, each genuine winding specialization lies in $\mathbb{C}[H]$: the phase-image pole lemma gives membership in the full group algebra, and separation by H-cosets gives its intersection with $\mathrm{Frac}\,\mathbb{C}[H]$. A reduced denominator of F_ξ with a nonconstant torus factor would remain an uncancelled denominator on all but finitely many winding slices, a contradiction. Its remaining factors depend only on v. None can vanish at v_0, because the original rational substitution at v_0 is well defined generically in z, its denominator being nonzero at $z = (1, \dots, 1)$. This proves (9). This argument uses no reality assumption on the phase lattice.

Proof of Theorem 11.1. Discard inactive blocks. The numbers E, η, L are algebraically independent by the joint independence theorem, since L is a nonconstant algebraic affine combination of the Bessel coordinates. Write an integral element of their field as $R(E, \eta, L)$ with $R = P/Q$ and coprime $P, Q \in k[Z, U, T]$.

A suitable one-dimensional phase restriction. Choose a homomorphism $\ell : H \to \mathbb{Z}$ avoiding the finitely many hyperplanes defined by a, $2t_i$, and $2t_i \pm 2t_l$. Replace ℓ by $-\ell$ if needed so that $M = \ell(a) > 0$. Replace each t_i by $-t_i$ when needed, using $\beta(-t_i) = \beta(t_i)$ and $C(-t_i) = -C(t_i)$ and changing the corresponding v_i. We may then assume

$$N_i = \ell(2t_i) > 0, \qquad N_i \neq N_l \quad (i \neq l).$$

Choose the common nonexceptional direction after these replacements. Restrict the phase characters by $e^{h(1-X)} \mapsto w^{\ell(h)}$. In the resulting formulas put

$$\begin{aligned} D_i(w,v) &= p_i + (v+\Delta_i)(w^{N_i}-1), \\ B_i(w,v) &= \beta_i - \frac{(v+\Delta_i)(v+\Delta_i-p_i)(w^{N_i}-1)}{D_i(w,v)}, \\ \Gamma_i(w,v) &= C_i - \frac{h_i(w^{N_i}-1)}{D_i(w,v)}, \\ \mathcal{E}(w) &= Ew^{-M}, \qquad \mathcal{U}(w,v) = \eta + (v+\Delta_a)(w^M-1), \\ \mathcal{L}(w,v) &= \ell_0 + \sum_i (u_i B_i + v_i \Gamma_i). \end{aligned}$$

The rational substitution remains injective, since its value at $w=1$ is the original tuple for every v. By (9),

$$R(\mathcal{E},\mathcal{U},\mathcal{L}) \in \mathbb{C}[w,w^{-1},v]. \tag{10}$$

It consequently has no pole on $\mathbb{C}^\times \times \mathbb{C}$.

A bounded affine-coordinate chart when a harmonic coefficient occurs. First suppose that some u_i is nonzero. The numbers N_i/p_i are nonzero and pairwise distinct: the N_i are distinct positive integers and $p_i/p_l \in \{1,-1\}$. Vandermonde invertibility shows that some integer r, $1 \le r \le m$, satisfies

$$A_r := \sum_i u_i (N_i/p_i)^r \neq 0.$$

Choose the least such r. Let v tend to positive real infinity and put $w = 1 + cv^{-1-1/r}$. Uniformly for c in a compact set,

$$B_i = \beta_i + \sum_{k=1}^{r} (-1)^k (N_i c/p_i)^k v^{1-k/r} + o(1), \qquad \Gamma_i \longrightarrow C_i. \tag{11}$$

Indeed, if $z_i = (v+\Delta_i)(w^{N_i}-1)/p_i$, then $z_i = (N_i c/p_i)v^{-1/r} + O(v^{-1-1/r})$ and $B_i - \beta_i = -(v+\Delta_i-p_i)z_i/(1+z_i)$. The geometric expansion through degree r has error $O(v^{-1/r})$; the fixed shifts contribute only $o(1)$. The denominators D_i tend uniformly to p_i. The minimality of r now gives

$$\mathcal{E} \longrightarrow E, \qquad \mathcal{U} \longrightarrow \eta, \qquad \mathcal{L} \longrightarrow L + (-1)^r A_r c^r. \tag{12}$$

Suppose that Q has positive degree in T. Coprimality over $k(Z,U)$ makes $\mathrm{Res}_T(P,Q)$ a nonzero rational function of Z,U. Joint independence of E,η shows that the specialized polynomials $P(E,\eta,T)$ and $Q(E,\eta,T)$ remain coprime and that $Q(E,\eta,T)$ retains its positive degree. Choose a root α of the latter. Then $P(E,\eta,\alpha) \neq 0$, and $\alpha \neq L$ because L is transcendental over $k(E,\eta)$. Choose $c_0 \neq 0$ such that $L + (-1)^r A_r c_0^r = \alpha$. On a sufficiently small disc about c_0, the limit denominator in (12) has an isolated zero, while the limit numerator is nonzero. Uniform convergence and Rouché's theorem produce, for all sufficiently large v, a zero of the substituted denominator in this disc at which its numerator and all D_i are nonzero. This is a pole at a point with $w \neq 0$, contrary to (10). Consequently Q has degree zero in T in this case.

The pure derivative-coordinate case. Now suppose that every $u_i = 0$, and choose i with $v_i \neq 0$. Set $u_i^* = \eta - Mp_i/N_i$. The three numbers E, u_i^*, L are algebraically independent. To prove this without assuming that the complex derivative coordinates are real, enlarge the coefficient field to

$$F_0 = k\big(E, C(t_l), C(\overline{t_l}) : 1 \le l \le m\big).$$

The coefficients of the Bessel functions are real, so this field is stable under complex conjugation. Repeated sign classes in the displayed family are identified using $C(-t) = -C(t)$. Joint

independence makes η transcendental over F_0. If u_i^* were algebraic over F_0, its conjugate would be algebraic as well, and their average η would be algebraic over F_0, a contradiction. Thus u_i^* is transcendental over F_0, and in particular over $k(E, L)$. The numbers E, L are independent, proving the assertion.

For $v \to +\infty$, put $b_i = v + \Delta_i$ and choose the branch near 1 of

$$w = \left(1 - \frac{p_i}{b_i} + \frac{c}{b_i^2}\right)^{1/N_i}.$$

Uniformly for c in a compact subset of $\mathbb{C}^\times$,

$$\begin{aligned} &D_i = c/b_i, && \Gamma_i \longrightarrow C_i + \frac{p_i h_i}{c}, \\ &D_l \longrightarrow p_l - \frac{N_l p_i}{N_i} \neq 0, && \Gamma_l \longrightarrow C_l \quad (l \neq i), \\ &\mathcal{E} \longrightarrow E, && \mathcal{U} \longrightarrow u_i^*, \qquad \mathcal{L} \longrightarrow L + \frac{v_i p_i h_i}{c}. \end{aligned}$$

If Q has positive degree in T, specialize its resultant with P at the independent pair (E, u_i^*). A root α of $Q(E, u_i^*, T)$ is not a root of $P(E, u_i^*, T)$, and $\alpha \neq L$ by the independence just proved. Take $c_0 = v_i p_i h_i/(\alpha - L) \neq 0$. On a small disc about c_0, the same uniform-convergence and Rouché argument produces a zero of the image denominator with nonzero numerator. All D_l are nonzero there for large v, including $D_i = c/b_i$, and $w \neq 0$. This contradicts (10). Thus, in both cases,

$$Q \in k[Z, U]. \tag{13}$$

Eliminating positive degree in the affine coordinate. Suppose that P has degree $d > 0$ in T, with nonzero leading coefficient $P_d(Z, U)$. Choose an active index i, that is, $(u_i, v_i) \neq (0, 0)$. On the divisor $D_i = 0$ take $w \to 1$, $w \neq 1$, and set

$$v = -\Delta_i - \frac{p_i}{w^{N_i} - 1}.$$

Along this divisor,

$$\begin{aligned} &\mathcal{E} \longrightarrow E, && \mathcal{U} \longrightarrow u_i^* := \eta - \frac{M p_i}{N_i}, \\ &D_l \longrightarrow p_l - \frac{N_l p_i}{N_i} \neq 0 && (l \neq i). \end{aligned}$$

The numbers E, u_i^* are algebraically independent over k. Indeed, E, η are real and independent; if u_i^* were algebraic over $k(E)$, complex conjugation would make its conjugate algebraic over the same field, and their average η would be algebraic over $k(E)$. Here Mp_i/N_i is purely imaginary, which is the only reality input in this proof. Hence

$$P_d(E, u_i^*) Q(E, u_i^*) \neq 0.$$

For w sufficiently close to but different from 1, both factors remain nonzero. At such a point $\mathcal{L}$ has a genuine simple pole along $D_i = 0$: its singular numerator is

$$-(u_i b_i (b_i - p_i) + v_i h_i)(w^{N_i} - 1), \qquad b_i = -\frac{p_i}{w^{N_i} - 1},$$

which is nonzero for w sufficiently close to 1: if $u_i \neq 0$, its quadratic term dominates as $|b_i| \to \infty$; otherwise $v_i h_i \neq 0$. Other blocks are regular there. Therefore $P(\mathcal{E}, \mathcal{U}, \mathcal{L})/Q(\mathcal{E}, \mathcal{U})$ has a pole of order d, again contradicting (10). It follows that $P \in k[Z, U]$ as well.

We have proved that the integral element belongs to $k(E, \eta)$. The one-point exponential–Ein exact intersection gives

$$\overline{\mathbf{E}} \cap k(E, \eta) = k[E, E^{-1}, \eta].$$

Conversely, all three displayed generators of this ring are E-values, so every element of it is integral over $\mathbf{E}$. This completes the proof. □

Corollary 11.2 (A mixed three-scale logarithm circuit)**.** *For*

$$\Delta = \beta(2) - 2\beta(4) + \beta(8) = \frac{K_0(2)}{I_0(2)} - 2\frac{K_0(4)}{I_0(4)} + \frac{K_0(8)}{I_0(8)},$$

and every positive algebraic a, one has

$$\overline{\mathbf{E}} \cap k(e^a, \mathrm{Ein}(a), \Delta) = k[e^a, e^{-a}, \mathrm{Ein}(a)].$$

In particular, a rational expression in these three independent numbers can be integral only if its reduced expression loses all dependence on Δ and is Laurent polynomial in e^a and polynomial in $\mathrm{Ein}(a)$.

Remark 11.3 (Scope). The coefficients of the affine Bessel combination are algebraic. The rational function subsequently applied to it may have arbitrary coefficients in $k(e^a, \mathrm{Ein}(a))$, as the theorem states. This is stronger than exclusion in $k(L)$ alone and covers any finite number of harmonic coordinates, including mutually dependent complex scales. It retains just one affine coordinate; it does not classify $k(e^a, \mathrm{Ein}(a), \beta(t_i), C(t_i) : 1 \le i \le m)$ for $m \ge 3$. The pure derivative-coordinate case is included, even at complex scales, by using the conjugation-stable enlarged field in the proof. For positive derivative scales that case was already implied by the existing arbitrary-derivative-scale theorem.

12 Affine Bessel exclusion over the positive exponential-integral field

Lemma 12.1 (A common imaginary translation of real independent numbers)**.** *Let $F \subseteq \mathbb{C}$ be a field of characteristic zero that is stable under complex conjugation. Let $x_1, \dots, x_n$ be real and algebraically independent over F. Let T be any real number, and let $c_1, \dots, c_n \in F \cap \mathbb{R}$. Then*

$$x_1 + iTc_1, \dots, x_n + iTc_n$$

are algebraically independent over F.

Proof. The case $c = 0$ is immediate. Otherwise choose an invertible matrix M over $F \cap \mathbb{R}$ sending the vector c to the first standard basis vector, and put $y = Mx$. The y_i are real and independent over F, while

$$M(x + iTc) = (y_1 + iT, y_2, \dots, y_n).$$

The field $F(y_2, \dots, y_n)$ is stable under conjugation. A dependence of the displayed tuple would make $y_1 + iT$ algebraic over that field, because $y_2, \dots, y_n$ are independent. Conjugation would also make $y_1 - iT$ algebraic over it. Their average y_1 would then be algebraic, contrary to the independence of the y_i. An invertible linear change of coordinates preserves algebraic independence, completing the proof. □

Put

$$K_{\exp} = \overline{\mathbb{Q}}(e^\rho : \rho \in \overline{\mathbb{Q}}), \qquad \mathcal{F}_+ = K_{\exp}(\mathrm{Ein}(a) : a \in \overline{\mathbb{Q}},\ a > 0).$$

Theorem 12.2 (Relative affine exclusion)**.** *Let $t_1, \dots, t_m \in k^\times$, $k = \overline{\mathbb{Q}}$, be distinct modulo sign, and let*

$$L = \ell_0 + \sum_{i=1}^{m} \bigl(u_i \beta(t_i) + v_i C(t_i)\bigr), \qquad \ell_0, u_i, v_i \in k,$$

be nonconstant. Then

$$\boxed{\overline{\mathbf{E}} \cap \mathcal{F}_+(L) = \overline{\mathbf{E}} \cap \mathcal{F}_+.}$$

Equivalently, if a rational function of L with coefficients in $\mathcal{F}_+$ is integral over $\mathbf{E}$, its reduced rational expression is constant in L.

Proof. One inclusion is immediate. An element of $\mathcal{F}_+(L)$ uses only finitely many distinct positive arguments $a_1, \dots, a_n$ and finitely many exponential values. Enlarge their exponential character lattice to contain the a_l and all $2t_i$, and choose a basis $\rho_1, \dots, \rho_d$. Put

$$E_j = e^{\rho_j}, \qquad \eta_l = \mathrm{Ein}(a_l).$$

The E_j are algebraically independent by the exponential theorem, and the joint theorem makes the η_l and L independent over $K_{\exp}$. Write the proposed integral element as $R(E, \eta, L) = P(E, \eta, L)/Q(E, \eta, L)$, with coprime

$$P, Q \in k[Z_1^{\pm 1}, \dots, Z_d^{\pm 1}, U_1, \dots, U_n, V].$$

The complex polynomial-winding argument applies to this larger finite field. It adds the exponential image $E_j e^{-\rho_j(1-X)}$ and, for each l, the image

$$\eta_l + (v + \Delta_l)(e^{a_l(1-X)} - 1).$$

All fixed branch constants are absorbed into the Δ_l. Choose and orient a phase-lattice homomorphism as in Theorem 11.1, so that the Bessel exponents $N_i > 0$ are pairwise distinct. Write

$$M_l = \ell(a_l), \qquad R_j = \ell(\rho_j).$$

Neither positivity nor nonvanishing of the M_l, R_j is needed. The folded images are

$$\mathcal{E}_j = E_j w^{-R_j}, \qquad \mathcal{U}_l = \eta_l + (v + \Delta_l)(w^{M_l} - 1),$$

and the Bessel images B_i, Γ_i are exactly those in Theorem 11.1. In particular,

$$R(\mathcal{E}, \mathcal{U}, \mathcal{L}) \in \mathbb{C}[w, w^{-1}, v]. \tag{14}$$

Suppose first that some $u_i \neq 0$. Let r be the first nonzero moment $\sum_i u_i (N_i/p_i)^r$, as in that theorem. On $w = 1 + cv^{-1-1/r}$ one has uniformly on compact c-sets

$$\mathcal{E} \longrightarrow E, \qquad \mathcal{U} \longrightarrow \eta, \qquad \mathcal{L} \longrightarrow L + Ac^r, \qquad A \neq 0.$$

If Q has positive degree in V, its nonzero resultant with P specializes nontrivially at (E, η). A root α of the specialized Q satisfies $P(E, \eta, \alpha) \neq 0$ and $\alpha \neq L$. A nonzero solution of $L + Ac^r = \alpha$, followed by Rouché's theorem, then gives a genuine pole in (14). All original Bessel denominators tend to their nonzero p_i. Hence Q is independent of V.

For the pure derivative case, choose i with $v_i \neq 0$. The finite C_i chart of Theorem 11.1 gives

$$\mathcal{E} \longrightarrow E, \qquad \mathcal{U}_l \longrightarrow \eta_l^* := \eta_l - \frac{M_l p_i}{N_i}, \qquad \mathcal{L} \longrightarrow L + \frac{v_i p_i h_i}{c}, \quad c \neq 0.$$

Consider the conjugation-stable field

$$F_0 = K_{\exp}(C(t_l), C(\overline{t_l}) : 1 \leq l \leq m),$$

with repeated sign classes identified. The real numbers $\eta_1, \dots, \eta_n$ are independent over F_0. Apply Lemma 12.1 with $T = \pi$ and $c_l = \mp M_l/N_i$, according as $p_i = \pm \pi i$. It follows that the η_l^* are independent over F_0. In particular the tuple (E, η^*, L) is algebraically independent over k. The same resultant argument applies at (E, η^*). For a root α one has $\alpha \neq L$, so the nonzero choice $c = v_i p_i h_i/(\alpha - L)$ and Rouché's theorem again contradict (14). Thus in every case Q has degree zero in V.

It remains to eliminate positive degree of P in V. Let $P_d(Z, U)$ be its nonzero leading coefficient, and choose any active block i. On its pole divisor, parameterized by

$$v = -\Delta_i - \frac{p_i}{w^{N_i} - 1}, \qquad w \to 1,$$

the other Bessel denominators tend to $p_l - N_l p_i/N_i \neq 0$, while

$$\mathcal{E} \longrightarrow E, \qquad \mathcal{U}_l \longrightarrow \eta_l^*.$$

Apply Lemma 12.1 over $K_{\exp}$. It makes $\eta_1^*, \dots, \eta_n^*$ independent over that field, and hence $P_d(E, \eta^*)Q(E, \eta^*) \neq 0$. At a point of this divisor sufficiently close to $w = 1$, the two factors remain nonzero. The affine coordinate $\mathcal{L}$ has a genuine simple pole there: its singular numerator is

$$-(u_i b_i (b_i - p_i) + v_i h_i)(w^{N_i} - 1), \qquad b_i = -p_i/(w^{N_i} - 1),$$

which cannot vanish near $w = 1$ for an active index. Thus a positive d produces a pole of order d in (14), impossible. Both P and Q are therefore independent of V, proving that the original element belongs to $\mathcal{F}_+$. □

Remark 12.3 (What this relative result leaves open). The theorem allows arbitrary coefficients from all algebraic exponential values and all positive-argument first exponential-integral values. It does not assert that $\overline{\mathbf{E}} \cap \mathcal{F}_+$ equals a specified polynomial ring, and it does not classify the full field of three or more independent harmonic Bessel coordinates at dependent scales. The restriction to positive Ein arguments supplies the reality used in Lemma 12.1; no analogous assertion for arbitrary complex Ein arguments is made here.

13 Derivative coordinates: a valid transfer and a limitation

Put $k = \overline{\mathbb{Q}}$ and retain the order-zero coordinates $\beta(t) = Y(t)/I_0(t)$ and $C(t) = I_1(t)/I_0(t)$.

Proposition 13.1 (All integer orders and derivative orders)**.** *For every integer $m \geq 0$, every $n \geq 0$, and every positive algebraic t, put*

$$C_{m,n}(t) = \frac{I_m^{(n+1)}(t)}{I_m^{(n)}(t)}.$$

The denominator is positive. There is a nondegenerate fractional-linear transformation $M_{m,n,t} \in \mathrm{PGL}_2(k)$ such that

$$C_{m,n}(t) = M_{m,n,t}(C(t)).$$

In particular, $k(C_{m,n}(t)) = k(C(t))$.

Proof. The modified Bessel recurrences

$$I_{m+1} = I_{m-1} - \frac{2m}{t} I_m, \qquad K_{m+1} = K_{m-1} + \frac{2m}{t} K_m$$

show that I_m and $\widetilde{K}_m = (-1)^m K_m$ are obtained from the respective pairs (I_0, I_0') and (K_0, K_0') by the same rational coefficient row. The initial pairs agree with this convention because $I_0' = I_1$

and $K_0' = -K_1$. Differentiation and the common order-zero equation preserve this property. Consequently there is a matrix $T_{m,n}(t) \in \mathrm{Mat}_2(\mathbb{Q}(t))$ with

$$\begin{pmatrix} I_m^{(n)} & \widetilde{K}_m^{(n)} \\ I_m^{(n+1)} & \widetilde{K}_m^{(n+1)} \end{pmatrix} = T_{m,n}(t) \begin{pmatrix} I_0 & K_0 \\ I_0' & K_0' \end{pmatrix}.$$

These matrices have no poles away from $t = 0$.

The power series of I_m has positive coefficients and nonnegative integer exponents, so $I_m^{(r)}(t) > 0$ for every $r \geq 0$ and $t > 0$. Moreover,

$$K_m(t) = \int_0^\infty e^{-t\cosh u} \cosh(mu)\, du$$

implies $(-1)^r K_m^{(r)}(t) > 0$. Thus the determinant on the left of the matrix identity has sign $(-1)^{m+n+1}$ and is nonzero. The order-zero Wronskian is also nonzero. Hence $T_{m,n}(t)$ is invertible for every positive t. At algebraic t, its entries are algebraic, and division of the second row value by the first gives the asserted nondegenerate fractional-linear transformation. □

Corollary 13.2 (Transfer of the attached winding theorem)**.** *Use Theorem 5.1, proved in [6]. For positive algebraic a and distinct positive algebraic t_1, t_2, and arbitrary nonnegative integers m_1, m_2, n_1, n_2, one has*

$$\overline{\mathbf{E}} \cap k(e^a, \mathrm{Ein}(a), \beta(t_1), C_{m_1,n_1}(t_1), \beta(t_2), C_{m_2,n_2}(t_2)) = k[e^a, e^{-a}, \mathrm{Ein}(a)].$$

Without the exponential-integral pair, the intersection equals k. All additional ratios $C_{m,n}(t_i)$ may be adjoined simultaneously without enlarging either field.

This is a coordinate-change consequence of the winding theorem. The new resultant theorem is not required for this positive-scale, integer-order statement. It supplies no additional algebraically independent coordinates at a fixed scale.

13.1 What the resultant theorem does supply

Let $\nu \in \mathbb{Q} \setminus \frac{1}{2}\mathbb{Z}$, $f = x^{-\nu} J_\nu(x)$, $c = 2\nu + 1$, and let A_n, B_n be the coefficient polynomials from [7] at $d = 0$:

$$f^{(n)} = x^{-n}(A_n f + x B_n f').$$

For $n \geq 1$, define

$$D_n(x) = A_n(x) B_{n+1}(x) - A_{n+1}(x) B_n(x).$$

Proposition 13.3 (Exact algebraicity locus for adjacent derivative ratios)**.** *The polynomial $D_n \in \mathbb{Q}[x]$ is even and monic of degree $2n$. For every nonzero algebraic t, the ratio $L_n(t) = f^{(n+1)}(t)/f^{(n)}(t)$ is defined, and*

$$L_n(t) \in k \quad \Longleftrightarrow \quad D_n(t) = 0.$$

If $D_n(t) \neq 0$, then $k(L_n(t)) = k(f'(t)/f(t))$.

Proof. The resultant theorem gives $f^{(n)}(t) \neq 0$ for every n. Writing $L = f'/f$, the ratio is

$$L_n(t) = \frac{A_{n+1}(t) + t B_{n+1}(t) L(t)}{t(A_n(t) + t B_n(t) L(t))}.$$

Siegel's theorem makes $L(t)$ transcendental. The displayed fractional-linear transformation is nondegenerate exactly when $D_n(t) \neq 0$, in which case it is invertible over k and its value is

transcendental. If $D_n(t) = 0$, its coefficient rows are proportional over k; the denominator row is nonzero, since $f^{(n)}(t) \neq 0$, and its value is algebraic.

For even $n = 2r$, the x-leading coefficients of A_n and B_{n+1} are both $(-1)^r$, while the other determinant product has smaller x-degree. For odd $n = 2r+1$, the leading coefficients of B_n and A_{n+1} are $(-1)^r$ and $(-1)^{r+1}$, respectively. In both cases D_n has leading term x^{2n}. Evenness follows from the recurrence. □

For example,

$$D_2(x) = x^2(x^2 - c).$$

At $\nu = 1/3$, $c = 5/3$ and $t = \sqrt{5/3}$, the nonvanishing theorem applies, yet

$$\frac{f'''(t)}{f''(t)} = -\sqrt{5/3}.$$

This also follows immediately by differentiating $f'' + (c/x)f' + f = 0$. It is a concrete counterexample to the proposed implication "all derivative denominators are nonzero, therefore all derivative ratios are transcendental or independent."

13.2 Why nonintegral orders do not inherit the winding line

For $\nu \notin \mathbb{Z}$, a fundamental pair for the normalized Bessel equation is

$$x^{-\nu} J_\nu(x), \qquad x^{-\nu} J_{-\nu}(x).$$

The first member is single valued at zero and the second has exponent -2ν. A counterclockwise loop at zero therefore acts diagonally with eigenvalues 1 and $e^{-4\pi i\nu}$. For rational ν, this monodromy has finite order. The same statement holds for the modified Bessel equation after rotating the argument.

In contrast, at order zero the second solution has logarithmic monodromy $K_0 \mapsto K_0 - 2\pi i I_0$. Its powers produce the unbounded additive parameter used by the winding-line construction. For rational nonintegral order, the finitely many values of the diagonal monodromy cannot justify interpolation to a continuous parameter, the large-winding parabolic limits, or the infinitely many line slopes in the integral-intersection argument.

The resultant theorem addresses zeros of derivative coefficients; it does not change this monodromy. A nonintegral-order analogue of the exact integral-intersection theorem would require a new arithmetic or geometric input. In particular, no statement about Euler's constant, Gompertz's constant, or the full four-constant independence target follows from combining these two mechanisms alone.

14 What remains open

The three-harmonic theorem gives the full rational field on three harmonic ratios, while the affine-projection theorems permit any number of blocks but retain just one affine coordinate. The relative theorem also enlarges the coefficient field to $\mathcal{F}_+$, without classifying $\overline{\mathbf{E}} \cap \mathcal{F}_+$ itself. These results do not give

$$\overline{\mathbf{E}} \cap k(\beta(t_1), C(t_1), \beta(t_2), C(t_2), \beta(t_3), C(t_3)) = k$$

for arbitrary dependent scales. Adjoining the exponential–Ein pair to all three harmonic coordinates is also not justified by combining the two theorems separately. For four harmonic coordinates the present higher-chart and logarithmic-boundary arguments both have additional cancellation problems.

The formal replacement argument of [6] concerns the unmarked family of phase embeddings and the stated E-value inputs. It is not a proof that every E-function method must fail to determine Euler's constant. Extra properties distinguishing the canonical parameter $s = \gamma$ are outside that formal replacement argument. None is established in this supplement.

Finally, the factorial series $\sum_{n\ge 0}(-1)^n n!$ diverges archimedeanly: the Euler–Gompertz constant is its Borel/Hardy sum, $\int_0^\infty e^{-u}/(1+u)\,du$. The existence of its separate p-adic sums does not identify those values with the complex sum. This clarification does not affect the phase-intersection proofs.

15 A closed derivative determinant and its exceptional set

The coefficient determinant in the preceding derivative discussion admits an explicit formula. This calculation is independent of every phase embedding and of the unpublished resultant input. It is an addition to the supplied manuscripts; no assertion of historical priority is intended.

Let c be a complex parameter. Define $A_0 = 1$, $B_0 = 0$ and

$$A_{n+1} = xA_n' - nA_n - x^2 B_n, \tag{15}$$

$$B_{n+1} = A_n + xB_n' + (1 - c - n)B_n. \tag{16}$$

These recurrences give $f^{(n)} = x^{-n}(A_n f + xB_n f')$ for every solution of $f'' + (c/x)f' + f = 0$ away from zero. Write $D_n = A_n B_{n+1} - A_{n+1}B_n$, and let $(a)_j$ denote the rising factorial, with $(a)_0 = 1$.

Theorem 15.1 (Closed determinant formula)**.** *For every integer $n \ge 0$,*

$$D_n(x) = \sum_{j=0}^{\lfloor n/2\rfloor} \frac{(-1)^j n!(c/2)_j}{(n-2j)!\,j!}\, x^{2n-2j}. \tag{17}$$

In particular, this identity holds in $\mathbb{Q}[c, x]$.

Proof. Fix c and choose independent local solutions f, g on a disc avoiding zero. Put $v_n = (f^{(n)}, g^{(n)})$ and $W_n = \det(v_n, v_{n+1})$. Differentiating $xf'' + cf' + xf = 0$ exactly n times gives, for $n \ge 1$,

$$xv_{n+2} + (c+n)v_{n+1} + xv_n + nv_{n-1} = 0.$$

Taking the determinant with v_{n+1} shows that

$$W_{n+1} = W_n + \frac{n}{x}\det(v_{n-1}, v_{n+1}) = W_n + \frac{n}{x}W_{n-1}'.$$

Abel's identity gives $W_0'/W_0 = -c/x$, and the original equation gives $W_1 = W_0$. Therefore $r_n = W_n/W_0$ satisfies

$$r_0 = r_1 = 1, \qquad r_{n+1} = r_n + \frac{n}{x}r_{n-1}' - \frac{nc}{x^2}r_{n-1}.$$

An induction now gives

$$r_n(x) = \sum_{j=0}^{\lfloor n/2\rfloor} \frac{(-1)^j n!(c/2)_j}{(n-2j)!\,j!}\, x^{-2j}.$$

For completeness, if $a_{n,j} = n!(c/2)_j/((n-2j)!j!)$ in its indicated range and is zero otherwise, the required coefficient identity is

$$a_{n+1,j} = a_{n,j} + n(c + 2j - 2)a_{n-1,j-1}.$$

It follows directly from $(c/2)_j = (c/2 + j - 1)(c/2)_{j-1}$. Finally the two derivative rows in the coefficient representation have determinant $x^{-2n}D_n$, so $W_n = x^{-2n}D_n W_0$. This proves the formula for each c and hence the polynomial identity. □

Corollary 15.2 (The exact nonzero degree)**.** *Suppose that $c \notin \{0, -2, -4, \dots\}$. The multiplicity of zero in D_n is exactly $2\lceil n/2 \rceil$. Thus D_n has exactly $2\lfloor n/2 \rfloor$ nonzero roots counted with multiplicity. If $c > 0$, none of those roots lies on the imaginary axis.*

Proof. The coefficient of the lowest displayed power in (17) is nonzero under the hypothesis on c, which gives the multiplicity and the remaining degree. For real $y \neq 0$ and $c > 0$,

$$\frac{D_n(iy)}{(iy)^{2n}} = \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{n!(c/2)_j}{(n-2j)!\, j!}\, y^{-2j} > 0.$$

□

For example,

$$\begin{aligned} D_2 &= x^2(x^2 - c), \\ D_3 &= x^4(x^2 - 3c), \\ D_4 &= x^4\big(x^4 - 6cx^2 + 3c(c+2)\big), \\ D_5 &= x^6\big(x^4 - 10cx^2 + 15c(c+2)\big). \end{aligned}$$

Under the nonvanishing and Siegel inputs used for the adjacent-ratio algebraicity criterion, these are precisely its exceptional loci. For $\nu \in \mathbb{Q} \setminus \frac{1}{2}\mathbb{Z}$ and $c = 2\nu + 1$, the exceptional algebraic arguments have total multiplicity $2\lfloor n/2 \rfloor$, rather than the degree bound $2n$ obtained from monicity alone. This is a multiplicity count, not a claim that all roots are distinct or real.

The polynomials $P_n(x; c) = x^{-n} D_n(x)$ also satisfy the formal generating identity

$$\sum_{n=0}^{\infty} P_n(x; c) \frac{z^n}{n!} = e^{xz}(1 + z^2)^{-c/2}.$$

In particular $P_n' = nP_{n-1}$. This is an Appell polynomial identity; it does not imply that the zeros are all real. For instance, the quadratic in x^2 in D_4 has discriminant $24c(c-1)$, so for $0 < c < 1$ its nonzero roots are not real or purely imaginary.

15.1 No degeneration at positive modified-Bessel arguments

Proposition 15.3 (All adjacent normalized modified derivatives)**.** *Let $\nu \in \mathbb{Q}$, $\nu > -\frac{1}{2}$, and put $F_\nu(t) = t^{-\nu} I_\nu(t)$ for $t > 0$. For every integer $n \geq 0$ and every positive algebraic t, the ratio*

$$L_{\nu,n}(t) = \frac{F_\nu^{(n+1)}(t)}{F_\nu^{(n)}(t)}$$

is defined and is a nondegenerate fractional-linear transformation over $\overline{\mathbb{Q}}$ of $F_\nu'(t)/F_\nu(t)$. Consequently

$$\overline{\mathbb{Q}}(L_{\nu,n}(t)) = \overline{\mathbb{Q}}\left(\frac{F_\nu'(t)}{F_\nu(t)}\right) = \overline{\mathbb{Q}}\left(\frac{I_{\nu+1}(t)}{I_\nu(t)}\right).$$

This proposition uses neither the phase embeddings nor the derivative resultant manuscript.

Proof. The function F_ν satisfies $F_\nu'' + (c/t)F_\nu' - F_\nu = 0$, where $c = 2\nu + 1 > 0$. Its series is a positive constant times

$$\sum_{m=0}^{\infty} \frac{t^{2m}}{4^m m!(\nu+1)_m}.$$

Every derivative is strictly positive on $(0,\infty)$, since infinitely many positive terms survive differentiation. Repeated use of the equation expresses $(F_\nu^{(n)}, F_\nu^{(n+1)})$ by a rational coefficient matrix applied to (F_ν, F_ν').

Rotating the ordinary equation by $x = it$ in Theorem 15.1, or repeating its Wronskian recurrence with the sign of the last term reversed, shows that the determinant of this matrix is

$$(-1)^n \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{n!(c/2)_j}{(n-2j)!\, j!}\, t^{-2j}.$$

It is nonzero for positive t. At algebraic t the matrix has algebraic entries, so its two row values give an invertible fractional-linear transformation. Finally the standard derivative recurrence gives $F_\nu'/F_\nu = I_{\nu+1}/I_\nu$. □

For $\nu \in \mathbb{Q} \setminus \frac{1}{2}\mathbb{Z}$ with $\nu > -\frac{1}{2}$, the published Siegel transcendence theorem for the base normalized Bessel logarithmic derivative (as stated and used in [26]) additionally makes every $L_{\nu,n}(t)$ transcendental. This arithmetic corollary requires only that classical theorem and the preceding proof; it does not require the derivative-nonvanishing resultant input. The field equality is a statement at one fixed order and scale. It does not identify fields at different nonintegral orders or supply the additive monodromy needed by the winding method.

Part II
Pisot floors, prime clocks, and local obstructions

16 The problem, the source boundary, and the new results

Let

$$A = \min\{x > 1 : \lfloor x^{3^k} \rfloor \text{ is prime for every } k \geq 1\}$$

be Mills' constant. Saito proved its irrationality in [13]. His later reduction [14, Theorem 1.7] states that if A is algebraic, some $\beta = A^{3^m}$ is a totally real cubic Pisot number whose two small conjugates satisfy

$$|\beta_3| < -\beta_2 \leq \min\{|\beta_3|^{17/23}, \beta^{-17/40}\}. \tag{18}$$

The trace is then the floor at all sufficiently large cubic-power levels. Excluding every such algebraic candidate would prove unconditional transcendence of A. Saito's Problem 1.1 in [12] asks the stronger question: does every cubic Pisot number have infinitely many composite cubic-power floors?

The word "question" matters here. The source papers contain explicit problems and alternatives, and an answer to a modified sequence does not automatically settle the original least constant. All source theorem numbers below refer to the specified arXiv versions, since journal numbering may differ.

16.1 What is already resolved and what remains

Source and item	Mathematical target	Status used in this paper
[13], Q1.6	Irrationality of the least constant with exponent 2^k	Not proved here.
[13], Q1.7	The cubic Pisot power alternative for Mills' constant	Not excluded in full.
[13], Q5.3	RH implies Mills transcendence	Already answered by Saito; his later Theorem 1.8 uses the Density Hypothesis.
[12], P1.1	All cubic Pisot floors at 3^n	Restricted closures and quantitative exceptions below; full problem remains.
[12], P1.7–1.8	Nonreversible clocks and shifted dyadic Fibonacci values	Already treated in the supplied older manuscripts [008] and [007].
[15], Q1.4–1.5	Unbounded multiplier and Wright-set dimensions	Already treated in supplied older [031], Theorems 14.3 and 15.1.

The last two rows report the source manuscript's descriptions of earlier research notes. Those notes are not supplied with this edition and their claimed resolutions are not independently certified here. They are not counted again as new resolutions. The RH implication is a published external result of Saito, not a consequence newly obtained from the present work.

16.2 Contribution relative to the supplied manuscripts

The source Pisot manuscript reports a prior inventory of 244 TeX manuscripts, including 38 in the Mills–Pisot topic. That archive is not part of the three-manuscript source set for this

edition. References to it below record the source manuscript's provenance, not a new audit of those 244 files. The core additions and their limitations are as follows.

1. Theorem 17.2 synchronizes finitely many shifts on one progression. The old single-shift positive-density statements alone do not supply this synchronization.
2. Theorem 19.1 extends the unshifted cubic coefficient sieve of round S37 to every shift and every odd degree, uniformly in the shift, with an improved explicit range of constants. The older [031] already contains qualitative density-one results; the quantitative rate and the stated extensions are the additions.
3. Theorem 20.1 adds three exact witnesses to the Pell cover of rounds S35–S36. It removes two further valuation layers, but leaves an infinite suborbit to be analyzed.
4. Theorem 21.4 refines the finite patterns of S28 by specifying exact selected prime valuations and their periodic recurrence. Prime-designated positions still require Schinzel's hypothesis H. The all-index irreducibility of S27 is reproduced as prior input, not counted as new.
5. Theorem 22.1 quantifies the persistence of Saito-balanced units in a fixed congruence coset, explaining why finite local exclusions and the real size conditions can coexist.
6. Theorems 24.3 and 24.5 turn an inherited cubic local obstruction into exclusions for all rational powers of Mills' constant in larger number fields. Theorem 23.1 records the existing cyclic norm restriction in a positive-density formulation for both floor branches.
7. Theorem 25.1 removes the seed assumption from finite-prime survival by allowing a fixed nonzero integral norm. It applies to every remaining totally real cubic field and every prescribed finite prime set. Corollary 25.2 places such survivors in any real interval after normalization.

The manuscript contains mathematical proofs and exact finite certificates; external priority remains unestablished. In particular, none of the remaining numbered global questions in the first four rows is claimed as a new complete resolution.

16.3 Conventions

A composite integer is positive and larger than one, with a proper prime divisor. Finite initial exceptions are harmless only after growth has been proved. Lower density always refers to the index n, not to the enormous integer value of the trace. A Pisot number is a real algebraic integer greater than one with every other conjugate strictly inside the unit circle. Traces count all conjugates with multiplicity when polynomial specializations are used.

17 Simultaneous composite windows from a three-adic limit

For a monic integral cubic $X^3 - uX^2 + vX - N$, let U_n be its 3^n-th root power sum, let V_n be the corresponding second symmetric function, and let $E_n = N^{3^n}$. Newton's identities give

$$(U_{n+1}, V_{n+1}, E_{n+1}) = (U_n^3 - 3U_nV_n + 3E_n,\ V_n^3 - 3E_nU_nV_n + 3E_n^2,\ E_n^3). \tag{19}$$

The norm coordinate must be advanced, except when $N = 0, 1, -1$.

Lemma 17.1 (Three-adic contraction)**.** *The sequence U_n has a limit $\lambda \in \mathbb{Z}_3$, with $\lambda - U_n \in 3^{n+1}\mathbb{Z}_3$. This limit depends only on $(u, v, N) \pmod 3$.*

Proof. Let F denote the polynomial map in (19). It is the identity modulo three. If $\mathbf{x} \equiv \mathbf{y} \pmod{3^r}$, $r \geq 1$, expansion gives $F(\mathbf{x}) \equiv F(\mathbf{y}) \pmod{3^{r+1}}$; every first derivative has coefficients

divisible by three, and higher terms have the requisite divisibility. Consecutive iterates therefore differ by a vector divisible by 3^{n+1} at step n. The same argument for two initial points congruent modulo three proves equality of their limits. □

Theorem 17.2 (A common progression for finitely many shifts)**.** *Suppose $U_n \to +\infty$ in the real topology. Let $\mathcal{H} \subset \mathbb{Z}$ be finite and assume*

$$\lambda + h \notin \{1, -1\} \qquad (h \in \mathcal{H}).$$

There are integers $n_0, D \geq 1$ such that all $U_{n_0+jD}+h$, $h \in \mathcal{H}$, are composite for every sufficiently large j.

Proof. Choose a common large n_0 so that $Q_h = U_{n_0} + h > 1$ for all shifts. When $3 \nmid \lambda + h$, also arrange

$$v_3(Q_h - 1) < n_0, \qquad v_3(Q_h + 1) < n_0.$$

The two limiting valuations are finite by hypothesis, so Lemma 17.1 permits this choice. If $3 \mid \lambda + h$, the prime three divides $U_n + h$ at every level and supplies a recurring divisor.

For any other shift factor Q_h into primes. If every prime factor q satisfied $v_3(q^2 - 1) \geq n_0$, then each such q would be 1 or -1 modulo 3^{n_0}. Their product, including multiplicities, would make $Q_h \equiv \pm 1 \pmod{3^{n_0}}$, a contradiction. Hence there is a prime $q_h \mid Q_h$ with

$$v_3(q_h^2 - 1) \leq n_0 - 1.$$

This prime is not three. Every nonzero root of the cubic modulo q_h has order dividing $M_h = \operatorname{lcm}(q_h - 1, q_h^2 - 1, q_h^3 - 1)$. Elementary lifting of valuations gives

$$v_3(M_h) \leq v_3(q_h^2 - 1) + 1 \leq n_0.$$

After powering to 3^{n_0}, all remaining root orders are prime to three. A period D_h for cubing on these groups therefore gives $q_h \mid U_{n_0+jD_h} + h$ for every $j \geq 0$. Zero roots and repeated roots cause no difficulty; the power sum counts multiplicities.

Take D to be the least common multiple of the finitely many D_h, with $D = 1$ when there are none. All chosen prime divisors recur on the same progression. Real growth makes them proper divisors for all sufficiently large j. Unlike a prime-seed argument, this construction works even when every initial value selected is already composite. □

Corollary 17.3 (Windows around cubic Pisot floors)**.** *Let β be a cubic Pisot number and let λ be its three-adic trace limit. If λ is not an ordinary integer, then for every $H \geq 0$ there is an arithmetic progression of indices on which, eventually, all*

$$\lfloor \beta^{3^n} \rfloor + h, \qquad |h| \leq H,$$

are composite simultaneously. Nonreal conjugates are permitted.

Proof. Apply Theorem 17.2 to all integer shifts in $[-H - 1, H]$. None can give $\lambda + h = \pm 1$. The sum of the two small conjugate powers tends to zero, so the floor is eventually either U_n or $U_n - 1$. The stated floor window lies inside the chosen trace window, regardless of changes in the sign of the small-conjugate sum. □

Separate positive-density statements for different shifts cannot be intersected without an argument. The common initial index and the prime-divisor construction above provide that missing step. The single-shift order-budget mechanism was already present in the older manuscript [24]; the simultaneous progression is the strengthening here.

Proposition 17.4 (Exact residue-class scope)**.** *The limit λ is an ordinary integer precisely when the residual cubic over $\mathfrak{F}_3$ either splits completely or is $(X-r)(X^2+1)$ with $r \in \mathfrak{F}_3$. Thus Corollary 17.3 applies to fourteen of the twenty-seven coefficient triples modulo three.*

Proof. For each nonzero residual root, take its Teichmüller lift, the unique prime-to-three root of unity with that residue. In a three-adic splitting field an integral root of positive valuation vanishes under repeated cubing; a unit root's principal-unit part tends to one. Frobenius permutes the multiset of residual roots, so λ is the sum of the Teichmüller lifts, with multiplicity.

A linear root contributes $0, 1$, or -1. The irreducible quadratic X^2+1 contributes two fourth roots of unity with sum zero. The other two irreducible quadratics are X^2+X-1 and X^2-X-1. Their lifted pairs have sum with square -2, so adding one linear contribution cannot give an integer.

For an irreducible cubic, the nonzero roots have order thirteen or twenty-six. The lifted trace is $\pm(\zeta^a+\zeta^{3a}+\zeta^{9a})$, where ζ is a primitive thirteenth root and $a \not\equiv 0 \pmod{13}$. This is not rational: otherwise a polynomial of degree at most twelve with at most four nonzero coefficients would vanish at ζ, in contradiction to its minimal polynomial $1+X+\cdots+X^{12}$. There are ten completely split monic cubics (multisets of three roots from a three-element set), three of the second integer-limit type, six with the other quadratic factors, and eight irreducible cubics. These exhaust the twenty-seven cases. □

Integer limit λ	$(u, v, N) \pmod 3$
-3	$(0,0,2)$
-2	$(1,1,0)$
-1	$(2,0,0),\ (2,1,2),\ (2,2,1)$
0	$(0,0,0),\ (0,1,0),\ (0,2,0)$
1	$(1,0,0),\ (1,1,1),\ (1,2,2)$
2	$(2,1,0)$
3	$(0,0,1)$

The complement of this table is the fourteen-class scope. This is a count of residue classes, not a density assertion about Pisot numbers under an unspecified ordering. For example, the plastic number, the real root of X^3-X-1, has irreducible reduction modulo three and satisfies the simultaneous-window conclusion.

For the unshifted trace problem only the six classes with $\lambda = \pm 1$ can escape Theorem 17.2. These include the surviving Mills alternatives. For the predecessor trace $U_n - 1$, only the four classes with $\lambda = 0$ or 2 can escape; all four have $N \equiv 0 \pmod 3$. In particular, whenever $3 \nmid N$ and the floor is eventually $U_n - 1$, its composite indices have positive lower density. This last single-shift consequence was already available in the earlier local obstruction.

18 Field criteria and a uniform density bound

The preceding simultaneous-return theorem has a useful field-theoretic consequence. We first record an elementary bound that also handles changing signs of nonreal conjugates.

Proposition 18.1 (A parity and trace criterion)**.** *Let β be a cubic Pisot number of odd norm. If $\mathrm{Tr}(\beta) \equiv 0 \pmod 3$, then*

$$\liminf_{M\to\infty} \frac{1}{M} \#\{0 \le n < M : \lfloor \beta^{3^n} \rfloor \text{ is composite}\} \ \ge\ \frac{1}{2}.$$

The two small conjugates may be nonreal.

Proof. Let U_n, V_n, E_n be the coefficient sequences in (19). Since the norm is odd, $E_n = 1$ modulo two. The map on $(U_n, V_n) \pmod 2$ is

$$(a, b) \longmapsto (a + ab + 1,\ b + ab + 1).$$

Its four states satisfy

$$00 \longmapsto 11, \qquad 11 \longmapsto 11, \qquad 01 \longleftrightarrow 10.$$

Consequently U_n is odd either at every sufficiently large index or at exactly one parity of the indices. At least half of all indices therefore have U_n odd. At the same time, (19) modulo three gives $U_n \equiv U_0 \equiv 0 \pmod 3$ for every n.

The sum of the two small conjugate powers is real and tends to zero. Thus, for every sufficiently large n, $\lfloor \beta^{3^n} \rfloor$ belongs to $\{U_n, U_n - 1\}$. When U_n is odd, the first possibility is divisible by three and the second is divisible by two. Growth makes these divisors proper. This proves the density bound without any assumption on the sign pattern of the small-conjugate sum. □

Theorem 18.2 (Cubic fields without a three-adic embedding)**.** *Let K be a cubic number field admitting no field embedding into $\mathbb{Q}_3$. Let $\beta \in \mathcal{O}_K$ be a cubic Pisot number with $3 \nmid N_{K/\mathbb{Q}}(\beta)$. Then the composite indices of $\lfloor \beta^{3^n} \rfloor$ contain, up to finitely many terms, an arithmetic progression. In particular they have positive lower density.*

More precisely, write f for the minimal polynomial of β. If $f \bmod 3$ is irreducible, every finite window about these floors is simultaneously composite on a progression, as in Corollary 17.3. Otherwise $f \bmod 3$ is a cube of a nonzero linear factor. In this second case, if the norm is also odd, the lower density is at least $1/2$.

Proof. A simple root of f mod 3 would lift by Hensel's lemma to a root in $\mathbb{Q}_3$. Since $K = \mathbb{Q}(\beta)$, that root would define an embedding $K \hookrightarrow \mathbb{Q}_3$, contrary to hypothesis. A monic cubic over $\mathbb{F}_3$ without a simple linear factor is either irreducible or $(X - a)^3$. The norm assumption makes $a \neq 0$ in the latter case.

In the irreducible case Proposition 17.4 gives a noninteger three-adic trace limit, and Corollary 17.3 proves the asserted window statement. In the remaining case the three Teichmüller lifts of the residual roots are all 1 or all -1. Thus the trace limit is $\lambda = 3$ or $\lambda = -3$, respectively. Neither λ nor $\lambda - 1$ equals 1 or -1. Apply Theorem 17.2 to the two shifts $\{0, -1\}$. Both U_n and $U_n - 1$ are composite on one eventual progression; the Pisot estimate again places the floor among these two integers. Finally $(X - a)^3$ has zero quadratic coefficient modulo three, so the odd-norm case satisfies Proposition 18.1. □

Corollary 18.3 (Totally ramified cubic fields)**.** *If three is totally ramified in a cubic number field K, then every cubic Pisot unit of K has composite floors at a set of indices of lower density at least $1/2$.*

Proof. The algebra $K \otimes_{\mathbb{Q}} \mathbb{Q}_3$ is a field of degree three, so K has no embedding into $\mathbb{Q}_3$. For any $\beta \in \mathcal{O}_K$, its three local conjugates have the same residue in the residue field $\mathbb{F}_3$. Its minimal polynomial therefore reduces to $(X - a)^3$ modulo three. A unit has nonzero residue and norm ± 1. The last assertion of Theorem 18.2 applies. □

For example, $K = \mathbb{Q}(r)$ with $r^3 - 3r - 1 = 0$ is a totally real cyclic cubic field: the polynomial is irreducible, has discriminant 81, and its translate $X^3 + 3X^2 - 3$ is Eisenstein at three. Hence the corollary covers every Pisot unit in this field. It also covers every Pisot unit in $\mathbb{Q}(\sqrt[3]{2})$, since $(X - 1)^3 - 2 = X^3 - 3X^2 + 3X - 3$ is Eisenstein at three. An explicit example in the latter field is $\beta = 1 + \sqrt[3]{2} + \sqrt[3]{4}$, whose minimal polynomial is $X^3 - 3X^2 - 3X - 1$. It has norm one and a unique real root greater than one; its conjugate pair has modulus $\beta^{-1/2} < 1$. Thus the density bound applies even though the floor correction need not have an eventually constant sign.

Completion of the norm-positive case in the field of discriminant 1524**.** Let r be the largest root of $X^3 - 7X^2 - X + 1$ and let $K = \mathbb{Q}(r)$. Showa round 20 proves that every norm-one unit $\beta \in K$ with $\mathrm{Tr}(\beta^{3^n}) \to +\infty$ has a positive lower density of composite trace terms. This is a prior result of the supplied archive, not a new trace theorem here. It also yields the following precise floor statement.

Corollary 18.4 (Conditional on the cited round-20 trace theorem)**.** *Every cubic Pisot unit of norm one in the field of discriminant* 1524 *has a positive lower density of composite floors* $\lfloor \beta^{3^n} \rfloor$.

Proof. The field is totally real. Since $\beta > 1$ and its norm is one, its two small conjugates have the same sign. If both are negative, the floor equals the trace eventually, and the cited round-20 theorem applies. If both are positive, the floor equals $U_n - 1$ eventually. The four-state parity calculation in the proof of Proposition 18.1 shows that U_n is odd at a set of indices of density at least $1/2$. At those sufficiently large indices $U_n - 1$ is a positive even composite integer. □

Scope and comparison with the supplied manuscripts. The source manuscript reports that [24] proves infinitely many composite floors for every nonreal cubic Pisot number, and gives positive-density results for fixed real shifts. Its projective return theorem already includes the cubic-residue norm refinement; that refinement is not new here. The absence of a three-adic embedding was already a necessary obstruction for a hypothetical prime trace tail in [25]. The new use here is the simultaneous treatment of the two possible floor corrections, together with the explicit density-1/2 criterion. In particular the statements above do not depend on an unjustified intersection of separate positive-density sets.

The discriminant-1524 corollary is a completion from traces to all norm-one Pisot floors using a prior field-specific theorem. It makes no assertion about norm-minus-one units in that field: round 17 exhibits such units that survive several local tests. Nor do these criteria exclude every totally real cubic Pisot candidate in Saito's Problem 1.1 or establish Mills transcendence.

19 Shifted traces: exact local counts and sparse exceptions

This section extends the unshifted cubic sieve of submitted round S37. The extensions are uniformity in the integer shift, both floor branches in totally real cubic fields, and arbitrary odd degree. We include the proof, since changing a trace to a shifted trace is not automatically legitimate in a fixed-norm argument.

For an odd integer $d \geq 3$, write

$$f_{\mathbf{a}}(X) = X^d - a_1 X^{d-1} + a_2 X^{d-2} - \cdots - a_d, \qquad P_n(\mathbf{a}) = \sum_{j=1}^{d} \alpha_j^{3^n}.$$

Here the roots are counted with multiplicity, and Newton's identities define $P_n \in \mathbb{Z}[a_1, \ldots, a_d]$ even at reducible specializations. Its total degree is 3^n, and its unique term of that degree is $a_1^{3^n}$. For a sequence of integers, a *recurring-divisor certificate* is a prime dividing all terms on an infinite arithmetic progression of indices. If the terms tend to positive infinity, such a certificate gives positive lower density of composite terms.

Theorem 19.1 (Uniform shifted sieve)**.** *Put* $c_0 = 1/(e\sqrt{2\log 3})$. *Fix an odd* $d \geq 3$ *and* $0 < c < c_0$. *Let* $\mathcal{B} \subset \mathbb{Z}^d$ *be any box with side lengths* $X_1, \ldots, X_d \geq 3$, *and put* $m = \min X_i$ *and* $V = \prod X_i$. *For every integer* h, *all but*

$$O_{d,c}(V \exp(-c\sqrt{\log m})) \tag{20}$$

coefficient vectors in $\mathcal{B}$ have a recurring-divisor certificate for $P_n(\mathbf{a}) + h$. The constant is independent of h and of the location of the box.

There is a constant $C_{d,c} > 0$ such that the certificate can be chosen with prime $p \leq \exp(C_{d,c}\sqrt{\log m})$ and progression period at most $\exp(dC_{d,c}\sqrt{\log m})$. Consequently, whenever $P_n(\mathbf{a}) \to +\infty$, the lower density of composite shifted terms is positive, and is at least the reciprocal of that period.

The theorem gives an upper bound on a set of possible exceptions; it does not prove that this set is empty. The assertion is about the full coefficient space. A thin fixed field, fixed norm, or selected one-parameter family requires a separate argument.

Lemma 19.2 (Exact shifted local count)**.** *For every prime $p \equiv 2 \pmod 3$, $n \geq 1$, and $t \in \mathfrak{F}_p$,*

$$\#\{\mathbf{a} \in \mathfrak{F}_p^d : P_n(\mathbf{a}) = t\} = p^{d-1}.$$

Repeated roots and zero constant terms are allowed.

Proof. The coefficient vectors are Frobenius-stable orbits of ordered root tuples under S_d. Twisted orbit counting gives the number in question as $d!^{-1} \sum_{\sigma \in S_d} \#\operatorname{Fix}(F\sigma)$, where the fixed tuples also satisfy the power-sum condition. This is an identity of integer counts, including orbits with stabilizers. All tuples under consideration lie in a common finite extension, so ordinary finite-set counting applies.

A cycle of length j in σ corresponds to an element $b \in \mathfrak{F}_{p^j}$ and contributes $\operatorname{Tr}_{\mathfrak{F}_{p^j}/\mathfrak{F}_p}(b^{3^n})$ to the power sum. Every permutation of an odd number of letters has an odd cycle. For such a cycle j, $3 \nmid p^j - 1$, so $b \mapsto b^{3^n}$ is a bijection of $\mathfrak{F}_{p^j}$. The field trace is surjective, with fibers of size p^{j-1}. Fixing all other cycles leaves exactly that many choices for this one, giving p^{d-1} fixed tuples for each permutation. Averaging proves the claim. Surjectivity of field trace does not require $p \nmid j$. □

Lemma 19.3 (Geometric separation)**.** *Over an algebraically closed field of characteristic different from three, $P_n - t$ is irreducible for every $n \geq 1$ and every t. For distinct indices i, j and any $s, t \in \mathfrak{F}_p$, $p \neq 3$,*

$$\#\{P_i = s,\ P_j = t\} \leq 3^{i+j} p^{d-2}.$$

Proof. Under the finite, surjective Vieta map the first polynomial pulls back to $Z_1^r + \cdots + Z_d^r - t$, $r = 3^n$. Regard this as a polynomial in Z_d over $k[Z_1, \ldots, Z_{d-1}]$. The nonconstant polynomial $Z_1^r + \cdots + Z_{d-1}^r - t$ has a prime factor with multiplicity one: it is squarefree, since any repeated factor would divide all its partial derivatives, which are multiples of the relatively prime monomials $Z_1^{r-1}, \ldots, Z_{d-1}^{r-1}$. Eisenstein at such a factor proves irreducibility of the pullback, and hence of $P_n - t$.

Different indices have different degrees, so the two hypersurfaces have no common component. Their intersection has pure dimension $d - 2$ and degree at most 3^{i+j}, by Bézout. An affine variety of dimension r and degree D has at most Dp^r rational points. One proof is induction on dimension: on each irreducible component choose a nonconstant coordinate, slice at its p rational values, and apply the degree bound for hyperplane sections. Constant coordinates are discarded. This argument also handles components defined only over an algebraic closure, since it bounds their $\mathfrak{F}_p$-points individually. These facts give the displayed bound. □

Lemma 19.4 (A uniform return budget)**.** *Fix $B \geq 1$ and put $b_d = \lfloor \log_3(d/2) \rfloor$. Suppose $p \equiv 2 \pmod 3$ and $v_3(p+1) \leq B$. For every coefficient vector modulo p, its power-sum sequence is periodic from index $B + b_d$, with period less than p^d.*

Proof. For an irreducible root factor of degree j, each nonzero root has order dividing p^j-1. For odd j this order is prime to three; for even $j=2k$, the lifting-the-exponent identity gives

$$v_3(p^{2k}-1)=v_3(p+1)+v_3(k)\le B+b_d.$$

After that many cubings all nonzero root powers have orders prime to three. Cubing is then a permutation of their finite groups. Zero roots stay zero and multiplicities do not change this argument. The least common multiple M of the remaining root orders is at most the product of p^j-1 over distinct nonzero irreducible factors, which is less than p^d. A common period is the order of 3 modulo M, with period one when $M=1$. □

Proof of Theorem 19.1. The eligible primes of Lemma 19.4 form fixed reduced residue classes modulo 3^{B+1} of density

$$\delta=\delta_B=\frac{1-3^{-B}}{2}.$$

Let $k_0=B+b_d$ and use the L consecutive levels $k_0,\dots,k_0+L-1$. At an eligible prime let Ω_p be the union of the loci $P_n+h=0$ at these levels, and put $\omega(p)=|\Omega_p|$. Exact counts and the first two inclusion–exclusion inequalities give

$$\frac{L}{p}-\frac{D_L}{p^2}\le\frac{\omega(p)}{p^d}\le\frac{L}{p},\qquad D_L=\sum_{k_0\le i<j<k_0+L}3^{i+j}<9^{k_0+L}.$$

For $p>y:=9^{k_0+L+1}$ we therefore have

$$\frac{L-1}{p}\le\frac{\omega(p)}{p^d}\le\frac{L}{p}. \tag{21}$$

These estimates are independent of the shift, even when it varies with the box.

We use the classical multidimensional large sieve in the form

$$|S|\ll_d\frac{(m+Q^2)^d}{H(Q)},\qquad H(Q)=\sum_{\substack{q\le Q\\ q\ \text{squarefree}}}\prod_{p\mid q}\frac{\omega(p)}{p^d-\omega(p)}, \tag{22}$$

for a set S in a cube of side m avoiding the forbidden classes. For completeness, its normalization can be seen directly. On $(\mathbb{Z}/p\mathbb{Z})^d$ take a mean-zero function equal to 1 on the allowed classes and to $-(p^d-\omega(p))/\omega(p)$ on the forbidden classes. Its normalized squared norm is $1/g(p)$, where $g(p)=\omega(p)/(p^d-\omega(p))$. The Chinese-remainder product, Parseval, and Cauchy–Schwarz imply

$$g(q)|S|^2\le\sum_{\substack{\mathbf{a}\bmod q\\ \gcd(a_1,\dots,a_d,q)=1}}\left|\sum_{\mathbf{x}\in S}e^{2\pi i\mathbf{a}\cdot\mathbf{x}/q}\right|^2.$$

Primitive frequencies with denominators at most Q are Q^{-2}-separated. The analytic large sieve bounds their combined energy by $O_d((m+Q^2)^d)|S|$, proving (22). This is a classical input, not a new sieve inequality; see [16].

Set $Q=\lfloor\sqrt{m}\rfloor$, $s=\sqrt{\log Q}$, $L=\lfloor as\rfloor$, and $z=\exp(bs)$, with fixed positive a,b such that

$$b>a\log 9,\qquad ab\delta<1.$$

Only eligible primes in $(y,z]$ are used. The prime number theorem in these fixed residue classes and partial summation give

$$\sum_{y<p\le z}^{*}\frac{1}{p}=\delta\log\frac{b}{a\log 9}+o(1),\qquad\sum_{p\le z}^{*}\frac{\log p}{p}=(\delta+o(1))\log z.$$

The star restricts to eligible primes. The moving endpoints cause no uniform-modulus issue: B is fixed first and both endpoints tend to infinity.

Expand the finite Euler product $W = \prod^*_{y<p\le z}(1+g(p))$ with its nonnegative squarefree weights. Equation (21) implies

$$\log W \ge \left(a\delta \log \frac{b}{a\log 9} + o(1)\right) s.$$

The weighted mean of $\log q$ in this expansion is exactly

$$\sum_{y<p\le z}^{*} \frac{g(p)}{1+g(p)} \log p \le (ab\delta + o(1)) \log Q.$$

Since $ab\delta < 1$, Markov's inequality leaves a fixed positive fraction of the weight on $q \le Q$. Thus $H(Q) \gg W$. Take

$$a = \frac{1}{e\sqrt{\delta \log 9}}, \qquad b = (1-\varepsilon)e\sqrt{\frac{\log 9}{\delta}}$$

with $\varepsilon > 0$ sufficiently small. As $\varepsilon \downarrow 0$, the coefficient in the lower bound for $\log H(Q)$, when expressed in units of $\sqrt{\log m}$, approaches $e^{-1}\sqrt{\delta/\log 3}$. Finally choose a fixed large B so that this exceeds the prescribed $c < c_0$. We obtain $H(Q) \gg_{d,c} \exp(c\sqrt{\log m})$.

The cube sieve proves (20) for a cube. A rectangular box is covered by at most $2^d V/m^d$ such cubes. A point not in the sifted set has a zero at one of the chosen levels modulo some $p \le z$. Lemma 19.4 supplies the claimed certificate and period bound. Bounded values of m are absorbed in the constants. □

Corollary 19.5 (Both totally real floor branches)**.** *Among coefficient triples in any box of minimum side m, the totally real cubic Pisot polynomials for which $\lfloor \beta^{3^n} \rfloor$ fails to be composite on a set of positive lower density number at most*

$$O_c(X_1X_2X_3 e^{-c\sqrt{\log m}}), \qquad c < c_0.$$

Proof. If the two small real conjugates are γ, η, their odd-power sum has a constant nonzero sign for all sufficiently large powers. The only potential exception is $\gamma = -\eta$, which would make β the integer trace, contradicting cubic irreducibility. Thus the floor is eventually P_n or $P_n - 1$. Apply the theorem to both shifts and take a union bound. The traces tend to infinity. □

Corollary 19.6 (Many shifts)**.** *For any finite set $\mathcal{H} \subset \mathbb{Z}$, the exceptional count for having a certificate for every $h \in \mathcal{H}$ is at most $O_{d,c}(|\mathcal{H}|Ve^{-c\sqrt{\log m}})$. In particular, for $0 < \eta < c < c_0$, one may take $|h| \le \exp(\eta\sqrt{\log m})$ and retain a saving $e^{-(c-\eta)\sqrt{\log m}}$. The progressions for different shifts in this corollary need not coincide. A genuinely simultaneous progression is proved separately in Section 17.*

20 Two further valuation layers on the twice-square Pell locus

Consider

$$g_u(X) = X^3 - uX^2 - X + 1, \qquad 4u^2 + 5u + 23 = 2w^2, \quad u \ge 2, \quad w > 0.$$

Let $P_n(u)$ be the sum of the 3^n-th powers of the roots of g_u. Signs at $-1, 0, 1, u, u+1$ give roots $-x, y, \beta$ with $0 < x, y < 1 < u < \beta < u+1$. Neither rational candidate ± 1 is a root. The norm is -1, and the second symmetric function -1 gives $\beta(y-x) = -1 + 1/\beta < 0$. Thus $x > y$ and $P_n(u) = \lfloor \beta^{3^n} \rfloor$ at every level. Write

$$T(u,w) = (17u + 12w + 10, 24u + 17w + 15), \qquad (a_j, r_j) = T^j(9,14), \quad (b_j, s_j) = T^j(1,4).$$

The Pell reduction in the supplied round 35 identifies the entire locus with a_j for $j \geq 0$ and b_j for $j \geq 1$. Round 36 excludes the first orbit and all second-orbit indices except $j \equiv 0, 4374 \pmod{13122}$. We have independently replayed the complete finite certificate for that assertion. The following three additional witnesses strengthen its conclusion.

Theorem 20.1. *If $(P_n(u))_{n\geq 0}$ is eventually prime on this Pell locus, then*

$$u = b_j, \qquad 39366 \mid j, \qquad v_3(u-1) \geq 10, \qquad u > 10^{59049}.$$

Every parameter excluded by this condition has a positive lower density of composite trace indices.

Proof. Put $\eta = 17 + 12\sqrt{2}$ and write $j = 2k$. The identity

$$(8b_j + 5) + 4s_j\sqrt{2} = (13 + 16\sqrt{2})\eta^j$$

shows that $\eta^{2m} = 1 \pmod{p}$ makes b_{2k} periodic modulo p with period dividing m. The following table consists of exact certificates; its column c is $b_{2a} \bmod p$.

p	m	a	c	n	$v_3((p-1)(p^2-1)(p^3-1))$
8279983	6561	2187	7051991	4863973	25
27398737	19683	6561	15563564	4929815	31
15037811	19683	13122	1777082	549606	9

In every row, p is prime, $\eta^{2m} = 1 \pmod{p}$, and $P_n(c) = 0 \pmod{p}$. The root-order return lemma makes each such zero recurrent, because n exceeds the last column. Thus the entire parameter-index class $k \equiv a \pmod{m}$ has infinitely many composite traces, along an arithmetic progression of trace indices.

For reproducibility, put $M_p = (p-1)(p^2-1)(p^3-1)$ and $E = 3^n \bmod M_p$. Binary polynomial powering modulo (g_c, p) gives $X^E = A + BX + CX^2$, with coefficient triples in the order of the table

$$(7459560, 4961071, 5552814), \qquad (21974350, 24684017, 2338784),$$
$$(13021067, 3973052, 8155287).$$

Each satisfies $3A + cB + (c^2+2)C = 0 \pmod{p}$. This verifies a large trace-index witness without constructing the integer trace. Primality of the three modest witness primes is certified by trial division up to their square roots.

The round 36 cover leaves $k \equiv 0, 2187 \pmod{6561}$. The first new row removes the latter class. The two remaining rows remove $k \equiv 6561, 13122 \pmod{19683}$, leaving $19683 \mid k$. Therefore $39366 \mid j$. The exact identity $v_3(b_j - 1) = 1 + v_3(j)$ for positive even j now gives $v_3(u-1) \geq 10$. On the Pell curve, $w > 4u/3$, so $T(u,w)_1 > 33u$. Hence

$$u = b_j > 33^j \geq 33^{39366} = 1089^{19683} > 10^{59049}.$$

□

Corollary 20.2 (Relative count on the complete Pell locus)**.** *Let $\mathcal{P}(X)$ count the parameters in the displayed Pell locus with $u \leq X$, and let $\mathcal{E}(X)$ count those with an eventually prime trace tower. Then*

$$\mathcal{P}(X) = \frac{2\log X}{\log(17 + 12\sqrt{2})} + O(1), \qquad \mathcal{E}(X) \leq \frac{\log X}{39366\log(17 + 12\sqrt{2})} + O(1).$$

In particular $\limsup_{X\to\infty} \mathcal{E}(X)/\mathcal{P}(X) \leq 1/78732$.

Proof. Each affine Pell orbit grows as a positive constant times $(17+12\sqrt{2})^j$. The theorem retains at most every 39366-th index of the second orbit and none of the first. □

Remark 20.3 (The remaining uniform step). The strengthened numerical and valuation bounds concern this special parameter family, not the numerical value of Mills' constant. The exceptional suborbit is not proved empty. Indeed, for any fixed modulus M, let L be the period of the orbit of $(1,4)$ under T modulo M. Such a period exists because the linear part of T has determinant one. At every positive index divisible by $39366L$, $b_j \equiv 1 \pmod{M}$. The trace pair $(1,-1)$ is an exact fixed point of the cubing recurrence, so $P_n(b_j) \equiv 1 \pmod{M}$ for every n. Thus an arbitrary prescribed finite divisibility sieve still leaves unboundedly many parameters in the surviving branch.

21 Uniform irreducibility and exact finite divisibility patterns

Provenance and improvement. The supplied Showa round 27 already proves uniform irreducibility in an explicit cyclic cubic unit family. It supersedes the earlier six-term conditional prefix result as an existence theorem, although the earlier six polynomials belonged to a different monomial family. Round 28 already realizes every finite prime/composite pattern under Schinzel's hypothesis H. We reproduce the decisive proofs and add two refinements: a general signed-specialization criterion with an exact three-block limitation, and arbitrary prescribed local multiplicities which recur unchanged on infinite progressions of trace indices. These refinements do not remove hypothesis H from any assertion of simultaneous prime values.

21.1 The balanced unit and its all-index irreducibility

Put

$$A = a^3+2a^2+3a+3, \qquad Q = a^2+a+1,$$
$$f_a(Z) = Z^3 - AZ^2 - a^2Z + 1, \qquad d(a) = (a+1)(a^2+3)(a^2+3a+3).$$

The discriminant is $d(a)^2$. In the cyclic root ordering set

$$s = \frac{ar+1}{1-Qr}, \qquad t = \frac{r-1}{Qr+a}, \qquad \beta_a = -r^5 s.$$

For integer $a \geq 8$, take r to be the largest root. The rational-root test and square discriminant show that the field is cyclic cubic. The trace seed is

$$\begin{aligned} U_0 &= a^{14}+9a^{13}+45a^{12}+161a^{11}+450a^{10}+1020a^9+1918a^8 \\ &\quad +3021a^7+3987a^6+4379a^5+3942a^4+2820a^3+1521a^2+558a+105, \\ V_0 &= -3a^6-15a^5-40a^4-75a^3-105a^2-90a-33, \end{aligned}$$

and

$$U_{n+1} = U_n^3 - 3U_nV_n - 3, \qquad V_{n+1} = V_n^3 + 3U_nV_n + 3. \tag{23}$$

These identities follow from $rst=-1$ and Newton's identities: $U_0 = -T_{5,1}$ and $V_0 = -T_{4,5}$, where $T_{ij} = r^is^j + s^it^j + t^ir^j$. Thus $U_n(a) = \mathrm{Tr}(\beta_a^{3^n})$.

Theorem 21.1 (Showa round 27, independently audited)**.** *Every U_n is monic and irreducible over $\mathbb{Q}$, of degree $14\cdot 3^n$, and $U_n(-1)=1$. For every integer $a \geq 8$ the number β_a is a cubic Pisot unit of norm -1 whose conjugates are $\beta_a, -X, Y$, with*

$$0 < Y < X < 1, \quad X^{40}\beta_a^{17} < 1, \quad X^{23} < Y^{17}, \qquad U_n(a) = \lfloor \beta_a^{3^n} \rfloor.$$

Proof. First every odd trace is positive at every real parameter. For real $a \neq -1$, the three roots of f_a are distinct and real. Their product is -1 and their pairwise sum is $-a^2 \leq 0$, so they have the form $p > q > 0, -w < 0$ with $pqw = 1$. If $a > -1$, then $f_a(1) < 0$ gives $q < 1 < p$. The cyclic fractional-linear map sends p to $-w$: its value cannot vanish or have a pole on this connected parameter interval, since its determinant is $(a+1)^2 \neq 0$, and it is negative at $a = 0$. The conjugates of β_a are therefore

$$P = p^5 w, \qquad R = w^5 q, \qquad -B = -q^5 p, \qquad B/P = q^6/p^3 < 1.$$

Their odd-power sum is positive. If $a < -1$, then $f_a(1) > 0$ and $A(a) < A(-1) = 1$ show that both positive roots lie below one: if both exceeded one, $A = p + q - 1/(pq) > 1$. In either cyclic orientation the negative conjugate has magnitude $B < 1$, while $PRB = 1$ forces a positive conjugate above one. The odd trace is again positive. At $a = -1$ the specialized conjugate multiset is $(1, 1, -1)$, and every odd trace is one.

Direct reduction gives

$$\overline{U}_0 = a^5(a-1)^3(a^2+1)^3 \quad \text{over } \mathbb{F}_3. \tag{24}$$

For $n \geq 1$ the three Hensel blocks of U_n are irreducible over $\mathbb{Q}_3$ and have degrees $5 \cdot 3^n, 3 \cdot 3^n, 6 \cdot 3^n$. Here is the valuation argument, including repeated residual factors. At a root ξ of U_n, monicity gives integrality and

$$v_3(U_j(\xi)) = 3^{j-n} \quad (0 \leq j < n).$$

The last step follows from $U_{n-1}^3 = 3(1 + U_{n-1}V_{n-1})$; descending steps compare valuations less than one with terms divisible by three. If a residual irreducible factor h has multiplicity m and degree e, write $U_0 = H^m B + 3C$ with $B(\xi)$ a unit. Then $v_3(H(\xi)) = 1/(m3^n)$. The ramification index is divisible by $m3^n$ and the residue degree by e. Their product equals the degree of the Hensel block, proving its irreducibility.

A monic rational factor F of U_n is integral and has no real root, so it has even degree and is positive on the real line. The only proper even subset sums of $5, 3, 6$ are 6 and 8. The corresponding residual factors are

$$(a^2+1)^{3^{n+1}}, \qquad a^{5 \cdot 3^n}(a-1)^{3^{n+1}}.$$

Both have value -1 modulo three at $a = -1$. On the other hand F and its monic complement have positive integer values there multiplying to $U_n(-1) = 1$, so $F(-1) = 1$. This contradiction proves irreducibility for every $n \geq 1$.

For the seed, the exact factorizations modulo two have degrees $6, 8$, and those modulo thirteen have degrees $5, 9$. For reproducibility, the factors are specified by ascending coefficient arrays:

$$\begin{array}{r|l} 2 & (1,0,1,1,0,1,1), \quad (1,0,0,1,0,1,1,0,1) \\ 13 & (12,7,8,8,4,1), \quad (12,7,2,11,12,9,6,4,5,1). \end{array}$$

Their products and irreducibility are certified by the Frobenius criterion $X^{p^d} = X \bmod F$ and $\gcd(F, X^{p^{d/q}} - X) = 1$ for primes $q \mid d$. The disjoint proper factor-degree sets prove rational irreducibility of U_0.

Finally, for $a \geq 8$ and $w = -s$,

$$a^3 < r < 2a^3, \quad a/Q < w < 1/a, \quad 0 < t < 1/a^2, \quad rwt = 1,$$

so

$$(4/5)a^{14} < \beta_a < 32a^{14}, \quad X = t^5 r < 2a^{-7}, \quad Y = w^5 t < a^{-7}.$$

Since $V_0(a) < -1$ and $\beta_a XY = 1$, $\beta_a(Y - X) = V_0(a) + 1/\beta_a < 0$, giving $X > Y$. Also $X^{40}\beta_a^{17} < 2^{125}a^{-42} \leq 1/2$, and $X^{23}/Y^{17} = X^{40}\beta_a^{17}$. Every odd-power trace is its positive dominant power minus a number strictly between zero and one, which proves the floor identities. □

21.2 A signed-specialization criterion and its exact limitation

Proposition 21.2. *Suppose integer polynomials U_0, V_0 satisfy (23), U_0 is monic,* $\deg V_0 < 2\deg U_0$*, and for some integer a_* one has $(U_0(a_*), V_0(a_*)) = (1, -1)$. Suppose each U_n is strictly positive on $\mathbb{R}$. Write $\overline{U}_0 = \prod_{i=1}^{r} h_i^{m_i}$ into distinct monic irreducible factors over $\mathbb{F}_3$, with* $\deg h_i = d_i$*. Give block i the label*

$$\lambda_i = (m_i d_i \bmod 2,\ h_i(a_*)^{m_i}) \in (\mathbb{Z}/2\mathbb{Z}) \times \mathbb{F}_3^{\times}.$$

If no nonempty proper subcollection of these labels has product $(0,1)$, then U_n is irreducible over $\mathbb{Q}$ for every $n \geq 1$. Moreover, this label criterion is possible only for $r \leq 3$. With $r = 3$, it holds exactly when the labels are the three different nonidentity elements of this group.

Proof. The root-valuation proof above gives one irreducible local block of degree $m_i d_i 3^n$ and reduction $h_i^{m_i 3^n}$ for each i. Since 3^n is odd, its parity and value at a_* retain precisely the label λ_i. A proper rational factor must have even degree and positive integer value one at a_*, contradicting the label condition. The product of all labels is $(0,1)$, because U_0 has even degree and value one at a_*. The group has three nonidentity elements, each of order two. A minimal product-one sequence cannot contain the identity (unless it has length one), or a repeated element (unless it has length two). Thus it has length at most three; the three-element case must consist of all three nonidentity elements. Conversely those three labels have product one and no proper product-one subcollection. □

This is a limitation of this particular factor obstruction, not a bound on the number of local factors of a globally irreducible polynomial. For the balanced seed the labels are $(1,-1), (1,+1), (0,-1)$, so the criterion is extremal. The four-block degree-31 seed of the earlier six-prefix argument cannot be handled by this exact criterion alone.

21.3 Simple forcing roots and exact local multiplicities

We next state a strengthening of round 28. All prime-value assertions below explicitly assume the usual integer-prime hypothesis H.

Lemma 21.3 (Forcing roots, from round 28)**.** *For every finite window $I = \{0, \dots, L\}$ and $j \in I$, a positive- Dirichlet-density set of primes $p \equiv 2 \pmod 3$ has a residue b such that*

$$U_j(b) = 0, \quad U_j'(b) \neq 0, \quad d(b) \neq 0, \qquad U_i(b) \neq 0 \quad (i \in I \setminus \{j\}).$$

Any prescribed finite set of primes may be excluded.

Proof. Let $K_j = \mathbb{Q}(\xi)$ for a root of U_j. It has an odd-degree completion above three. For $j \geq 1$ this follows from the local blocks of degrees $5 \cdot 3^j$ and $3 \cdot 3^j$. For $j = 0$, Hensel lifts the residual factor a^5 to a degree-five block; at least one of its irreducible constituents has odd degree. Thus $\mathbb{Q}(\sqrt{-3})$ is not a subfield of K_j, since its completion at three has degree two and would embed in every completion of K_j.

In a Galois extension containing the splitting field of U_j and $\mathbb{Q}(\sqrt{-3})$, choose an automorphism fixing ξ and acting nontrivially on the quadratic field. It exists by the preceding noncontainment. Its conjugacy class has a fixed root and the same nontrivial quadratic character. Chebotarev gives a positive-density set of primes with a root of U_j and $p \equiv 2 \pmod 3$. Exclude the finitely many divisors of $\operatorname{disc}(U_j)$, $\operatorname{Res}(U_j, d)$, and $\operatorname{Res}(U_j, U_i)$ for $i \neq j$. All these integers are nonzero by irreducibility and unequal degrees. The asserted simple root and all other nonvanishing conditions follow. □

Theorem 21.4 (Prescribed exact multiplicities with periodic recurrence)**.** *Fix a finite window $I = P \sqcup C$, a positive integer M, and for each $j \in C$ a nonempty finite list of positive integers $e_{j,1}, \dots, e_{j,r_j}$. There are pairwise distinct primes $p_{j,k} \geq 5$ with $p_{j,k} \equiv 2 \pmod 3$, all coprime to M, and an arithmetic progression $\mathcal{A}$ of integer parameters with the following unconditional properties for every sufficiently large $a \in \mathcal{A}$:*

1. *$a \equiv -1 \pmod M$, and every odd-power trace of β_a is 1 modulo M;*
2. *for $i \in I$,*
$$v_{p_{j,k}}(U_i(a)) = \begin{cases} e_{j,k}, & i = j, \\ 0, & i \neq j; \end{cases}$$
3. *with*
$$T_{j,k} = \operatorname{ord}_{p_{j,k}^{e_{j,k}}(p_{j,k}^3-1)}(3),$$
one has
$$v_{p_{j,k}}(U_{j+hT_{j,k}}(a)) = e_{j,k} \qquad (h \geq 0).$$

Assuming hypothesis H, infinitely many $a \in \mathcal{A}$ additionally make every $U_i(a)$, $i \in P$, prime. Every prescribed composite position j then has at least r_j different prime divisors, with exactly the specified multiplicities for the chosen divisors. If P is empty, no conjecture is required.

Proof. For each pair (j,k) choose a fresh prime $p \geq 5$ and simple forcing root (p,b) from Lemma 21.3. Hensel's lemma lifts b uniquely to a root modulo p^{e+1}, where $e = e_{j,k}$. Among the p lifts of the root modulo p^e, choose any of the other $p-1$ classes modulo p^{e+1}. Call it c. Taylor expansion and $U_j'(b) \neq 0$ give $v_p(U_j(c)) = e$; every other designated $U_i(c)$ is a unit modulo p. CRT now imposes $a \equiv -1 \pmod M$ and $a \equiv c_{j,k} \pmod{p_{j,k}^{e_{j,k}+1}}$ simultaneously. The resulting progression has modulus

$$D = M \prod_{j \in C} \prod_{k=1}^{r_j} p_{j,k}^{e_{j,k}+1}.$$

This proves the exact multiplicities within the finite window.

We verify recurrence at the same precision. Fix a chosen p, e and consider

$$\mathcal{B} = (\mathbb{Z}/p^{e+1}\mathbb{Z})[Z]/(f_a(Z)).$$

Because $p \nmid d(a)$, its reduction is etale. The square discriminant forces splitting type $1+1+1$ or 3, so every residue-field unit is killed by $p^3 - 1$. The kernel of reduction on $\mathcal{B}^\times$ is $1 + p\mathcal{B}$ and is killed by p^e, by induction with the binomial theorem. Hence every unit of $\mathcal{B}$ is killed by $E = p^e(p^3-1)$. The root transformation defining s has invertible denominator: a simultaneous zero of $1 - QZ$ and $aZ + 1$ would force $(a+1)^2 = 0 \bmod p$, excluded by $d(a) \neq 0$. Thus $\beta = -r^5 s$ is a unit in this algebra, and its traces agree with the integer polynomials. Since $p \equiv 2 \pmod 3$, $\gcd(E, 3) = 1$. For $T = \operatorname{ord}_E(3)$,

$$U_{n+T}(a) \equiv U_n(a) \pmod{p^{e+1}} \qquad (n \geq 0).$$

The nonzero residue of $U_j(a)/p^e$ modulo p is therefore retained at every $j + hT$, proving the exact recurring valuations.

Write the CRT progression as $a = A_0 + Dz$. For each $i \in P$, the polynomial $U_i(A_0 + Dz)$ is irreducible, primitive, and has positive leading coefficient. Their product has no fixed prime divisor: primes dividing M give values one; each forcing prime gives nonzero values at every prime-designated index; at a prime not dividing D choose z so that $A_0 + Dz \equiv -1$.

Hypothesis H consequently supplies infinitely many simultaneous prime values. All other claims are unconditional congruences. Finally, every odd trace is an integer polynomial in a and has value one at $a = -1$, proving the modulo-M assertion. Taking a large ensures the Pisot bounds and makes each forced value larger than its chosen prime divisors. □

The theorem prescribes valuations of selected divisors, not a complete prime factorization: unselected prime factors may also occur. It strengthens the finite-pattern statement in the archive while retaining its essential finite-window quantifier and its explicit conjectural prime-value assumption.

21.4 Audit boundaries and remaining arithmetic question

The single-block norm-minus-one family of round 22 also has all-index local irreducibility, but its index-zero trace is a fourth power; its prime-prefix statement deliberately begins at the next index. The cyclic unit family above is stronger for conditional prime-prefix existence and starts at index zero. Round 23 is compatible with the present recurrence theorem: its isolated divisors are chosen at primes congruent to one modulo three, while the recurring divisors above lie at primes congruent to two modulo three. In round 23 the prime and unit may depend on the isolated index; the result does not say that every trace has such a divisor in one fixed tower.

Round 26 proves a separate polynomial-family obstruction: fixed nonzero norm and square discriminant force the polynomial coefficient functions to be algebraically dependent and, over $\mathbb{C}$, to factor through one common polynomial. Its logarithmic two-form argument and the normalization-to-an-affine-line argument are valid; they do not prevent the one-parameter construction used here. We do not use a conjectural cyclic lift of the round 22 family, the multivariate Hilbert construction of round 24, or an unverified primality certificate from round 25 as inputs to the proofs above.

Uniform irreducibility is an unconditional result already in the archive. Arbitrarily long prime prefixes remain conditional on H; exact prescribed local divisibility is unconditional. None implies one fixed algebraic number whose entire infinite trace tower is prime. Nor does any identify the least-prime path defining Mills' constant. The full cubic Pisot compositeness problem and unconditional Mills transcendence remain open within these arguments.

22 Finite local conditions and logarithmic unit lattices

The new non-Mills material does not supply an additional proved route to Mills transcendence. A useful transfer is nevertheless possible: the unit-lattice viewpoint gives a quantitative explanation of why finitely many local conditions and Saito's strict conjugate bounds can coexist. It identifies a genuine limitation of that combination of tests without assuming that any trace is prime.

Let K be a totally real cubic field, with real embeddings $\sigma_1, \sigma_2, \sigma_3$. Fix a unit $w \in \mathcal{O}_K^\times$ with

$$\sigma_1(w) > 0, \qquad \sigma_2(w) < 0, \qquad \sigma_3(w) > 0.$$

Its norm is -1. Let $\mathfrak{m}$ be a nonzero integral ideal, and set

$$E^+(\mathfrak{m}) = \{u \in \mathcal{O}_K^\times : \sigma_i(u) > 0 \ (1 \le i \le 3),\ u \equiv 1 \pmod{\mathfrak{m}}\}.$$

Its two-coordinate logarithmic lattice is

$$\Lambda_{\mathfrak{m}} = \{(\log \sigma_1(u), \log \sigma_2(u)) : u \in E^+(\mathfrak{m})\} \subset \mathbb{R}^2.$$

Write $\Delta_{\mathfrak{m}} > 0$ for its covolume in these coordinates. This convention avoids an extraneous factor of $\sqrt{3}$ from embedding the lattice into the plane $x + y + z = 0$ in $\mathbb{R}^3$.

Theorem 22.1 (Quantitative persistence in every admissible unit coset)**.** *Fix $0 < a < 1/2$. For $H > 1$, let $B_{w,\mathfrak{m},a}(H)$ count the units $\beta \in wE^+(\mathfrak{m})$ such that, on writing*

$$B = \sigma_1(\beta), \qquad r = \sigma_2(\beta), \qquad s = \sigma_3(\beta),$$

one has

$$1 < B \le H, \qquad 0 < s < -r < 1, \qquad -r < B^{-a}.$$

Then

$$B_{w,\mathfrak{m},a}(H) = \frac{1-2a}{4\Delta_{\mathfrak{m}}}(\log H)^2 + O_{K,w,\mathfrak{m},a}(\log H + 1). \tag{25}$$

Every counted B is a cubic Pisot number of norm -1. In addition,

$$-r < s^{a/(1-a)}, \qquad \beta \equiv w \pmod{\mathfrak{m}}.$$

In particular, for $a = 17/40$, the two strict bounds

$$-r < \min(B^{-17/40}, s^{17/23})$$

hold for quadratically many units in logarithmic height in every such coset.

Proof. The subgroup $E^+(\mathfrak{m})$ has finite index in the unit group: positivity imposes only a finite sign condition, and reduction modulo $\mathfrak{m}$ maps units to the finite group $(\mathcal{O}_K/\mathfrak{m})^\times$. Dirichlet's unit theorem therefore makes $\Lambda_{\mathfrak{m}}$ a full lattice of rank two. Positivity removes its possible torsion, so its logarithm map is injective.

The logarithms of the coset $wE^+(\mathfrak{m})$ form the translate

$$(\log \sigma_1(w), \log|\sigma_2(w)|) + \Lambda_{\mathfrak{m}}.$$

For a point of this translate put $x = \log B$ and $y = \log(-r)$. Because $|\operatorname{N}\beta| = 1$, one has $\log s = -x - y$. The asserted inequalities are exactly

$$0 < x \le \log H, \qquad -x/2 < y < -ax. \tag{26}$$

Indeed $y > -x/2$ is $-r > s$, and $y < -ax < 0$ gives $-r < 1$; then $-x - y < -x/2 < 0$, so $s < 1$ as well. The triangle in (26) has area

$$\int_0^{\log H} (1/2 - a)x\,dx = \frac{1-2a}{4}(\log H)^2$$

and boundary length $O_a(\log H)$. Counting translates of a fixed fundamental parallelogram gives its area divided by $\Delta_{\mathfrak{m}}$, with error $O(\log H + 1)$: only cells meeting a fixed-width neighborhood of the boundary can contribute to the error. Strict versus non-strict boundaries changes the count by at most the same order. This proves (25).

The dominant embedding exceeds one and the other two have modulus less than one, so B is Pisot. It cannot be rational, since a rational unit is ± 1; its degree is therefore three. Finally $B(-r)s = 1$, and the inequality $-r < B^{-a}$ is equivalent to $(-r)^{1-a} < s^a$, proving the additional bound. The congruence is the one defining the coset. □

Corollary 22.2 (Persistence of every fixed finite trace sieve)**.** *Take $\mathfrak{m} = M\mathcal{O}_K$ with integer $M \ge 2$. Every unit counted in Theorem 22.1 satisfies*

$$\operatorname{Tr}_{K/\mathbb{Q}}(\beta^j) \equiv \operatorname{Tr}_{K/\mathbb{Q}}(w^j) \pmod{M} \qquad (j \ge 1).$$

Thus, if a seed w has $\gcd(\operatorname{Tr}(w^{3^n}), M) = 1$ at every level $n \ge 0$, then all the counted balanced Pisot units have the same avoidance property at every level.

Proof. From $\beta - w \in M\mathcal{O}_K$ one obtains $\beta^j - w^j \in M\mathcal{O}_K$ for every positive integer j. The trace of an algebraic integer is an integer, so tracing gives the stated congruence. The last assertion follows separately at every level; no primality assertion is inferred from coprimality with M. □

The seed hypothesis in this corollary is essential. The theorem does not assert that an arbitrary finite sieve has an avoiding seed in a specified field. Instead it proves that, once a finite residue class is admissible, the archimedean restrictions alone cannot eliminate it: there are asymptotically $c(\log H)^2$ members satisfying the strict conjugate bounds. The regulator enters only through the positive covolume $\Delta_{\mathfrak{m}}$ and changes the leading constant, not the existence of these candidates. This is stronger than an unquantified ray construction but does not close the remaining norm-minus-one cyclic Pisot case.

Example 22.3 (An explicit nonempty congruence coset). Let r be the largest root of $X^3 - 25X^2 - 4X + 1$ and order its other roots as $s < 0 < t$. The rational root test excludes ± 1, the discriminant is 273^2, and root brackets give

$$25 < r < 26, \qquad -0.294 < s < -0.292, \qquad 0.135 < t < 0.137.$$

Thus $K = \mathbb{Q}(r)$ is cyclic cubic and r has signs $(+, -, +)$ and norm -1. Direct root enumeration of the polynomial over $\mathbb{F}_2, \mathbb{F}_5, \mathbb{F}_{17}$ shows it irreducible in all three cases; these primes are inert. The integral element $v = (r^2 + 2)/3$ satisfies

$$v^3 - 213v^2 + 290v - 99 = 0.$$

This polynomial reduces to $X(X-1)(X+1)$ modulo three, so 3 splits completely. The residues of r at its three primes are $(-1, 1, 1)$ in some order, as follows from its norm -1 and trace $25 \equiv 1 \pmod 3$.

Put $w = r^{66619}$, where $66619 = 7 \cdot 31 \cdot 307$. At 2, the order of r divides 7; at 5 and 17, the finite-field norm identities give $r^{31} = -1$ and $r^{307} = -1$, respectively. The relevant quotients of 66619 are odd. Consequently, for every odd integer $j \geq 1$,

$$\operatorname{Tr}(w^j) \equiv \begin{cases} 1 & \pmod 2, \\ 1 & \pmod 3, \\ 2 & \pmod 5, \\ 14 & \pmod{17}. \end{cases}$$

Applying Theorem 22.1 to $wE^+(510\mathcal{O}_K)$ gives asymptotically $3(\log H)^2/(80\Delta_{510\mathcal{O}_K})$ strictly Saito-balanced Pisot units whose entire cubic-power trace towers avoid $2, 3, 5, 17$. This asserts finite-prime avoidance, not primality.

23 A complete cyclic cubic norm class

Submitted round S14 already proves that an eventually prime, growing trace tower in a cyclic cubic field must have norm congruent to zero or minus one modulo three [11, Theorem 1]. Thus the norm class excluded below is not newly discovered here. Combining that restriction with the trace-limit classification and cyclic Frobenius-return argument of [24], we record a positive-density formulation for the actual Pisot floors, including both possible real floor branches. This is a synthesis of existing ingredients, not a new return mechanism or a claim of historical priority.

Theorem 23.1 (Cyclic cubic norms congruent to one)**.** *Let $K/\mathbb{Q}$ be a cyclic cubic extension, and let $\beta \in K$ be a cubic Pisot number. If*

$$N = \mathrm{N}_{K/\mathbb{Q}}(\beta) \equiv 1 \pmod 3,$$

then the set

$$\mathcal{C}_\beta = \{n \geq 0 : \lfloor \beta^{3^n} \rfloor \text{ is composite}\}$$

has positive lower density. In particular, this conclusion holds for every cyclic cubic Pisot unit of norm $+1$.

Proof. A cyclic cubic field is totally real, since a nontrivial complex conjugation would have order two in a group of order three. Write the conjugates of β as $\beta, \gamma_2, \gamma_3$, and put

$$U_n = \mathrm{Tr}_{K/\mathbb{Q}}(\beta^{3^n}), \qquad f(X) = X^3 - uX^2 + vX - N.$$

The two small conjugates have different absolute values: equality would give $\gamma_2 = -\gamma_3$ and hence $\beta = u \in \mathbb{Z}$, contrary to degree three. Thus their odd-power sum has a fixed nonzero sign. For all sufficiently large n,

$$\lfloor \beta^{3^n} \rfloor = U_n - s, \qquad s \in \{0, 1\},$$

where $s = 1$ exactly when the small conjugate of larger absolute value is positive. Also $U_n \to +\infty$. Let $\lambda = \lim_{n\to\infty} U_n$ in $\mathbb{Z}_3$.

If $s = 1$, Proposition 17.4 and its explicit table show that either equality $\lambda = 0$ or $\lambda = 2$ would force $3 \mid N$. Since $N \equiv 1 \pmod 3$, neither equality is possible. Theorem 17.2, with the single shift $h = -1$, supplies an eventual arithmetic progression of composite floors.

Suppose now that $s = 0$. If $\lambda \notin \{1, -1\}$, Theorem 17.2 with $h = 0$ gives the same conclusion. We handle the two remaining limits separately.

If $\lambda = 1$, the residue classification gives

$$f(X) \equiv (X - 1)(X^2 + 1) \pmod 3,$$

because $N \equiv 1 \pmod 3$. Its root 1 is simple, so Hensel's lemma gives an embedding $K = \mathbb{Q}(\beta) \hookrightarrow \mathbb{Q}_3$. In a Galois extension all primes over three have the same local degree; the degree-one completion therefore implies that three splits completely in K. The three conjugates of β now lie in $\mathbb{Q}_3$ and are units, because $3 \nmid N$. Each has Teichmüller part 1 or -1. Their sum is $\lambda = 1$, so these three parts must be $1, 1, -1$ in some order. Their product gives $N \equiv -1 \pmod 3$, a contradiction. Equivalently, the displayed separable reduction has factorization type $1 + 2$, which cannot occur at an unramified prime in a cyclic cubic field.

It remains to treat $\lambda = -1$. Lemma 17.1 gives $U_n \equiv -1 \pmod 3$ for every n. If only finitely many U_n are prime, then every sufficiently late floor is composite, and the density is one. Otherwise choose a sufficiently late prime $p = U_r$ with

$$p \nmid 3N \operatorname{disc}(f).$$

This is possible since $U_n \to +\infty$. Frobenius in the cyclic cubic field has cycle type $1 + 1 + 1$ or 3. Consequently all roots of f modulo p lie in $\mathfrak{F}_{p^3}$ and are nonzero. Since $p \equiv -1 \pmod 3$,

$$\gcd(3, p^3 - 1) = 1.$$

Choose $D \geq 1$ with $3^D \equiv 1 \pmod{p^3 - 1}$. For every residual root a and every $j \geq 0$ this gives $a^{3^{r+jD}} = a^{3^r}$. Summing over the three roots yields

$$U_{r+jD} \equiv U_r = 0 \pmod p \qquad (j \geq 0).$$

For sufficiently large j the floor equals $U_{r+jD} > p$, so this is an arithmetic progression of composite floors. Every progression obtained above has positive lower density, completing the proof. □

Remark 23.2 (What remains in the cyclic Mills branch). For a hypothetical cyclic cubic shadow of the least Mills constant, the dominant small conjugate is negative. An eventually prime trace tail therefore forces $\lambda = 1$: other limits are excluded by Theorem 17.2 or by the last part of the proof above. If its norm is prime to three, complete splitting at three then forces $N \equiv -1 \pmod 3$. In particular, a unit shadow must have norm -1. The norm--1 cyclic units constructed elsewhere in this paper show why Theorem 23.1 does not settle the entire cyclic case or unconditional Mills transcendence.

24 Excluding rational powers of Mills' constant from number fields

Restricted cubic results already give complete exclusion statements for other, substantially larger classes of algebraic expressions. Throughout this section A is the least Mills constant defined in the introduction. The cubic three-adic embedding obstruction below was already proved in the supplied manuscript [25]. We include a short proof from Theorem 17.2 and Proposition 17.4 to make the subsequent field deductions self-contained. The ambient-field and cyclotomic consequences are deductions from this obstruction; no historical priority is asserted for them.

Lemma 24.1 (Prime traces force a three-adic embedding)**.** *Let β be a cubic Pisot number, with minimal polynomial $f(X) = X^3 - uX^2 + vX - N$, and put $U_n = \mathrm{Tr}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta^{3^n})$. If U_n is prime for every sufficiently large n, then its three-adic limit satisfies $\lambda \in \{1, -1\}$, and $\mathbb{Q}(\beta)$ admits an embedding into $\mathbb{Q}_3$. No restriction on the norm N is required.*

Proof. The Pisot property gives $U_n = \beta^{3^n} + o(1)$, so $U_n \to \infty$. If $\lambda \notin \{1, -1\}$, apply Theorem 17.2 with the single shift 0. It gives infinitely many composite U_n, contrary to the hypothesis. Write $\lambda = \varepsilon$, where $\varepsilon \in \{1, -1\}$. The six corresponding coefficient classes in Proposition 17.4 satisfy

$$u \equiv \varepsilon, \qquad N \equiv \varepsilon v \pmod 3, \qquad f(X) \equiv (X - \varepsilon)(X^2 + v) \pmod 3.$$

If $v \not\equiv 2 \pmod 3$, the residual root ε is simple. If $v \equiv 2 \pmod 3$, the residual polynomial is $(X - \varepsilon)^2(X + \varepsilon)$, whose root $-\varepsilon$ is simple. Hensel's lemma therefore gives a root of f in $\mathbb{Z}_3$. Since f is irreducible over $\mathbb{Q}$, evaluation at this root embeds $\mathbb{Q}[X]/(f) = \mathbb{Q}(\beta)$ into $\mathbb{Q}_3$. □

Lemma 24.2 (Pisot powers retain their field)**.** *If β is a Pisot number and r is a positive integer, then $\mathbb{Q}(\beta^r) = \mathbb{Q}(\beta)$.*

Proof. An embedding of $\mathbb{Q}(\beta)$ into $\mathbb{C}$ that fixes $\mathbb{Q}(\beta^r)$ sends β to a conjugate γ with $\gamma^r = \beta^r$. Hence $|\gamma| = \beta > 1$. The defining Pisot condition leaves only $\gamma = \beta$. Separability in characteristic zero says that the number of such embeddings is $[\mathbb{Q}(\beta) : \mathbb{Q}(\beta^r)]$, which is therefore one. □

Theorem 24.3 (An obstruction in every ambient number field)**.** *Let $K \subseteq \mathbb{C}$ be a number field. If $A^q \in K$ for some $q \in \mathbb{Q} \setminus \{0\}$, then K contains a totally real cubic subfield F admitting an embedding into $\mathbb{Q}_3$. Consequently, if K has no such subfield, then*

$$\boxed{A^q \notin K \qquad (q \in \mathbb{Q} \setminus \{0\}).}$$

Here rational powers of $A > 1$ denote their positive real values.

Proof. Replacing q by $-q$ if necessary, write $q = r/s$ with positive integers r, s. The inclusion $A^{r/s} \in K$ makes A algebraic. Saito's reduction [14, Theorem 1.7] supplies $\beta = A^{3^m}$, a totally

real cubic Pisot number satisfying (18). Its two small conjugates have negative sum at every sufficiently large odd power: the negative one has strictly larger absolute value. Thus

$$\lfloor \beta^{3^n} \rfloor = \mathrm{Tr}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta^{3^n})$$

for every sufficiently large n. These integers are prime by the definition of A. Lemma 24.1 therefore gives a $\mathbb{Q}_3$-embedding of $F = \mathbb{Q}(\beta)$. Finally,

$$\beta^r = (A^{r/s})^{s3^m} \in K, \qquad \mathbb{Q}(\beta^r) = F$$

by Lemma 24.2, proving $F \subseteq K$. □

Corollary 24.4 (Cubic fields and radical expressions)**.** *If a cubic number field K has no embedding into $\mathbb{Q}_3$, then no nonzero rational power of A belongs to K. In particular this holds when 3 is inert or totally ramified in K. Equivalently, no positive element of such a field, raised to a nonzero rational power, can equal A.*

Proof. Any cubic subfield of the cubic field K is K itself, so Theorem 24.3 applies. Inertness and total ramification both exclude a prime of K over 3 with ramification index and residue degree equal to one, which an embedding into $\mathbb{Q}_3$ would require. □

Theorem 24.5 (A Galois prime-count obstruction)**.** *Let $K/\mathbb{Q}$ be a finite Galois extension and let $g_3(K)$ be the number of prime ideals of K above 3. If $3 \nmid g_3(K)$, then $A^q \notin K$ for every $q \in \mathbb{Q} \setminus \{0\}$.*

Proof. Suppose otherwise and let $F \subseteq K$ be the cubic subfield from Theorem 24.3. Put $G = \mathrm{Gal}(K/\mathbb{Q})$ and $H = \mathrm{Gal}(K/F)$, so $[G : H] = 3$. The $\mathbb{Q}_3$-embedding supplies a prime $\mathfrak{p}$ of F with completion $F_\mathfrak{p} = \mathbb{Q}_3$. Choose a prime $\mathfrak{P}$ of K above $\mathfrak{p}$, and let $D \subseteq G$ be its decomposition group. The image of F in $K_\mathfrak{P}$ lies in $\mathbb{Q}_3$, so every member of D fixes F and $D \subseteq H$. The usual transitive action on the primes above 3 gives

$$g_3(K) = [G : D] = [G : H][H : D] = 3[H : D],$$

contrary to the hypothesis. □

Remark 24.6 (The exact abelian scope of the local obstruction). If $K/\mathbb{Q}$ is finite abelian, it contains a cyclic cubic subfield in which 3 splits completely if and only if $3 \mid g_3(K)$. Indeed, its decomposition group D is normal. An index-three subgroup H containing D exists precisely when the finite abelian group G/D has order divisible by three. Its fixed field is cyclic cubic, is totally real because its degree is odd, and has trivial decomposition group at 3. This equivalence concerns the existence of a subfield with the required local property; it does not assert that a Mills candidate exists in that subfield.

Corollary 24.7 (Cyclotomic exclusions)**.** *Write $N = 3^b m$, where $b \geq 0$ and $(m, 3) = 1$. If*

$$3 \nmid \frac{\varphi(m)}{\mathrm{ord}_m(3)},$$

then $A^q \notin \mathbb{Q}(\zeta_N)$ for every nonzero rational q. Here ζ_N is a primitive Nth root of unity, and $\varphi(1) = \mathrm{ord}_1(3) = 1$ by convention. In particular, for every nonzero rational q,

$$A^q \notin \bigcup_{a,b \geq 0} \mathbb{Q}(\zeta_{7^a 3^b}), \qquad A^q \notin \bigcup_{a,b \geq 0} \mathbb{Q}(\zeta_{13^a 3^b}).$$

Proof. In $\mathbb{Q}(\zeta_m)$ the prime 3 is unramified and its Frobenius has order $\operatorname{ord}_m(3)$, so it has $\varphi(m)/\operatorname{ord}_m(3)$ primes above it. The 3-power cyclotomic factor is totally ramified at 3. Locally, adjoining this totally ramified factor to the unramified completion multiplies the local degree by $\varphi(3^b)$, just as the global degree is multiplied by $\varphi(3^b)$. Thus it does not change the number of primes:

$$g_3(\mathbb{Q}(\zeta_{3^b m})) = \frac{\varphi(m)}{\operatorname{ord}_m(3)}.$$

Theorem 24.5 proves the first assertion.

For $a \geq 1$, elementary lifting of multiplicative orders gives

$$\operatorname{ord}_{7^a}(3) = 6 \cdot 7^{a-1}, \qquad \operatorname{ord}_{13^a}(3) = 3 \cdot 13^{a-1}.$$

Indeed, the orders modulo 7 and 13 are 6 and 3, respectively, and $v_7(3^6 - 1) = v_{13}(3^3 - 1) = 1$. The corresponding values of g_3 are 1 and 4, neither divisible by three. The case $a = 0$ has $g_3 = 1$. Every element of either displayed union lies in one finite stage, proving the two exclusions. □

Corollary 24.8 (A quotient obstruction)**.** *Let $K/\mathbb{Q}$ be finite Galois. If $\operatorname{Gal}(K/\mathbb{Q})$ has no quotient isomorphic to C_3 or S_3, then $A^q \notin K$ for every nonzero rational q. This applies, in particular, when the Galois group is perfect, or is S_d with $d \geq 5$.*

Proof. If a cubic subfield F existed, its normal closure would lie in K and have a transitive cubic Galois group, necessarily C_3 or S_3. Restriction would give the prohibited quotient. Apply Theorem 24.3. Quotients of a perfect group are perfect, whereas C_3 and S_3 are not. For $d \geq 5$, the only proper nontrivial quotient of S_d is C_2. □

Remark 24.9 (Composita and the remaining cubic problem). These exclusions cannot be combined by simply taking composita. For example, $\mathbb{Q}(\zeta_7)$ and $\mathbb{Q}(\zeta_{13})$ separately satisfy Theorem 24.5, but their compositum has

$$g_3(\mathbb{Q}(\zeta_{91})) = \frac{72}{\operatorname{lcm}(6,3)} = 12.$$

By Remark 24.6, that compositum contains a cyclic cubic subfield in which 3 splits completely.

The section excludes entire fields and all rational powers at once, without a unit or norm hypothesis. Nevertheless, it does not eliminate the cubic shadows already satisfying $\lambda = \pm 1$, a $\mathbb{Q}_3$-embedding, and Saito's real-conjugate bounds. In the cyclic case such a shadow must have 3 completely split. Excluding every remaining Mills shadow would suffice for unconditional transcendence of A, even without solving the stronger question for every cubic Pisot number. No such final exclusion is obtained here.

25 The exact field boundary for a fixed finite set of prime divisors

The unit-coset result above assumes an admissible seed. Allowing a fixed nonzero integral norm, rather than requiring norm minus one, removes that assumption in every remaining totally real cubic field. The resulting statement concerns divisibility by a fixed finite set of primes; it makes no assertion that the surviving integers are prime.

For a cubic algebraic integer β, write

$$\lambda(\beta) = \lim_{n \to \infty} \operatorname{Tr}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta^{3^n}) \quad \text{in } \mathbb{Z}_3.$$

Its existence and its dependence on the residual polynomial are the three-adic convergence and classification in Proposition 17.4 and its preceding construction.

Theorem 25.1 (Every admissible field survives every fixed finite prime sieve)**.** *Let K be a totally real cubic field, with real embeddings $\sigma_1, \sigma_2, \sigma_3$. The following are equivalent.*

1. *There is a field embedding $K \hookrightarrow \mathbb{Q}_3$.*
2. *For every finite set P of rational primes, there are infinitely many $\beta \in \mathcal{O}_K$ such that, on writing*
$$B = \sigma_1(\beta), \qquad \sigma_2(\beta) = -\rho, \qquad \sigma_3(\beta) = s,$$
one has
$$B > 1, \qquad 0 < s < \rho < 1, \qquad \rho < \min\{B^{-17/40}, s^{17/23}\}, \tag{27}$$
$$\lambda(\beta) = 1, \qquad \gcd\left(\lfloor B^m \rfloor, \prod_{p \in P} p\right) = 1 \quad \text{for every positive odd integer } m. \tag{28}$$

Here every B in (2) is a cubic Pisot number generating K. For fixed K, P, the elements in (2) may be chosen in one coset $\alpha E^+(M\mathcal{O}_K)$, where
$$M = \prod_{p \in P \cup \{3\}} p, \qquad \mathrm{N}_{K/\mathbb{Q}}(\alpha) = -N, \qquad N \geq 1, \qquad \gcd(N, M) = 1.$$

Thus their norm is the same integer $-N$. With Δ_M denoting the covolume of the logarithmic lattice of $E^+(M\mathcal{O}_K)$ in its first two embedding coordinates, the number of these elements satisfying (27) and $B \leq H$ is
$$\frac{3}{80\Delta_M}(\log H)^2 + O_{K,\alpha,M}(\log H + 1). \tag{29}$$

The entire statement remains true with $\lambda(\beta) = -1$.

Proof. Assume first that $K \hookrightarrow \mathbb{Q}_3$. Then the degree-three étale algebra $K \otimes_{\mathbb{Q}} \mathbb{Q}_3$ is either $\mathbb{Q}_3^3$ or $\mathbb{Q}_3 \times E$, with $E/\mathbb{Q}_3$ a quadratic field. The corresponding decomposition of $\mathcal{O}_K \otimes \mathbb{Z}_3$ allows the following prescriptions modulo three.

Local algebra	Prescribed residues	Residual polynomial
$\mathbb{Q}_3^3$	$(1, 1, -1)$	$(X-1)^2(X+1)$
$\mathbb{Q}_3 \times E$, E ramified	$(-1, 1)$	$(X+1)(X-1)^2$
$\mathbb{Q}_3 \times E$, E unramified	$(1, i)$, $i^2 = -1$ in $\mathfrak{F}_9$	$(X-1)(X^2+1)$

In the ramified case the residue 1 is the scalar element of $\mathcal{O}_E/3\mathcal{O}_E$, so its multiplication operator has characteristic polynomial $(X-1)^2$, including multiplicity. In the unramified case $i \notin \mathfrak{F}_3$, and its two residual conjugates are i and $-i$. Every listed element is a unit modulo three.

For each $p \in P \setminus \{3\}$ prescribe the scalar residue $1 \in \mathcal{O}_K/p\mathcal{O}_K$. The Chinese remainder theorem gives a nonempty coset $\alpha_0 + M\mathcal{O}_K$ satisfying all these prescriptions. Under the three real embeddings, $M\mathcal{O}_K$ is a full lattice in $\mathbb{R}^3$. Every translate of such a lattice meets the open orthant $(+,-,+)$: a sufficiently large multiple of a vector in that orthant has a lattice-coset point within a fixed distance. Choose α in this intersection. Its norm is $-N$ for a positive integer N, and all prescribed residues are units, giving $\gcd(N, M) = 1$.

Every element $\beta \in \alpha E^+(M\mathcal{O}_K)$ has the same signs, norm, and residues as α. Consequently, for every positive integer m and every $p \in P \setminus \{3\}$,
$$\mathrm{Tr}(\beta^m) \equiv 3 \pmod{p}.$$

For every positive odd m, the three-adic prescriptions instead give $\mathrm{Tr}(\beta^m) \equiv 1 \pmod 3$: the residual roots are either $1, 1, -1$ or $1, i, -i$, with multiplicity. Their Teichmüller lifts have sum one, so the classification in Proposition 17.4 gives $\lambda(\beta) = 1$ whenever β generates K.

It remains to count the elements with the required real conjugates. The subgroup $E^+(M\mathcal{O}_K)$ has finite index in the unit group. By Dirichlet's unit theorem [10, Chapter 5], its first two logarithmic coordinates therefore form a full rank-two lattice of covolume Δ_M. Positivity removes the torsion kernel. The corresponding coordinates of $\alpha E^+(M\mathcal{O}_K)$ form a translate of this lattice. Put $x = \log B$, $y = \log \rho$, and $L = \log N \geq 0$. The norm identity gives

$$\log s = L - x - y.$$

For $a = 17/40$, the inequalities $1 < B \leq H$, $0 < s < \rho < 1$, and $\rho < B^{-a}$ are exactly

$$0 < x \leq \log H, \qquad \frac{L-x}{2} < y < -ax. \tag{30}$$

This region has area

$$\frac{1-2a}{4}(\log H)^2 + O_N(\log H + 1)$$

and boundary length $O_N(\log H + 1)$. Counting fundamental parallelograms, as in Theorem 22.1, gives (29). The leading constant is positive.

For every counted point, the dominant embedding exceeds one and the other two have modulus less than one. Thus B is Pisot and cannot be rational; because $[K : \mathbb{Q}] = 3$, it generates K. The additional bound in (27) follows from the integer norm:

$$\rho^{40} < B^{-17} \leq N^{17}B^{-17} = (\rho s)^{17}, \qquad \text{so} \qquad \rho < s^{17/23}.$$

Finally, if m is positive and odd, then $-\rho^m + s^m \in (-1, 0)$. Therefore

$$\lfloor B^m \rfloor = \operatorname{Tr}_{K/\mathbb{Q}}(\beta^m),$$

and the preceding residual calculations prove (28).

Conversely, a single element from (2) generates K and has $\lambda = 1$. Proposition 17.4 shows that its minimal polynomial modulo three is one of

$$X^2(X-1), \qquad (X-1)(X^2+1), \qquad (X-1)^2(X+1).$$

Each has a simple linear root over $\mathfrak{F}_3$. Hensel's lemma lifts that root to $\mathbb{Q}_3$, and the resulting root of the irreducible minimal polynomial defines an embedding $K \hookrightarrow \mathbb{Q}_3$.

For the variant with $\lambda = -1$, negate only the prescribed residues at three. The real sign choice is still made independently in the same open orthant. The odd traces are now -1 modulo three, and the Teichmüller sum is -1; all other arguments are unchanged. The reverse implication follows by negating the three residual polynomials in the preceding paragraph. □

Corollary 25.2 (Normalized survivors in every real interval)**.** *Fix an admissible field K, a finite prime set P, and a nonempty open interval $I \subset (1, \infty)$. For every sufficiently large integer k, there are elements β from Theorem 25.1 such that*

$$B^{1/3^k} \in I, \qquad \gcd\left(\left\lfloor (B^{1/3^k})^{3^{k+n}} \right\rfloor, \prod_{p \in P} p\right) = 1 \quad (n \geq 0).$$

If I is contained in an open interval on which a prescribed finite list of prime floors $\lfloor t^{3^j} \rfloor$ is constant, these normalized survivors have that same finite prime prefix.

Proof. Choose $1 < a < b$ with $(a, b) \subset I$. Subtracting (29) at $H = b^{3^k}$ and $H = a^{3^k}$ gives

$$\frac{3}{80\Delta_M} 3^{2k}((\log b)^2 - (\log a)^2) + O_{K,\alpha,M,a,b}(3^k),$$

which is positive for all sufficiently large k. Any resulting $a^{3^k} < B \le b^{3^k}$ can be obtained with strict upper endpoint as well, by choosing a smaller inner interval. The displayed tail identity is simply $(B^{1/3^k})^{3^{k+n}} = B^{3^n}$. The assertion about the finite prefix follows from membership in I. □

Remark 25.3 (Scope of the obstruction). The quantifiers in Theorem 25.1 are

$$\text{for every finite } P, \text{ there exists a fixed } N \text{ and infinitely many } \beta.$$

The integer N and the coset may depend on P; no one element is asserted to avoid every prime. The normalized construction likewise asserts neither an all-prime tail nor a prime condition at the intervening indices between a fixed prefix and its prescribed tail. What is ruled out is a field-uniform proof using only Saito's strict conjugate bounds, the limiting trace condition, and divisibility by one fixed finite set of primes. The theorem does not rule out adaptive choices of divisors depending on β, other arithmetic information, or an argument using the least-prime property specific to the minimum defining Mills' constant.

26 The remaining Mills alternative

The results yield several rigorous forms of progress, but each has a precise quantifier. The simultaneous-window theorem closes a residue-defined subclass. The coefficient sieve controls the size of a possible exceptional set in boxes. The Pell certificates exclude entire arithmetic classes of parameters in one family. The conditional construction realizes every prescribed *finite* pattern, and the unit-lattice theorem preserves *fixed finite* local data. The new residual-field theorem supplies its own seed in every three-adically admissible cubic field; this still preserves a finite-divisor quantifier, not primality. None of these statements exchanges a finite quantifier for an infinite one.

Three specific obstructions remain.

1. At a nonintegral three-adic limit, the prime-factor order budget works for every fixed shift. At $\lambda = \pm 1$, its key distance from ± 1 is zero for the unshifted trace. These are exactly the residual classes in which a new argument is required.

2. A coefficient set of stretched-exponentially small proportion can still contain a particular algebraic candidate. An ambient box estimate gives no automatic exclusion on a fixed field or on the exponentially sparse Pell orbit. The separate Pell calculation is needed precisely for this reason.

3. An infinite collection of algebraic numbers, each with a long finite prime prefix, need not contain one number with an infinite prime tower. Nor does it determine the minimum defining Mills' constant. The minimum requires a global prime-chain argument in addition to local polynomial information.

Consequently the remaining full targets are unchanged: Saito's cubic Pisot Problem 1.1, the exclusion of the Mills cubic alternative, and irrationality of the least square-clock constant. What is established here is a stronger collection of restricted theorems, quantitative bounds, and exact exclusions, with reusable proofs.

A concrete remaining target using minimality. For a surviving cubic candidate let $T_n = \mathrm{Tr}(\beta^{3^n})$. A hypothetical algebraic least Mills constant requires both eventual primality of T_n and, for every sufficiently large n,

$$T_{n+1} = \mathrm{nextprime}(T_n^3).$$

The latter property is an inherited observation from the supplied manuscript [010]. To see why it holds, the prime-interval bound used in [14] extends every sufficiently large prime q to a prime child strictly between q^3 and $(q+1)^3-1$. An earlier prime child of T_n could therefore be continued to an infinite nested prime chain below the defining minimum, a contradiction. Consequently it would suffice to prove the following restricted, still unproved assertion: every surviving candidate either has infinitely many composite T_n, or admits arbitrarily large n with a prime strictly between T_n^3 and T_{n+1}. This target uses the least-prime selection rule that the fixed-prime survival theorem does not encode. It would settle Mills transcendence without requiring composite floors for all cubic Pisot numbers.

Verification boundary. The present edition replays the complete Python certificate below and adds independent checks of the new structural extensions. The following additional audit description is inherited from the source manuscript: its central symbolic arguments were checked in parallel. The residual-field proofs were checked independently, including the ramified local algebra at three, all-rational-power quantifiers, and the distinction between a finite-prime sieve and an all-prime tower. Exact arithmetic replayed the two finite-field factorizations of the balanced seed, the old 26 second-orbit Pell witnesses and five first-orbit witnesses, the three new witnesses, and the completeness of the new residue cover. Additional exhaustive small-prime checks test the shifted local counts; they are consistency checks, not substitutes for the all-degree proof. No large numerical prime certificate from an earlier submission is needed for the new theorems. A compact standard-library certificate is included below.

27 Prime clocks in arbitrary degree and polynomial observables

The simultaneous-return argument extends beyond cubic traces. The degree enters only through a finite-field order budget. This section is independent of the phase-embedding hypotheses used in the Euler–Bessel part of the paper.

Let ℓ be a prime, let $d \ge 1$, and let $\alpha_1, \dots, \alpha_d$ be the roots, counted with multiplicity, of a monic polynomial in $\mathbb{Z}[X]$. Define

$$\boldsymbol{a}_n = (a_{1,n}, \dots, a_{d,n}), \qquad a_{j,n} = e_j(\alpha_1^{\ell^n}, \dots, \alpha_d^{\ell^n}),$$

where e_j is the jth elementary symmetric polynomial. All $a_{j,n}$ are integers. Put

$$r_\ell = \begin{cases} 2, & \ell = 2, \\ \ell - 1, & \ell > 2, \end{cases} \qquad b_{\ell,d} = \lfloor \log_\ell d \rfloor.$$

Lemma 27.1 (Coefficient contraction at every prime clock)**.** *There is an integral polynomial map $F_\ell : \mathbb{Z}^d \to \mathbb{Z}^d$ such that $\boldsymbol{a}_{n+1} = F_\ell(\boldsymbol{a}_n)$ and, as a polynomial identity modulo ℓ,*

$$(F_\ell)_j(X_1, \dots, X_d) \equiv X_j^\ell \pmod{\ell}.$$

The sequence $\boldsymbol{a}_n$ converges to a vector $\boldsymbol{\Lambda} \in \mathbb{Z}_\ell^d$, and

$$\boldsymbol{\Lambda} - \boldsymbol{a}_n \in \ell^{n+1}\mathbb{Z}_\ell^d.$$

The limit depends only on $\boldsymbol{a}_0 \pmod{\ell}$.

Proof. The fundamental theorem of symmetric polynomials supplies the integral map. In characteristic ℓ, the identity $e_j(T_1^\ell, \dots, T_d^\ell) = e_j(T_1, \dots, T_d)^\ell$ proves the displayed polynomial congruence. Consequently every first partial derivative of $(F_\ell)_j$ is divisible by ℓ. If $\boldsymbol{x} \equiv \boldsymbol{y}$

(mod ℓ^s) with $s \geq 1$, the linear terms of the integral Taylor expansion are divisible by ℓ^{s+1} and all higher terms by ℓ^{2s}. Thus

$$F_\ell(\boldsymbol{x}) \equiv F_\ell(\boldsymbol{y}) \pmod{\ell^{s+1}}.$$

Fermat's congruence gives $F_\ell(\boldsymbol{a}_0) \equiv \boldsymbol{a}_0 \pmod{\ell}$. Iteration yields $\boldsymbol{a}_{n+1} - \boldsymbol{a}_n \in \ell^{n+1}\mathbb{Z}^d$ and the asserted limit and error bound. Applying the same contraction to two initial vectors congruent modulo ℓ proves the last assertion. □

Lemma 27.2 (A uniform residual-order bound)**.** *Let $q \neq \ell$ be a rational prime. Then*

$$v_\ell\Big(\operatorname{lcm}_{1\leq f\leq d}(q^f - 1)\Big) \leq v_\ell(q^{r_\ell} - 1) + b_{\ell,d}.$$

Proof. For odd ℓ, put $h = \operatorname{ord}_{\mathbb{F}_\ell^\times}(q)$. If $h \nmid f$, the relevant valuation is zero. Otherwise the lifting-the-exponent identity gives

$$v_\ell(q^f - 1) = v_\ell(q^h - 1) + v_\ell(f/h) = v_\ell(q^{\ell-1} - 1) + v_\ell(f/h),$$

which has the required bound. For $\ell = 2$, the prime q is odd. If f is odd then $v_2(q^f - 1) = v_2(q - 1) \leq v_2(q^2 - 1)$. If f is even, the same elementary identity gives $v_2(q^f - 1) = v_2(q^2 - 1) + v_2(f) - 1$. Taking the maximum over $1 \leq f \leq d$ proves the assertion, with a harmless slack in the dyadic bound. □

Theorem 27.3 (Simultaneous composite polynomial observables)**.** *Let $P_1, \ldots, P_s \in \mathbb{Z}[X_1, \ldots, X_d]$, and suppose*

$$Q_j(n) := P_j(\boldsymbol{a}_n) \longrightarrow +\infty \qquad (1 \leq j \leq s).$$

Write $\lambda_j = P_j(\boldsymbol{\Lambda})$. If

$$\lambda_j^{r_\ell} \neq 1 \qquad (1 \leq j \leq s),$$

there are integers $n_0 \geq 0$ and $D \geq 1$ for which every $Q_j(n_0 + mD)$ is composite for all sufficiently large integers m. In particular the simultaneous composite indices have positive lower density.

Proof. If $\ell \mid \lambda_j$, then $\ell \mid Q_j(n)$ at every index, because the coefficient vector is constant modulo ℓ. This supplies a fixed recurring prime for that observable.

For the remaining indices j, define the finite integer $K_j = v_\ell(\lambda_j^{r_\ell} - 1)$. Choose one sufficiently large n_0 so that all $Q_j(n_0) > 1$, so that

$$v_\ell(Q_j(n_0)^{r_\ell} - 1) = K_j, \qquad n_0 \geq K_j + b_{\ell,d}$$

for every unit limit, and so that the limits have stabilized to these valuations. Such a choice follows from Lemma 27.1 and the assumed real growth.

Fix one unit limit. Some prime divisor q_j of $Q_j(n_0)$ satisfies

$$v_\ell(q_j^{r_\ell} - 1) \leq K_j.$$

Indeed, if every prime divisor had valuation at least $K_j + 1$, raising the prime factorization of $Q_j(n_0)$ to the power r_ℓ would give $Q_j(n_0)^{r_\ell} \equiv 1 \pmod{\ell^{K_j+1}}$, a contradiction. Here $q_j \neq \ell$ because $Q_j(n_0)$ is an ℓ-adic unit.

Reduce the original degree-d polynomial modulo q_j. Every nonzero residual root belongs to $\mathbb{F}_{q_j^f}$ for some $f \leq d$. By Lemma 27.2, the ℓ-primary part of its multiplicative order divides ℓ^{n_0}. Its ℓ^{n_0}th power therefore has order prime to ℓ. There is a common $D_j \geq 1$ such that raising

all these powered roots to ℓ^{D_j} fixes each of them. Zero roots remain zero, and repeated roots retain their multiplicities. Hence the entire coefficient vector satisfies

$$\boldsymbol{a}_{n_0+mD_j} \equiv \boldsymbol{a}_{n_0} \pmod{q_j} \qquad (m \geq 0),$$

and $q_j \mid Q_j(n_0 + mD_j)$.

Take D to be the least common multiple of the finitely many D_j, using $D = 1$ if all observables have ℓ-divisible limits. All selected primes recur on this common progression. Since every $Q_j(n)$ tends to $+\infty$, each selected divisor is eventually proper. The progression has density $1/D$ in the nonnegative indices. □

Corollary 27.4 (Pisot floor windows in every degree)**.** *Let β be any Pisot number, of any degree, and let $T_n = \mathrm{Tr}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta^{\ell^n})$. Write $\lambda = \lim_n T_n \in \mathbb{Z}_\ell$. For an integer $H \geq 0$, assume*

$$(\lambda + h)^{r_\ell} \neq 1 \qquad (-H - 1 \leq h \leq H).$$

There is an arithmetic progression of indices on which, eventually, every integer

$$\lfloor \beta^{\ell^n} \rfloor + h, \qquad |h| \leq H,$$

is composite. For $\ell = 2$ or $\ell = 3$, a single sufficient condition for this conclusion at every fixed H is $\lambda \notin \mathbb{Z}$.

Proof. Apply Theorem 27.3 to the linear observables $P_h(\boldsymbol{X}) = X_1 + h$ for $-H-1 \leq h \leq H$. The Pisot property gives $T_n - \beta^{\ell^n} \to 0$ in $\mathbb{R}$, and $T_n \to +\infty$. Thus the floor belongs to $\{T_n - 1, T_n\}$ for every sufficiently large n, regardless of the sign of the error. Both possible floor windows are contained in the selected trace window. If $\ell = 2$ or 3, then $r_\ell = 2$, so the equation $x^{r_\ell} = 1$ has only the two solutions 1 and -1; equality at an integer shift would force $\lambda \in \mathbb{Z}$. □

Example 27.5 (A quartic window theorem). Let β be the real root greater than one of

$$f(X) = X^4 - X^3 - X^2 - X - 1.$$

For every fixed H, all integers in the radius-H window around $\lfloor \beta^{3^n} \rfloor$ are simultaneously composite on an eventual arithmetic progression of indices.

Here are complete verifications of the hypotheses. Since $f(1) < 0 < f(2)$, a root lies in $(1, 2)$. The identity $(X - 1)f(X) = X^5 - 2X^4 + 1$ and Rouché's theorem show that for every $r > 1$ sufficiently close to one, the latter polynomial has exactly four roots in $|X| < r$: on that circle, $r^5 + 1 < 2r^4$. One of these roots is 1, so f has three roots in $|X| < r$. It has no root on $|X| = 1$, since equality in $X^5 + 1 = 2X^4$ would force both $X^5 = 1$ and $X^4 = 1$, hence $X = 1$, while $f(1) \neq 0$. Its other three roots therefore lie strictly in the unit disc. A nontrivial monic integral factor without the root β would have a nonzero integral constant term of absolute value less than one. Thus f is irreducible and β is a quartic Pisot unit.

Modulo three,

$$f(X) = (X - 1)(X^3 - X + 1),$$

and the cubic factor has no root in $\mathbb{F}_3$. The three-adic trace limit is the sum of the Teichmüller lifts of the four residual roots. To see this directly, work in an unramified splitting field: each unit root is its Teichmüller lift times a principal unit, whose 3^nth power tends to one; Frobenius permutes the multiset of Teichmüller lifts. The linear factor contributes 1. The cubic factor contributes

$$\varepsilon(\zeta^a + \zeta^{3a} + \zeta^{9a}), \qquad \varepsilon \in \{1, -1\}, \quad a \not\equiv 0 \pmod{13},$$

where ζ is a primitive thirteenth root of unity in the unramified cubic extension of $\mathbb{Q}_3$. This sum is not rational. Otherwise a polynomial of degree at most twelve, with at most four nonzero coefficients, would vanish at ζ, contradicting its minimal polynomial $1 + X + \cdots + X^{12}$ over $\mathbb{Q}$. Thus $\lambda \notin \mathbb{Z}$, and Corollary 27.4 applies.

Proposition 27.6 (An explicit quartic progression)**.** *For the quartic Pisot number in Example 27.5, the three integers*

$$\lfloor \beta^{3^n} \rfloor - 1, \qquad \lfloor \beta^{3^n} \rfloor, \qquad \lfloor \beta^{3^n} \rfloor + 1$$

are all composite for every sufficiently large $n \equiv 4 \pmod{68}$.

Proof. Write $S_m = \sum_{i=1}^{4} \alpha_i^m$ and $T_n = S_{3^n}$. Newton's identities give

$$S_0 = 4, \quad S_1 = 1, \quad S_2 = 3, \quad S_3 = 7, \qquad S_{m+4} = S_{m+3} + S_{m+2} + S_{m+1} + S_m,$$

and hence the exact value $T_4 = S_{81} = 121808162096862030748321$. The following table records recurring divisors of four consecutive trace shifts; the period refers to the index n after its initial transient.

Shift h	Prime q	$T_4 \bmod q$	Period
-2	17	2	68
-1	3	1	1
0	11	0	4
1	2	1	4

Here is a finite verification of the periods. Modulo two, $f(X) = \Phi_5(X)$, so the nonzero root orders divide five and $3^4 \equiv 1 \pmod 5$. Modulo eleven,

$$f(X) = (X^2 + X - 3)(X^2 - 2X + 4).$$

The roots lie in $\mathbb{F}_{11^2}^{\times}$; after one cubing their orders divide forty, and $3^4 \equiv 1 \pmod{40}$. Modulo seventeen,

$$f(X) = (X - 6)(X^3 + 5X^2 - 5X + 3),$$

where the cubic has no root in $\mathbb{F}_{17}$ and is irreducible. The root orders thus divide $17^3 - 1 = 4912$, and $3^{68} \equiv 1 \pmod{4912}$. Finally the Frobenius identity gives $T_n \equiv S_1 = 1 \pmod 3$ for every n. All four listed primes therefore divide their indicated shifts when $n \equiv 4 \pmod{68}$. The shifts grow to infinity, making the divisors proper, and the Pisot floor belongs to $\{T_n - 1, T_n\}$ eventually. Its radius-one window is consequently contained in these four composite trace shifts. □

Remark 27.7 (What the extension does and does not exclude). The theorem enlarges the degree, the prime clock, and the class of observables without assuming prime initial values. Its excluded roots of unity are genuine limitations of this argument: for odd ℓ, an ℓ-adic unit may satisfy $\lambda^{\ell-1} = 1$ without being the rational number 1 or -1. Therefore the noninteger-limit simplification in Corollary 27.4 is asserted only for clocks two and three. This extension does not exclude the residual cubic alternatives with trace limit 1 or -1, and hence does not settle the full cubic Pisot question or Mills' transcendence. No external priority claim for the general extension is made here.

28 The predecessor branch survives every fixed modulus

The three-adic embedding criterion for a negative dominant small conjugate does not extend unchanged to the other floor branch. In fact the predecessor branch admits the following uniform construction in every totally real cubic field. The modulus may contain arbitrary prime powers, and the conclusion holds at every positive integer exponent, rather than only at cubic-power exponents.

Theorem 28.1 (Constant invertible floor residues in every field)**.** *Let K be a totally real cubic field with ordered embeddings $\sigma_1, \sigma_2, \sigma_3$, let $M \geq 2$, and fix $0 < a < 1/2$. There exist a positive integer C, a positive integer M' divisible by M, and a residue $c_M \in (\mathbb{Z}/M\mathbb{Z})^\times$ such that infinitely many*

$$\beta \in CE^+(M'\mathcal{O}_K)$$

have embeddings $B = \sigma_1(\beta)$, $r = \sigma_2(\beta)$, $s = \sigma_3(\beta)$ satisfying

$$B > 1, \qquad 0 < s < r < B^{-a}, \qquad r + s < 1.$$

Each such B is a cubic Pisot number generating K, has norm C^3 and three-adic trace limit zero, and satisfies

$$\lfloor B^m \rfloor = \mathrm{Tr}_{K/\mathbb{Q}}(\beta^m) - 1 \equiv c_M \pmod{M} \qquad (m \geq 1). \tag{31}$$

The residue can be chosen to be -1 modulo the 2- and 3-primary parts of M, and 2 modulo every other prime-power factor of M. If $\Delta_{M'}$ is the two-coordinate logarithmic covolume of $E^+(M'\mathcal{O}_K)$, the number of elements in this coset with $B \leq H$ and the displayed inequalities is

$$\frac{1-2a}{4\Delta_{M'}}(\log H)^2 + O_{K,C,M',a}(\log H + 1).$$

Proof. Put $M' = \mathrm{lcm}(M, 6)$. By the Chinese remainder theorem, choose $C > 0$ such that $C \equiv 0$ modulo the full 2- and 3-primary parts of M', and $C \equiv 1$ modulo every other prime-power factor of M'. Every element of $CE^+(M'\mathcal{O}_K)$ is totally positive, has norm C^3, and is congruent to the scalar C modulo $M'\mathcal{O}_K$. Consequently, for every $m \geq 1$,

$$\mathrm{Tr}(\beta^m) - 1 \equiv \begin{cases} -1 \pmod{p^e}, & p \in \{2,3\},\ p^e \mid M', \\ 2 \pmod{p^e}, & p \geq 5,\ p^e \mid M'. \end{cases}$$

All these residues are units. Since $3 \mid C$, the residual characteristic polynomial at three is X^3, and the three-adic trace limit is zero.

Write $x = \log B$, $y = \log r$, and $L = 3\log C$. The third embedding has logarithm $L - x - y$. Apart from the extra condition $r + s < 1$, the required region is

$$0 < x \leq \log H, \qquad (L - x)/2 < y < -ax.$$

Its area is $(1-2a)(\log H)^2/4 + O_{C,a}(\log H + 1)$ and its boundary length is $O_{C,a}(\log H + 1)$. Its logarithmic coset is a translate of a rank-two lattice of covolume $\Delta_{M'}$. Counting fundamental parallelograms gives the stated asymptotic. Moreover $r + s < 2B^{-a} < 1$ once $B > 2^{1/a}$, so imposing this extra condition removes only a bounded portion of the region and does not alter the asymptotic.

The resulting dominant embedding is greater than one, and its two other conjugates lie in $(0, 1)$. It cannot be rational, and hence has degree three. Finally $0 < r^m + s^m \leq r + s < 1$ for every $m \geq 1$. The integer trace is $B^m + r^m + s^m$, which proves both the floor identity and its congruence in (31). □

Corollary 28.2 (No field-uniform finite divisor cover in this branch)**.** *For every totally real cubic field K and every finite set P of rational primes, infinitely many cubic Pisot numbers in K have $\gcd(\lfloor B^m \rfloor, \prod_{p \in P} p) = 1$ for every $m \geq 1$. This remains true in fields having no embedding into $\mathbb{Q}_3$.*

Proof. Take M divisible by every member of P and apply Theorem 28.1. □

The norm is fixed within each construction but may depend on M; it is divisible by three. Thus the corollary does not contradict the prime-to-three norm hypothesis in Theorem 18.2. It also does not assert primality of any value: the surviving floors may have prime divisors outside the prescribed finite set. In particular it neither disproves Saito's problem nor eliminates its remaining cases.

28.1 An eventual-compositeness refinement of the parity criterion

Proposition 28.3. *Let β be a cubic Pisot number with minimal polynomial X^3-uX^2+vX-N, and suppose $3 \mid u$. If either*

$$N \equiv 1 \pmod 2, \quad u \equiv v \pmod 2, \qquad \text{or} \qquad N \equiv 0, \quad u \equiv 1, \quad v \equiv 0 \pmod 2,$$

then $\lfloor \beta^{3^n} \rfloor$ is composite for every sufficiently large n. Nonreal conjugates are allowed. If instead N is odd and $u \not\equiv v \pmod 2$, the same argument guarantees composite floors on one parity class of indices, and hence lower density at least $1/2$.

Proof. The trace U_n remains zero modulo three. With odd norm the coefficient pair modulo two follows $00 \mapsto 11 \mapsto 11$ and $01 \leftrightarrow 10$; thus equal initial parities make U_n odd eventually, and opposite initial parities make it odd on one parity class. With even norm the map is $(a,b) \mapsto (a+ab,b)$, so the state $(1,0)$ is fixed. In each case U_n is odd at the asserted indices. The Pisot estimate puts the sufficiently late floor in $\{U_n, U_n-1\}$. At those indices U_n is divisible by three and U_n-1 is divisible by two; both are eventually larger than these divisors. □

29 Conclusions and remaining exact targets

The composite-window theorem now applies in every degree and for every prime clock, subject to its explicit local-limit exclusion. The finite winding theorem replaces infinitely many exact specializations by a degree-dependent finite test in the mixed two-block setting, under the same independence hypotheses as its intersection theorem. The derivative determinant is obtained by a direct differential-algebra calculation. None of these statements removes the surviving $\lambda = \pm 1$ cubic Mills branch, supplies the missing joint-independence proof, or identifies the canonical winding parameter arithmetically. Those remain separate requirements for stronger conclusions.

A The inherited Pell reduction and the exact finite cover

For self-contained use of the new certificate we record the earlier reduction and cover. The substitutions $x = 8u+5$, $y = 4w$ transform $4u^2+5u+23 = 2w^2$ into

$$x^2 - 2y^2 = -343, \qquad x \equiv 5 \pmod 8, \quad y \equiv 0 \pmod 4.$$

Multiplying $x+y\sqrt{2}$ by powers of $3-2\sqrt{2}$ reduces it uniquely to $[\sqrt{343}, (3+2\sqrt{2})\sqrt{343})$. This bounds $0 < x < 53$, $0 < y < 40$. Enumeration gives exactly $(7,14),(13,16),(25,22),(35,28)$. Their four-term residue cycles modulo eight retain only the first seed at odd powers of $3+2\sqrt{2}$, and the second at even powers. These become the two orbits from $(9,14)$ and $(1,4)$ under T. The latter initial point is omitted because $u=1$ is a reducible degeneration.

For the first orbit a_j, the residues have periods two, ten, and five modulo $3, 19, 29$ respectively:

3	$0, 1$
19	$9, 8, 17, 1, 18, 4, 5, 15, 12, 14$
29	$9, 12, 13, 15, 24.$

Even indices give trace zero modulo three. The five odd classes $j = 1, 3, 5, 7, 9 \pmod{10}$ are covered by $(p, c, n) = (29, 12, 41), (29, 15, 5), (19, 4, 17), (29, 13, 1), (19, 14, 7)$. Every listed zero is beyond its root-order return budget. For the second orbit odd j give trace zero modulo three. Writing $j = 2k$, the following old rows cover all k except $0, 2187 \pmod{6561}$; a pair (a, n) means a recurring zero at trace index n whenever $k \equiv a \pmod m$.

p	m	(a, n)
5	3	$(1, 3)$
7	3	$(2, 8)$
179	9	$(3, 52)$
199	9	$(6, 126)$
107	27	$(9, 4), (18, 25)$
487	81	$(27, 366), (54, 1333)$
4373	729	$(162, 2379), (486, 7323), (567, 3228)$
20411	729	$(81, 7662), (405, 34139)$
8747	2187	$(243, 7161), (324, 2814), (648, 105), (729, 830)$
		$(1053, 2330), (1701, 3456), (1782, 4165), (2106, 12465)$
52489	2187	$(1377, 54889)$
39367	6561	$(3645, 45679), (4374, 15588)$
629857	2187	$(972, 647491), (1458, 678551)$

The three new rows of Theorem 20.1 reduce the survivor set to $k \equiv 0 \pmod{19683}$.

To verify the valuation formula used there, set $(17 + 12\sqrt{2})^{2k} = C_k + D_k\sqrt{2}$. Then

$$b_{2k} - 1 = \frac{13(C_k - 1) + 32D_k}{8}, \quad D_{3k} = D_k(3 + 8D_k^2).$$

For $3 \nmid k$, reduction modulo nine gives $v_3(D_k) = 1$. Repeated cubing gives $v_3(D_k) = 1 + v_3(k)$ for every $k > 0$. Since $C_k^2 - 2D_k^2 = 1$ and $C_k \equiv 1 \pmod 3$, $v_3(C_k - 1) = 2v_3(D_k)$. The term involving D_k therefore determines the numerator's valuation, proving $v_3(b_j - 1) = 1 + v_3(j)$ for even $j > 0$.

B A reproducible exact certificate

The following Python 3 code uses only the standard library. It checks primality by trial division, modular periods in the quadratic ring, cubic trace zeros by binary polynomial powering, and the full residue cover. It never constructs a trace of millions-digit exponent. The algebraic reasons why these finite checks imply infinite progressions are given in the proofs above.

```
from math import isqrt

rows=[(5,3,[(1,3)]),(7,3,[(2,8)]),(179,9,[(3,52)]),(199,9,[(6,126)]),(107,27,[(9,4),(18,25)])
     ,(487,81,[(27,366),(54,1333)]),(4373,729,[(162,2379),(486,7323),(567,3228)]),(20411,729,[(81,7662)
     ,(405,34139)]),(8747,2187,[(243,7161),(324,2814),(648,105),(729,830),(1053,2330),(1701,3456)
     ,(1782,4165),(2106,12465)]),(52489,2187,[(1377,54889)]),(39367,6561,[(3645,45679),(4374,15588)])
     ,(629857,2187,[(972,647491),(1458,678551)])]
def v3(n):
 a=0
 while n%3==0 and n: a+=1;n//=3
 return a

def qm(x,y,p):return ((x[0]*y[0]+2*x[1]*y[1])%p,(x[0]*y[1]+x[1]*y[0])%p)
def qp(x,n,p):
 r=(1,0)
 while n:
  if n&1:r=qm(r,x,p)
  x=qm(x,x,p);n//=2
 return r

def b(j,p):
 x,y=qm((13,16),qp((17,12),j,p),p)
 return ((x-5)*pow(8,-1,p))%p

def pm(a,b,u,p):
 c=[0]*5
 for i,x in enumerate(a):
  for j,y in enumerate(b):c[i+j]=(c[i+j]+x*y)%p
 for k in (4,3):
```

```
  v=c[k];c[k]=0;c[k-1]=(c[k-1]+u*v)%p;c[k-2]=(c[k-2]+v)%p;c[k-3]=(c[k-3]-v)%p
 return c[:3]
def pp(a,n,u,p):
 r=[1,0,0]
 while n:
  if n&1:r=pm(r,a,u,p)
  a=pm(a,a,u,p);n//=2
 return r

def trace(n,u,p):
 M=(p-1)*(p*p-1)*(p**3-1); E=pow(3,n,M); A,B,C=pp([0,1,0],E,u,p)
 return (3*A+u*B+(u*u+2)*C)%p
for p,m,items in rows:
 assert p>=2 and all(p%d for d in range(2,isqrt(p)+1)),p
 assert qp((17,12),2*m,p)==(1,0),(p,m)
 for a,n in items:
  assert n>=v3((p-1)*(p*p-1)*(p**3-1)),(p,n)
  assert trace(n,b(2*a,p),p)==0,(p,m,a,n,b(2*a,p))
surv=[k for k in range(6561) if all(all(k%m!=a for a,n in items) for p,m,items in rows)]
assert surv==[0,2187],surv
for p,c,n in [(29,12,41),(29,15,5),(19,4,17),(29,13,1),(19,14,7)]:
 assert trace(n,c,p)==0
 assert n>=v3((p-1)*(p*p-1)*(p**3-1))
print('All 26 second-orbit witnesses and 5 first-orbit witnesses verified by exact arithmetic.')
print('Surviving k residues modulo6561:',surv)

# Three additional witnesses: (prime, parameter period, class, trace index).
new=[(8279983,6561,2187,4863973),
     (27398737,19683,6561,4929815),
     (15037811,19683,13122,549606)]
for p,m,a,n in new:
 assert all(p%d for d in range(2,isqrt(p)+1))
 assert qp((17,12),2*m,p)==(1,0)
 assert n>=v3((p-1)*(p*p-1)*(p**3-1))
 assert trace(n,b(2*a,p),p)==0
remaining=[k for k in range(19683)
 if all(all(k%m!=a for a,n in items) for p,m,items in rows)
 and all(k%m!=a for p,m,a,n in new)]
assert remaining==[0]
# Reduced Pell seeds and first-orbit parameter periods.
seeds=[(x,y) for x in range(1,53) for y in range(1,40)
       if x*x-2*y*y==-343]
assert seeds==[(7,14),(13,16),(25,22),(35,28)]
def orbit(seed,p,length):
 u,w=seed;out=[]
 for j in range(length):
  out.append(u%p);u,w=(17*u+12*w+10)%p,(24*u+17*w+15)%p
 assert (u,w)==(seed[0]%p,seed[1]%p)
 return out
assert orbit((9,14),3,2)==[0,1]
assert orbit((9,14),19,10)==[9,8,17,1,18,4,5,15,12,14]
assert orbit((9,14),29,5)==[9,12,13,15,24]
assert orbit((1,4),3,2)==[1,0]
print('All certificates verified; surviving k modulo19683:',remaining)
```

C Source map and attribution

The earlier archive cited by the source manuscript is research material, not a collection of independently published papers. Round identifiers below refer to `Race244_Topic02(1).zip`; S means Showa. The earlier numbered manuscripts refer to the previously supplied 36-paper collection.

Source	Use and status
S12, S27	Local factor blocks and uniform irreducibility, reproduced and audited.
S28	Finite prime/composite patterns under H; exact multiplicities are added here.
S35, S36	Complete Pell reduction and inherited finite cover; three new rows added.
S37	Fixed-norm trace-zero counts and unshifted quantitative sieve; extended to all shifts and odd degrees.
S14	Inherited all-norm cyclic prime-trace restrictions; used in the positive-density floor synthesis.
S20	Prior norm-one trace theorem in discriminant 1524; used only in the explicitly identified field corollary.
Heisei R05	A cyclic cubic seed for finite-sieve survival; the coset-counting theorem is proved here.
Old [007], [008]	Previously resolved dyadic Fibonacci and nonreversible recurrence problems.
Old [009], [031]	Earlier local obstructions, the three-adic embedding lemma, nonreal case, qualitative density and dimension results.

The source manuscript reports that other topics were screened for useful connections. The pure-cubic sixth-power observation recovers results already subsumed by the old nonreal-cubic argument. No proved identity in the Gieseking, integral-period, or elliptic-resolvent notes supplies a missing prime-divisor theorem for a surviving Mills candidate. No such speculative connection is used as a proof input.

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